MAT120_SUMMER_2024_SKY

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or

SCAN the following QR Code:

You can share this link with the following foreword to anyone or Keep it confidential before end of this semester.

FOREWORD

"These resources are only for BRACU students of MAT 120 course instructed by me, some of these are my personal comments and not a universal and complete guide for
INTEGRAL CALCULUS AND DIFFERENTIAL EQUATION.
The reference books are better to refer to anyone for a complete view."

-

Sakal Roy
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Academic Counselor,
BdMOC,
Former Senior Math Olympiad Consultant,
PEDP4, DPE, MoPME,
Author, Primary and Secondary Mathematics Textbook,
NCTB

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COURSE INFO

% Will be provided by Course Teacher

* Will be finalized by Course Coordinator

Will be provided by Course Teacher and assessed by Student Tutor

Course ID: MAT 120 **Course Name:** Integral Calculus & Differential Equations

Faculty Name: SAKAL ROY **Initial:** SKY

Semester: SUMMER-2024 **Section:** 5 , 6

Email: Official: ext.sakal.roy@bracu.ac.bd **Personal:** sakalroy10@gmail.com

% Class Info:

Section	Day	Class Time 🕑	Room No
5	SAT & THU	2:00 PM - 3:20 PM	10H-37C
6	SAT & THU	3:30 PM - 4:50 PM	10-H37C

% My Consultation Info:

Consultation Room No:

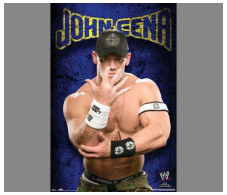
4 th Floor, MNS, Cubicle/Open Space for faculties (1 st Row, Check My NAME and INITIAL:SKY)

Consultation Hour:

🕑 SAT : 11:30 AM - 1:00 PM

🕑 MON : 12:30 PM - 2:00 PM

Contact in my personal/official email in if:



Student Tutor(ST):

Sec: 5 : **Name:** MALIHA MUSSARRAT, **Email:** maliha.mussarrat@g.bracu.ac.bd

ST Consultation Info: (This info is from previous semester, this years has not been announced yet, that's why it's yellow marked, will be updated when ST lets you know)

Consultation Room No: UB 6 0 6 0 1 (Building 6)

Consultation Hour:

🕑 SUN : 3:30 PM - 4:50 PM

🕑 TUE : 3:30 PM - 4:50 PM

🕑 WED : 11:00 AM - 12:20 PM

🕑 WED : 2:00 PM - 3:20 PM

🕑 THU : 8:00 AM - 9:20 AM

Sec: 6 : **Name:** Sajida Hossain, **Email:** sajida.hossain@g.bracu.ac.bd

ST Consultation Info: (This info is from previous semester, this years has not been announced yet, that's why it's yellow marked, will be updated when ST lets you know)

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🕑 WED : 2:00 PM - 3:20 PM

🕑 THU : 8:00 AM - 9:20 AM

*** Lecture Note: https://shorturl.at/ensGI**

*** Resources (Textbooks, Exercise Sheets with Practice Problems): https://shorturl.at/htyV9**

% The links will lead to a Google Drive. The content of these Google Drive will be referred for ASSIGNMENT, QUIZ because they are important for MID and FINAL EXAM also. Let me know if you have any problem accessing any of the material provided here.

Among the reference text books provided in the G-drive, I'll personally recommend **"CALCULUS: EARLY TRANSCENDENTALS by Anton, Bivens, Davis 10th Edition "**.

ANSWERS CAN BE FOUND HERE:

calculus-early-transcendentals-10th-ed-howard-anton-iril-bivens-stephen-davis-solution2.pdf

Surprisingly, I found the authors of this book share quite similar tastes with me in terms of explaining. So if you enjoy my class lectures, you can follow this reference book. I hope you can connect with this book easily and enjoy it while preparing.

% Group Communication:

The slack channel links are given below. Find your section and join only there accordingly.

Use your personal mail in case you have a problem accessing your BRACU g-suite email address.

- **Section 5 : Join using the link:**
  https://join.slack.com/t/mat120sec05/shared_invite/zt-2ksm5a74w-zwPoa8trwm9CMSjz~W1HdQ

- **Section 6 : Join using the link:**
  https://join.slack.com/t/mat120sec06/shared_invite/zt-2kjbqa9ys-BIdGcFPkmeP8qB6RMHLqzQ

* **The mark distribution for MAT 120 in this semester:**

Attendance	5 %
Assignment	10 %
Quiz	15 %
Midterm	20 %
Final	30 %
Lab	20 %
Total	**100 %**

%Tentative dates and topics of Quiz and Assignment:

% Number of Quiz: 3 (best 2 will be counted) # Number of Assignment: 4 (best 3 will be counted)

Quiz/ Assignment	Quiz Date & Time/ Assignment Submission Deadline	Instruction
Assignment 1	**Submission Deadline:** TBA	**Topics:** Mathematical Reasoning **Submission Method:** Group Presentation
Assignment 2	**Submission Deadline:** TBA	**Topics:** TBA **Submission Method:** TBA
Assignment 3	**Submission Deadline:** TBA	**Topics:** TBA **Submission Method:** TBA
Assignment 4	**Submission Deadline:** TBA	**Topics:** TBA **Submission Method:** TBA
Quiz 1	**Date:** TBA **Time**: Class Time	**Topics:** Mathematical Reasoning **Submission Method:** Oral Presentation
Quiz 2	**Date:** TBA **Time**: TBA	**Topics:** TBA **Submission Method:** TBA
Quiz 3	**Date:** TBA **Time**: TBA	**Topics:** TBA **Submission Method:** TBA

*** Mid-term Exam Modality: Central**

Total marks: 40 Time: 1 hour 20 minutes

There'll be 6 questions from the 6 topics given below.

You've to answer any 5 questions.

Each Question will have 8 Marks. So in total 5 x 8 = 40 Marks.

Your Obtained Marks will be Converted into 20 marks later.

Topics:

- **Integration using Riemann Sums**
- **Integration and Antiderivatives, Indefinite & Definite integrals, Integrals with variable limits, Integration by Substitution**
- **Integration by parts, Reduction formula, Improper Integrals, Infinite integrals**
- **Integration by Trigonometric Substitution, Integration of Rational Function**
- **Beta and Gamma Function**
- **Length of a curve, Surface area of revolution**

***Points to be noted for MID-TERM Preparation:**

- **1 question will contain/focus on 1 set of topics only, that means there'll be no question which consists of multiple set of topics.**

- **You can use the theorems and definitions noted in Reference Books/Lecture Note Directly for corresponding question/topic. No need to prove any theorem unless specifically asked in the question. There'll be no question in which you need to prove any mathematical theorem.**
- **Practise all the problems of the Practice Sheets. If you're stuck in some problem see Reference Textbook (HOWARD ANTON will be enough in most cases). Use my given online solvers if needed.**
- **Use my given further resources and reference Book by HOWARD ANTON if you need to revisit the topics since they're mostly related to my class lectures.**
- **You may use the lecture notes only to get a list of the basic formulas which you can directly use and to find at least 1 worked out example for techniques needed for Practice Sheet Problems.**
- **DON'T try to go through everything from the Lecture Note or reference Book which are not needed to solve problems mentioned from Reference Text Books and problems in the Practice Sheets. Everything will not appear in the MID-TERM exam so focus only which parts you need. If you've confusion about any specific topics, feel free to DM me in Slack (or post it in general channel in Slack). You can email me in my official/personal email for any kind of query regarding the exam. But first go through my provided Further resources, they'll solve your problems easily and make you confident.**
- **I'll take 1 or 2 EXTRA ONLINE SESSION before your exam to clear any confusion. I'll announce the time/date in GOOGLE CLASSROOM/SLACK.**

- **Last but not the least DON'T MISS THE CLASSES. Best wishes.**
- **Here are a few resources that might help in last moment preparation:**
  - **Choosing what integration methods to use - Part I**
  - **Choosing what integration methods to use - Challenging Example!**
  - **MATH PROF vs TRICKY INTEGRALS**

*** Final Exam Modality: Central**

Total marks: 30 Time: 1 hour 20 minutes

There'll be 6 questions from the 6 topics given below.

You've to answer any 5 questions.

Each Question will have 6 Marks. So in total 5 x 6 = 30 Marks.

No conversion will happen to the FINAL EXAM marks.

Topics:

- **Basic concept of Double integral, Double integral in Cartesian and Polar Coordinate system**
- **Concept of Triple Integral, Triple integral in Cartesian, Cylindrical and Spherical Coordinate system.**
- **Introduction to Differential Equation, Separable Variables, Linear Equations, Integrating Factor**
- **Exact Equation, A non-Exact DE made Exact.**
- **Higher order DE, homogeneous and inhomogeneous cases.**
- **Modeling with 1^{st} and 2^{nd} order DE**

***Points to be noted for FINAL Preparation:**

-

%FURTHER RESOURCES:

Calculus Index:

Calculus

Calculus I

MAT137 - YouTube

Dr. Trefor Bazett - YouTube

Calculus I (Limits, Derivative, Integrals) **Full Course** - YouTube

Calculus II (Integration Methods, Series, Parametric/Polar, Vectors) **Full Course** - YouTube

Multivariable Calculus (Vectors, Curves, Partial Derivatives, Multiple Integrals, Optimization, etc) **Full Course ** - YouTube

Calculus IV: Vector Calculus (Line Integrals, Surface Integrals, Vector Fields, Greens' Thm, Divergence Thm, Stokes Thm, etc) **Full Course** - YouTube

Ordinary Differential Equations (ODEs) - YouTube

Prof Ghrist Math - YouTube

Calculus BLUE Vol 1 : Vectors & Matrices (MULTIVARIABLE) - YouTube

Calculus GREEN Vol 2 : Integration (SINGLE VARIABLE) - YouTube

Calculus BLUE Vol 3 : Integrals (MULTIVARIABLE) - YouTube

Multivariable Calculus Theory - YouTube

CALCULUS 3 CH 1 DOUBLE INTEGRALS - YouTube

NiLTime - YouTube

ClearMath - YouTube

Essence of calculus - YouTube

blackpenredpen - YouTube

Exact Equations (Nagle Sect2.4) - YouTube

Differential Equations: Special Integrating Factors (Nagle Sect2.5) - YouTube

Calculus 1 - Ximera

7. The definition of integral - YouTube
8. The Fundamental Theorem of Calculus - YouTube
9. Integration methods - YouTube
12. Improper integrals - YouTube

Calculus -- The foundation of modern science

This Is the Calculus They Won't Teach You

Mathematics in Movies

Online Solver:

Math Calculators on eMathHelp

Calculators - Calculus I - eMathHelp

Calculators - Calculus II - eMathHelp

Calculators - Calculus III - eMathHelp

Calculators - Differential Equations - eMathHelp

Free online calculators to solve mathematical tasks

Graphing Calculator - GeoGebra

3D Calculator - GeoGebra

MathDF

Integral Calculator

- **BEFORE INTEGRATION**

lim x->0 sinx /x , circular reasoning

link to derivation of area formulas

Area:

Square, Rectangle,

Parallelogram,

Triangle,Rhombus,

Circle, Circle Sector

Surface Area and Volume:
Cylinder,
Pyramid,
Cone,
Sphere

READ:

Geometry

Metric Volume

Conversion of Volume

https://www.mathsisfun.com/search/search.html?submit=&query=volume&search=1#ff

- **Integration using Riemann Sums, Integration and Antiderivatives, Indefinite & Definite integrals, Integrals with variable limits, Infinite Integral, Integration by Substitution**

Integration using Riemann Sums

READ:
Integral Approximations
Integral Approximations Calculator and Graph
Sigma Notation
Partial Sums
Riemann Sums

Calculus I - Summation Notation
Calculus I - Area Problem

How to calculate sum of series:sum_(n=0)^oor^nsin(nx) and sum_(n=0)^oor^ncos(nx)? (i don't have specifications for r). | Socratic

READ:

Georg Friedrich Bernhard Riemann (1826–1866) German mathematician. Bernhard Riemann, as he is commonly known, was the son of a Protestant minister. He received his elementary education from his father and showed brilliance in arithmetic at an early age. In 1846 he enrolled at Göttingen University to study theology and philology, but he soon transferred to mathematics. He studied physics under W. E. Weber and mathematics under Carl Friedrich Gauss, whom some people consider to be the greatest mathematician who ever lived. In 1851 Riemann received his Ph.D. under Gauss, after which he remained at Göttingen to teach. In 1862, one month after his marriage, Riemann suffered an attack of pleuritis, and for the remainder of his life was an extremely sick man. He finally succumbed to tuberculosis in 1866 at age 39.

An interesting story surrounds Riemann's work in geometry. For his introductory lecture prior to becoming an associate professor, Riemann submitted three possible topics to Gauss. Gauss surprised Riemann by choosing the topic Riemann liked the least, the foundations of geometry. The lecture was like a scene from a movie. The old and failing Gauss, a giant in his day, watching intently as his brilliant and youthful protégé skillfully pieced together portions of the old man's own work into a complete and beautiful system. Gauss is said to have gasped with delight as the lecture neared its end, and on the way home he marveled at his student's brilliance. Gauss died shortly thereafter. The results presented by Riemann that day eventually evolved into a fundamental tool that Einstein used some 50 years later to develop relativity theory.

In addition to his work in geometry, Riemann made major contributions to the theory of complex functions and mathematical physics. The notion of the definite integral, as it is presented in most basic calculus courses, is due to him. Riemann's early death was a great loss to mathematics, for his mathematical work was brilliant and of fundamental importance.

[*Image: http://commons.wikimedia.org/wiki/File:Georg_Friedrich_Bernhard_Riemann.jpeg*]

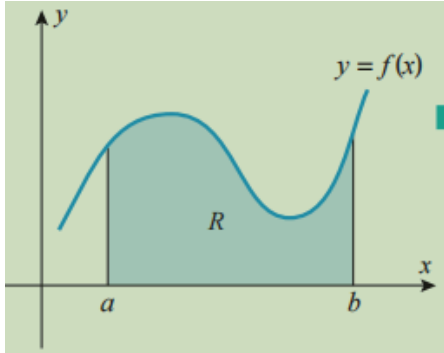
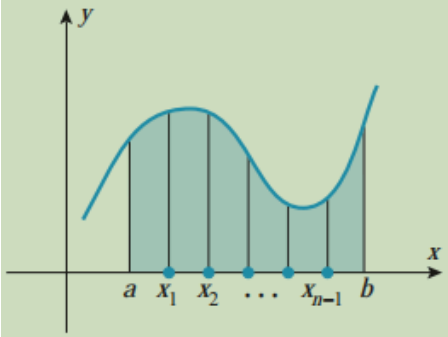


Figure 5.4.2

A DEFINITION OF AREA

We now turn to the problem of giving a precise definition of what is meant by the "area under a curve." Specifically, suppose that the function f is continuous and nonnegative on the interval $[a, b]$, and let R denote the region bounded below by the x-axis, bounded on the sides by the vertical lines $x = a$ and $x = b$, and bounded above by the curve $y = f(x)$ (Figure 5.4.2). Using the rectangle method of Section 5.1, we can motivate a definition for the area of R as follows:



▲ Figure 5.4.3

- Divide the interval $[a, b]$ into n equal subintervals by inserting $n - 1$ equally spaced points between a and b, and denote those points by

$$x_1, x_2, \ldots, x_{n-1}$$

(Figure 5.4.3). Each of these subintervals has width $(b - a)/n$, which is customarily denoted by

$$\Delta x = \frac{b - a}{n}$$

- Over each subinterval construct a rectangle whose height is the value of f at an arbitrarily selected point in the subinterval. Thus, if

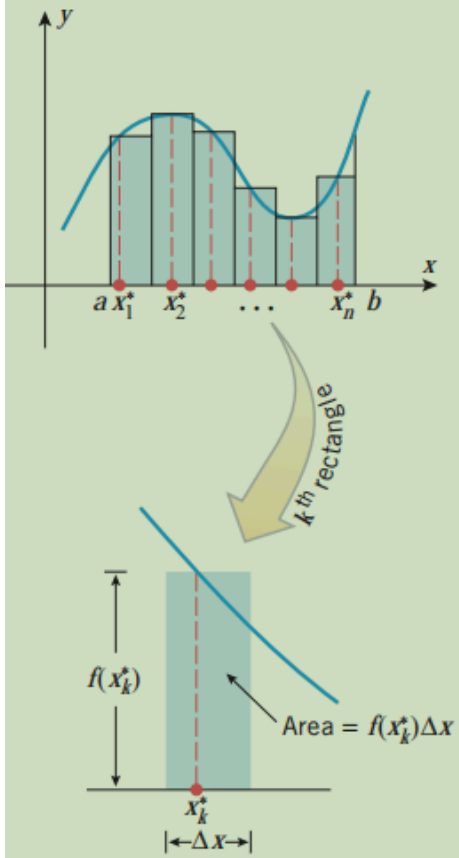


Figure 5.4.4

$$x_1^*, x_2^*, \ldots, x_n^*$$

denote the points selected in the subintervals, then the rectangles will have heights $f(x_1^*), f(x_2^*), \ldots, f(x_n^*)$ and areas

$$f(x_1^*)\Delta x, \quad f(x_2^*)\Delta x, \ldots, \quad f(x_n^*)\Delta x$$

(Figure 5.4.4).

- The union of the rectangles forms a region R_n whose area can be regarded as an approximation to the area A of the region R; that is,

$$A = \text{area}(R) \approx \text{area}(R_n) = f(x_1^*)\Delta x + f(x_2^*)\Delta x + \cdots + f(x_n^*)\Delta x$$

(Figure 5.4.5). This can be expressed more compactly in sigma notation as

$$A \approx \sum_{k=1}^{n} f(x_k^*)\Delta x$$

- Repeat the process using more and more subdivisions, and define the area of R to be the "limit" of the areas of the approximating regions R_n as n increases without bound. That is, we define the area A as

$$A = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x$$

In summary, we make the following definition.

Activate
Go to Settin

The limit in (2) is interpreted to mean that given any number $\epsilon > 0$ the inequality

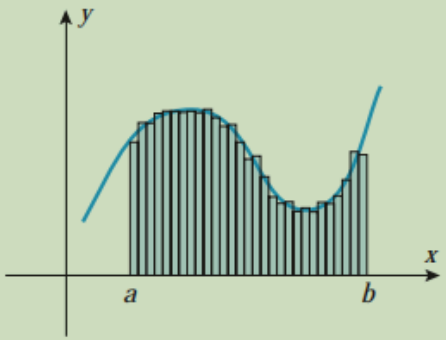
$$\left| A - \sum_{k=1}^{n} f(x_k^*)\Delta x \right| < \epsilon$$

holds when n is sufficiently large, no matter how the points x_k^* are selected.

5.4.3 DEFINITION (*Area Under a Curve*) If the function f is continuous on $[a, b]$ and if $f(x) \geq 0$ for all x in $[a, b]$, then the ***area*** A under the curve $y = f(x)$ over the interval $[a, b]$ is defined by

$$A = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x \tag{2}$$

REMARK There is a difference in interpretation between $\lim_{n \to +\infty}$ and $\lim_{x \to +\infty}$, where n represents a positive integer and x represents a real number. Later we will study limits of the type $\lim_{n \to +\infty}$ in detail, but for now suffice it to say that the computational techniques we have used for limits of type $\lim_{x \to +\infty}$ will also work for $\lim_{n \to +\infty}$.



▲ **Figure 5.4.5** $\text{area}(R_n) \approx \text{area}(R)$

The values of $x_1^*, x_2^*, \ldots, x_n^*$ in (2) can be chosen arbitrarily, so it is conceivable that different choices of these values might produce different values of A. Were this to happen, then Definition 5.4.3 would not be an acceptable definition of area. Fortunately, this does not happen; it is proved in advanced courses that if f is continuous (as we have assumed), then the same value of A results no matter how the x_k^* are chosen. In practice they are chosen in some systematic fashion, some common choices being

- the left endpoint of each subinterval
- the right endpoint of each subinterval
- the midpoint of each subinterval

To be more specific, suppose that the interval $[a, b]$ is divided into n equal parts of length $\Delta x = (b - a)/n$ by the points $x_1, x_2, \ldots, x_{n-1}$, and let $x_0 = a$ and $x_n = b$ (Figure 5.4.6). Then,

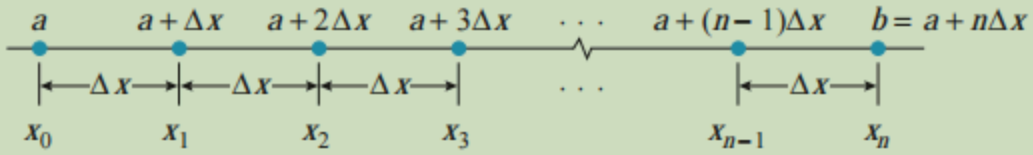
$$x_k = a + k\Delta x \quad \text{for } k = 0, 1, 2, \ldots, n$$

Thus, the left endpoint, right endpoint, and midpoint choices for $x_1^*, x_2^*, \ldots, x_n^*$ are given by

$$x_k^* = x_{k-1} = a + (k - 1)\Delta x \qquad \text{Left endpoint} \tag{3}$$

$$x_k^* = x_k = a + k\Delta x \qquad \text{Right endpoint} \tag{4}$$

$$x_k^* = \tfrac{1}{2}(x_{k-1} + x_k) = a + \left(k - \tfrac{1}{2}\right)\Delta x \qquad \text{Midpoint} \tag{5}$$



▶ **Figure 5.4.6**

► **Example 4** Use Definition 5.4.3 with x_k^* as the right endpoint of each subinterval to find the area between the graph of $f(x) = x^2$ and the interval $[0, 1]$.

Solution. The length of each subinterval is

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$$

so it follows from (4) that

$$x_k^* = a + k\Delta x = \frac{k}{n}$$

Thus,

$$\sum_{k=1}^{n} f(x_k^*)\Delta x = \sum_{k=1}^{n} (x_k^*)^2 \Delta x = \sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3} \sum_{k=1}^{n} k^2$$

$$= \frac{1}{n^3}\left[\frac{n(n+1)(2n+1)}{6}\right]$$

Part (*b*) of Theorem 5.4.2

$$= \frac{1}{6}\left(\frac{n}{n} \cdot \frac{n+1}{n} \cdot \frac{2n+1}{n}\right) = \frac{1}{6}\left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right)$$

from which it follows that

$$A = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x = \lim_{n \to +\infty} \left[\frac{1}{6}\left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right) \right] = \frac{1}{3}$$

Observe that this is consistent with the results in Table 5.1.2 and the related discussion in Section 5.1. ◀

In the solution to Example 4 we made use of one of the "closed form" summation formulas from Theorem 5.4.2. The next result collects some consequences of Theorem 5.4.2 that can facilitate computations of area using Definition 5.4.3.

What pattern is revealed by parts (*b*)–(*d*) of Theorem 5.4.4? Does part (*a*) also fit this pattern? What would you conjecture to be the value of

$$\lim_{n \to +\infty} \frac{1}{n^m} \sum_{k=1}^{n} k^{m-1} \, ?$$

5.4.4 THEOREM

(*a*) $\lim_{n \to +\infty} \frac{1}{n} \sum_{k=1}^{n} 1 = 1$

(*b*) $\lim_{n \to +\infty} \frac{1}{n^2} \sum_{k=1}^{n} k = \frac{1}{2}$

(*c*) $\lim_{n \to +\infty} \frac{1}{n^3} \sum_{k=1}^{n} k^2 = \frac{1}{3}$

(*d*) $\lim_{n \to +\infty} \frac{1}{n^4} \sum_{k=1}^{n} k^3 = \frac{1}{4}$

▶ **Example 5** Use Definition 5.4.3 with x_k^* as the midpoint of each subinterval to find the area under the parabola $y = f(x) = 9 - x^2$ and over the interval $[0, 3]$.

Solution. Each subinterval has length

$$\Delta x = \frac{b-a}{n} = \frac{3-0}{n} = \frac{3}{n}$$

so it follows from (5) that

$$x_k^* = a + \left(k - \frac{1}{2}\right)\Delta x = \left(k - \frac{1}{2}\right)\left(\frac{3}{n}\right)$$

Thus,

$$f(x_k^*)\Delta x = [9 - (x_k^*)^2]\Delta x = \left[9 - \left(k - \frac{1}{2}\right)^2\left(\frac{3}{n}\right)^2\right]\left(\frac{3}{n}\right)$$

$$= \left[9 - \left(k^2 - k + \frac{1}{4}\right)\left(\frac{9}{n^2}\right)\right]\left(\frac{3}{n}\right)$$

$$= \frac{27}{n} - \frac{27}{n^3}k^2 + \frac{27}{n^3}k - \frac{27}{4n^3}$$

from which it follows that

$$A = \lim_{n\to+\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x$$

$$= \lim_{n\to+\infty} \sum_{k=1}^{n} \left(\frac{27}{n} - \frac{27}{n^3}k^2 + \frac{27}{n^3}k - \frac{27}{4n^3}\right)$$

$$= \lim_{n\to+\infty} 27\left[\frac{1}{n}\sum_{k=1}^{n} 1 - \frac{1}{n^3}\sum_{k=1}^{n} k^2 + \frac{1}{n}\left(\frac{1}{n^2}\sum_{k=1}^{n} k\right) - \frac{1}{4n^2}\left(\frac{1}{n}\sum_{k=1}^{n} 1\right)\right]$$

$$= 27\left[1 - \frac{1}{3} + 0\cdot\frac{1}{2} - 0\cdot 1\right] = 18$$

Theorem 5.4.4 ◀

5.4.5 DEFINITION (***Net Signed Area***) If the function f is continuous on $[a, b]$, then the ***net signed area*** A between $y = f(x)$ and the interval $[a, b]$ is defined by

$$A = \lim_{n\to+\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x \tag{9}$$

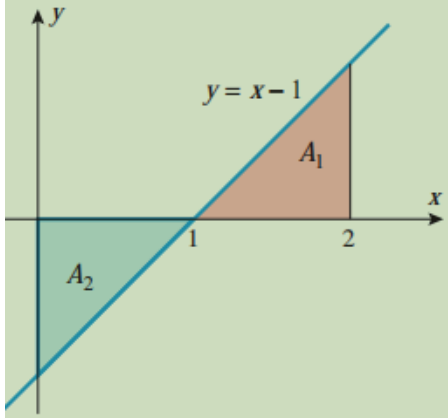


Figure 5.4.11

Figure 5.4.11 shows the graph of $f(x) = x - 1$ over the interval $[0, 2]$. It is geometrically evident that the areas A_1 and A_2 in that figure are equal, so we expect the net signed area between the graph of f and the interval $[0, 2]$ to be zero.

▶ Example 7 Confirm that the net signed area between the graph of $f(x) = x - 1$ and the interval $[0, 2]$ is zero by using Definition 5.4.5 with x_k^* chosen to be the left endpoint of each subinterval.

Solution. Each subinterval has length

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{n} = \frac{2}{n}$$

so it follows from (3) that

$$x_k^* = a + (k-1)\Delta x = (k-1)\left(\frac{2}{n}\right)$$

Thus,

$$\sum_{k=1}^{n} f(x_k^*)\Delta x = \sum_{k=1}^{n} (x_k^* - 1)\Delta x = \sum_{k=1}^{n} \left[(k-1)\left(\frac{2}{n}\right) - 1\right]\left(\frac{2}{n}\right)$$

$$= \sum_{k=1}^{n} \left[\left(\frac{4}{n^2}\right)k - \frac{4}{n^2} - \frac{2}{n}\right]$$

from which it follows that

$$A = \lim_{n\to+\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x = \lim_{n\to+\infty} \left[4\left(\frac{1}{n^2}\sum_{k=1}^{n} k\right) - \frac{4}{n}\left(\frac{1}{n}\sum_{k=1}^{n} 1\right) - 2\left(\frac{1}{n}\sum_{k=1}^{n} 1\right)\right]$$

$$= 4\left(\frac{1}{2}\right) - 0\cdot 1 - 2\cdot 1 = 0 \qquad \text{Theorem 5.4.4}$$

This confirms that the net signed area is zero. ◀

WATCH:

- **7.1 Preview of the definition of integral**
- **7.2 "Sigma notation" for sums**
- **7.10 Riemann sums**
- **The Definite Integral Part I: Approximating Areas with rectangles**
- **The Definite Integral Part II: Using Summation Notation to Define the Definite Integral**
- **The Definite Integral Part III: Evaluating From The Definition**
- **"Reverse" Riemann Sums | Finding the Definite Integral Given a Sum**
- চলো ছবিতে অনুভব করি প্রথম n সংখ্যক স্বাভাবিক সংখ্যার বর্গ, ঘন এর যোগফলের সূত্র | চমক হাসান
- অসীম গুণোত্তর ধারার আনন্দ - ১ | **The Joy of Geometric Series -1/2 | Chamok Hasan**
- **Proof of Sum Cos(kx) | Two Methods**

READ: **Limits to Infinity**

Use the following online solver:
[Riemann Sum Calculator for a Function - eMathHelp]

PRACTISE: HOWARD ANTON: 5.4: Riemann Sum: 27-30, 35-40, 41-44, 45-48

27–30 Divide the specified interval into $n = 4$ subintervals of equal length and then compute

$$\sum_{k=1}^{4} f(x_k^*)\Delta x$$

with x_k^* as (a) the left endpoint of each subinterval, (b) the midpoint of each subinterval, and (c) the right endpoint of each subinterval. Illustrate each part with a graph of f that includes the rectangles whose areas are represented in the sum. ■

27. $f(x) = 3x + 1;\ [2, 6]$

28. $f(x) = 1/x;\ [1, 9]$

29. $f(x) = \cos x;\ [0, \pi]$

30. $f(x) = 2x - x^2;\ [-1, 3]$

35–40 Use Definition 5.4.3 with x_k^* as the *right* endpoint of each subinterval to find the area under the curve $y = f(x)$ over the specified interval. ■

35. $f(x) = x/2;\ [1, 4]$

36. $f(x) = 5 - x;\ [0, 5]$

37. $f(x) = 9 - x^2;\ [0, 3]$

38. $f(x) = 4 - \frac{1}{4}x^2;\ [0, 3]$

39. $f(x) = x^3;\ [2, 6]$

40. $f(x) = 1 - x^3;\ [-3, -1]$

41–44 Use Definition 5.4.3 with x_k^* as the *left* endpoint of each subinterval to find the area under the curve $y = f(x)$ over the specified interval. ■

41. $f(x) = x/2;\ [1, 4]$

42. $f(x) = 5 - x;\ [0, 5]$

43. $f(x) = 9 - x^2;\ [0, 3]$

44. $f(x) = 4 - \frac{1}{4}x^2;\ [0, 3]$

45–48 Use Definition 5.4.3 with x_k^* as the *midpoint* of each subinterval to find the area under the curve $y = f(x)$ over the specified interval. ■

45. $f(x) = 2x;\ [0, 4]$

46. $f(x) = 6 - x;\ [1, 5]$

47. $f(x) = x^2;\ [0, 1]$

48. $f(x) = x^2;\ [-1, 1]$

Integration and Antiderivatives, Indefinite & Definite integrals, Integrals with variable limits, Integration by Substitution

Indefinite Integral

READ:
Introduction to Integration
Calculus I - Constant of Integration

Graphical Intro to Derivatives and Integrals

Integration Rules
Calculus I - Computing Indefinite Integrals

READ:

INTEGRATION FORMULAS

Integration is essentially educated guesswork—given the derivative f of a function F, one tries to guess what the function F is. However, many basic integration formulas can be obtained directly from their companion differentiation formulas. Some of the most important are given in Table 5.2.1.

Table 5.2.1

INTEGRATION FORMULAS

DIFFERENTIATION FORMULA	INTEGRATION FORMULA	DIFFERENTIATION FORMULA	INTEGRATION FORMULA
1. $\frac{d}{dx}[x] = 1$	$\int dx = x + C$	8. $\frac{d}{dx}[-\csc x] = \csc x \cot x$	$\int \csc x \cot x \, dx = -\csc x + C$
2. $\frac{d}{dx}\left[\frac{x^{r+1}}{r+1}\right] = x^r \quad (r \neq -1)$	$\int x^r \, dx = \frac{x^{r+1}}{r+1} + C \quad (r \neq -1)$	9. $\frac{d}{dx}[e^x] = e^x$	$\int e^x \, dx = e^x + C$
3. $\frac{d}{dx}[\sin x] = \cos x$	$\int \cos x \, dx = \sin x + C$	10. $\frac{d}{dx}\left[\frac{b^x}{\ln b}\right] = b^x \quad (0 < b, b \neq 1)$	$\int b^x \, dx = \frac{b^x}{\ln b} + C \quad (0 < b, b \neq 1)$
4. $\frac{d}{dx}[-\cos x] = \sin x$	$\int \sin x \, dx = -\cos x + C$	11. $\frac{d}{dx}[\ln \lvert x \rvert] = \frac{1}{x}$	$\int \frac{1}{x} \, dx = \ln \lvert x \rvert + C$
5. $\frac{d}{dx}[\tan x] = \sec^2 x$	$\int \sec^2 x \, dx = \tan x + C$	12. $\frac{d}{dx}[\tan^{-1} x] = \frac{1}{1+x^2}$	$\int \frac{1}{1+x^2} \, dx = \tan^{-1} x + C$
6. $\frac{d}{dx}[-\cot x] = \csc^2 x$	$\int \csc^2 x \, dx = -\cot x + C$	13. $\frac{d}{dx}[\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}$	$\int \frac{1}{\sqrt{1-x^2}} \, dx = \sin^{-1} x + C$
7. $\frac{d}{dx}[\sec x] = \sec x \tan x$	$\int \sec x \tan x \, dx = \sec x + C$	14. $\frac{d}{dx}[\sec^{-1} \lvert x \rvert] = \frac{1}{x\sqrt{x^2-1}}$	$\int \frac{1}{x\sqrt{x^2-1}} \, dx = \sec^{-1} \lvert x \rvert + C$

PROPERTIES OF THE INDEFINITE INTEGRAL

Our first properties of antiderivatives follow directly from the simple constant factor, sum, and difference rules for derivatives.

5.2.3 THEOREM *Suppose that $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$, respectively, and that c is a constant. Then:*

(a) *A constant factor can be moved through an integral sign; that is,*

$$\int cf(x)\,dx = cF(x) + C$$

(b) *An antiderivative of a sum is the sum of the antiderivatives; that is,*

$$\int [f(x) + g(x)]\,dx = F(x) + G(x) + C$$

(c) *An antiderivative of a difference is the difference of the antiderivatives; that is,*

$$\int [f(x) - g(x)]\,dx = F(x) - G(x) + C$$

The statements in Theorem 5.2.3 can be summarized by the following formulas:

$$\int cf(x)\,dx = c\int f(x)\,dx \tag{4}$$

$$\int [f(x)+g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx \tag{5}$$

$$\int [f(x)-g(x)]\,dx = \int f(x)\,dx - \int g(x)\,dx \tag{6}$$

WATCH:

- **8.1 Antiderivatives**
- **What's an anti-derivative?**
- **Solving for the constant in the general anti-derivative**

Use the following online solver:

Integral Calculator
Integral Calculator - eMathHelp
Integral Calculator

PRACTISE: HOWARD ANTON: 5.2 :Indefinite Integral: 15-22, 25, 27-31

15–34 Evaluate the integral and check your answer by differentiating. ■

15. $\int x(1+x^3)\,dx$

16. $\int (2+y^2)^2\,dy$

17. $\int x^{1/3}(2-x)^2\,dx$

18. $\int (1+x^2)(2-x)\,dx$

19. $\int \dfrac{x^5+2x^2-1}{x^4}\,dx$

20. $\int \dfrac{1-2t^3}{t^3}\,dt$

21. $\int \left[\dfrac{2}{x}+3e^x\right]dx$

22. $\int \left[\dfrac{1}{2t}-\sqrt{2}e^t\right]dt$

25. $\int \sec x(\sec x+\tan x)\,dx$

27. $\int \dfrac{\sec\theta}{\cos\theta}\,d\theta$

28. $\int \dfrac{dy}{\csc y}$

29. $\int \dfrac{\sin x}{\cos^2 x}\,dx$

30. $\int \left[\phi+\dfrac{2}{\sin^2\phi}\right]d\phi$

31. $\int [1+\sin^2\theta\csc\theta]\,d\theta$

Integration by Substitution

READ:

Integration by Substitution

Calculus I - Substitution Rule for Indefinite Integrals

Calculus I - More Substitution Rule

READ:

u-SUBSTITUTION

The method of substitution can be motivated by examining the chain rule from the viewpoint of antidifferentiation. For this purpose, suppose that F is an antiderivative of f and that g is a differentiable function. The chain rule implies that the derivative of $F(g(x))$ can be expressed as

$$\frac{d}{dx}[F(g(x))] = F'(g(x))g'(x)$$

which we can write in integral form as

$$\int F'(g(x))g'(x)\,dx = F(g(x)) + C \tag{1}$$

or since F is an antiderivative of f,

$$\int f(g(x))g'(x)\,dx = F(g(x)) + C \tag{2}$$

For our purposes it will be useful to let $u = g(x)$ and to write $du/dx = g'(x)$ in the differential form $du = g'(x)\,dx$. With this notation (2) can be expressed as

$$\int f(u)\,du = F(u) + C \tag{3}$$

The process of evaluating an integral of form (2) by converting it into form (3) with the substitution

$$u = g(x) \quad \text{and} \quad du = g'(x)\,dx$$

is called the ***method of u-substitution***. Here our emphasis is *not* on the interpretation of the expression $du = g'(x)\,dx$. Rather, the differential notation serves primarily as a useful "bookkeeping" device for the method of u-substitution. The following example illustrates how the method works.

▶ **Example 1** Evaluate $\int (x^2+1)^{50} \cdot 2x\,dx$.

Solution. If we let $u = x^2 + 1$, then $du/dx = 2x$, which implies that $du = 2x\,dx$. Thus, the given integral can be written as

$$\int (x^2+1)^{50} \cdot 2x\,dx = \int u^{50}\,du = \frac{u^{51}}{51} + C = \frac{(x^2+1)^{51}}{51} + C \quad \blacktriangleleft$$

It is important to realize that in the method of u-substitution you have control over the choice of u, but once you make that choice you have no control over the resulting expression for du. Thus, in the last example we *chose* $u = x^2 + 1$ but $du = 2x\,dx$ was *computed.* Fortunately, our choice of u, combined with the computed du, worked out perfectly to produce an integral involving u that was easy to evaluate. However, in general, the method of u-substitution will fail if the chosen u and the computed du cannot be used to produce an integrand in which no expressions involving x remain, or if you cannot evaluate the resulting integral. Thus, for example, the substitution $u = x^2, du = 2x\,dx$ will not work for the integral

$$\int 2x \sin x^4\,dx$$

because this substitution results in the integral

$$\int \sin u^2 \, du$$

which still cannot be evaluated in terms of familiar functions.

In general, there are no hard and fast rules for choosing u, and in some problems no choice of u will work. In such cases other methods need to be used, some of which will be discussed later. Making appropriate choices for u will come with experience, but you may find the following guidelines, combined with a mastery of the basic integrals in Table 5.2.1, helpful.

Guidelines for u-Substitution

Step 1. Look for some composition $f(g(x))$ within the integrand for which the substitution

$$u = g(x), \quad du = g'(x)\,dx$$

produces an integral that is expressed entirely in terms of u and its differential du. This may or may not be possible.

Step 2. If you are successful in Step 1, then try to evaluate the resulting integral in terms of u. Again, this may or may not be possible.

Step 3. If you are successful in Step 2, then replace u by $g(x)$ to express your final answer in terms of x.

▶ **Example 11** Evaluate $\int x^2\sqrt{x-1}\,dx$.

Solution. The composition $\sqrt{x-1}$ suggests the substitution

$$u = x - 1 \quad \text{so that} \quad du = dx \tag{4}$$

From the first equality in (4)

$$x^2 = (u+1)^2 = u^2 + 2u + 1$$

so that

$$\int x^2\sqrt{x-1}\,dx = \int (u^2 + 2u + 1)\sqrt{u}\,du = \int (u^{5/2} + 2u^{3/2} + u^{1/2})\,du$$

$$= \tfrac{2}{7}u^{7/2} + \tfrac{4}{5}u^{5/2} + \tfrac{2}{3}u^{3/2} + C$$

$$= \tfrac{2}{7}(x-1)^{7/2} + \tfrac{4}{5}(x-1)^{5/2} + \tfrac{2}{3}(x-1)^{3/2} + C \quad \blacktriangleleft$$

▶ **Example 12** Evaluate $\int \cos^3 x\, dx$.

Solution. The only compositions in the integrand that suggest themselves are

$$\cos^3 x = (\cos x)^3 \quad \text{and} \quad \cos^2 x = (\cos x)^2$$

However, neither the substitution $u = \cos x$ nor the substitution $u = \cos^2 x$ work (verify). In this case, an appropriate substitution is not suggested by the composition contained in the integrand. On the other hand, note from Equation (2) that the derivative $g'(x)$ appears as a factor in the integrand. This suggests that we write

$$\int \cos^3 x\, dx = \int \cos^2 x \cos x\, dx$$

and solve the equation $du = \cos x\, dx$ for $u = \sin x$. Since $\sin^2 x + \cos^2 x = 1$, we then have

$$\int \cos^3 x\, dx = \int \cos^2 x \cos x\, dx = \int (1 - \sin^2 x) \cos x\, dx = \int (1 - u^2)\, du$$

$$= u - \frac{u^3}{3} + C = \sin x - \frac{1}{3}\sin^3 x + C \quad \blacktriangleleft$$

▶ Example 13 Evaluate $\int \frac{dx}{a^2 + x^2}\, dx$, where $a \neq 0$ is a constant.

Solution. Some simple algebra and an appropriate u-substitution will allow us to use Formula 12 in Table 5.2.1.

$$\int \frac{dx}{a^2 + x^2} = \int \frac{a(dx/a)}{a^2(1 + (x/a)^2)} = \frac{1}{a} \int \frac{dx/a}{1 + (x/a)^2} \qquad \begin{array}{l} u = x/a \\ du = dx/a \end{array}$$

$$= \frac{1}{a} \int \frac{du}{1 + u^2} = \frac{1}{a} \tan^{-1} u + C = \frac{1}{a} \tan^{-1} \frac{x}{a} + C \quad \blacktriangleleft$$

The method of Example 13 leads to the following generalizations of Formulas 12, 13, and 14 in Table 5.2.1 for $a > 0$:

$$\int \frac{du}{a^2 + u^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + C \tag{5}$$

$$\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \frac{u}{a} + C \tag{6}$$

$$\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \sec^{-1} \left| \frac{u}{a} \right| + C \tag{7}$$

▶ **Example 14** Evaluate $\int \frac{dx}{\sqrt{2-x^2}}$.

Solution. Applying (6) with $u = x$ and $a = \sqrt{2}$ yields

$$\int \frac{dx}{\sqrt{2-x^2}} = \sin^{-1}\frac{x}{\sqrt{2}} + C \quad \blacktriangleleft$$

WATCH:

9.1 Integration by substitution - The theory

9.2 Integration by substitution - Examples

Intro to Substitution - Undoing the Chain Rule

Adjusting the Constant in Integration by Substitution

Use the following online solver:

Integral Calculator

Integral Calculator - eMathHelp

Integral Calculator

PRACTISE: HOWARD ANTON: 5.3: Integration by Substitution: 15-68

15–56 Evaluate the integrals using appropriate substitutions.

15. $\int (4x-3)^9\,dx$

16. $\int x^3\sqrt{5+x^4}\,dx$

17. $\int \sin 7x\,dx$

18. $\int \cos\frac{x}{3}\,dx$

19. $\int \sec 4x\tan 4x\,dx$

20. $\int \sec^2 5x\,dx$

21. $\int e^{2x}\,dx$

22. $\int \frac{dx}{2x}$

23. $\int \frac{dx}{\sqrt{1-4x^2}}$

24. $\int \frac{dx}{1+16x^2}$

25. $\int t\sqrt{7t^2+12}\,dt$

26. $\int \frac{x}{\sqrt{4-5x^2}}\,dx$

27. $\int \frac{6}{(1-2x)^3}\,dx$

28. $\int \frac{x^2+1}{\sqrt{x^3+3x}}\,dx$

29. $\int \frac{x^3}{(5x^4+2)^3}\,dx$

30. $\int \frac{\sin(1/x)}{3x^2}\,dx$

31. $\int e^{\sin x} \cos x \, dx$

32. $\int x^3 e^{x^4} \, dx$

33. $\int x^2 e^{-2x^3} \, dx$

34. $\int \dfrac{e^x + e^{-x}}{e^x - e^{-x}} \, dx$

35. $\int \dfrac{e^x}{1 + e^{2x}} \, dx$

36. $\int \dfrac{t}{t^4 + 1} \, dt$

37. $\int \dfrac{\sin(5/x)}{x^2} \, dx$

38. $\int \dfrac{\sec^2(\sqrt{x})}{\sqrt{x}} \, dx$

39. $\int \cos^4 3t \sin 3t \, dt$

40. $\int \cos 2t \sin^5 2t \, dt$

41. $\int x \sec^2(x^2) \, dx$

42. $\int \dfrac{\cos 4\theta}{(1 + 2\sin 4\theta)^4} \, d\theta$

43. $\int \cos 4\theta \sqrt{2 - \sin 4\theta}\, d\theta$

44. $\int \tan^3 5x \sec^2 5x\, dx$

45. $\int \dfrac{\sec^2 x\, dx}{\sqrt{1 - \tan^2 x}}$

46. $\int \dfrac{\sin\theta}{\cos^2\theta + 1}\, d\theta$

47. $\int \sec^3 2x \tan 2x\, dx$

48. $\int [\sin(\sin\theta)] \cos\theta\, d\theta$

49. $\int \dfrac{dx}{e^x}$

50. $\int \sqrt{e^x}\, dx$

51. $\int \dfrac{dx}{\sqrt{x}\, e^{(2\sqrt{x})}}$

52. $\int \dfrac{e^{\sqrt{2y+1}}}{\sqrt{2y+1}}\, dy$

53. $\int \dfrac{y}{\sqrt{2y+1}}\, dy$

54. $\int x\sqrt{4 - x}\, dx$

55. $\int \sin^3 2\theta\, d\theta$

56. $\int \sec^4 3\theta\, d\theta$ [*Hint:* Apply a trigonometric identity.]

57–60 Evaluate each integral by first modifying the form of the integrand and then making an appropriate substitution, if needed. ■

57. $\int \dfrac{t+1}{t}\, dt$

58. $\int e^{2\ln x}\, dx$

59. $\int [\ln(e^x) + \ln(e^{-x})]\, dx$

60. $\int \cot x\, dx$

61–62 Evaluate the integrals with the aid of Formulas (5), (6), and (7). ■

61. (a) $\displaystyle\int \frac{dx}{\sqrt{9-x^2}}$ (b) $\displaystyle\int \frac{dx}{5+x^2}$ (c) $\displaystyle\int \frac{dx}{x\sqrt{x^2-\pi}}$

62. (a) $\displaystyle\int \frac{e^x}{4+e^{2x}}\,dx$ (b) $\displaystyle\int \frac{dx}{\sqrt{9-4x^2}}$ (c) $\displaystyle\int \frac{dy}{y\sqrt{5y^2-3}}$

63–65 Evaluate the integrals assuming that n is a positive integer and $b \neq 0$. ■

63. $\displaystyle\int (a+bx)^n\,dx$ **64.** $\displaystyle\int \sqrt[n]{a+bx}\,dx$

65. $\displaystyle\int \sin^n(a+bx)\cos(a+bx)\,dx$

66. Use a CAS to check the answers you obtained in Exercises 63–65. If the answer produced by the CAS does not match yours, show that the two answers are equivalent. [*Suggestion: Mathematica* users may find it helpful to apply the Simplify command to the answer.]

FOCUS ON CONCEPTS

67. (a) Evaluate the integral $\int \sin x \cos x \, dx$ by two methods: first by letting $u = \sin x$, and then by letting $u = \cos x$.

(b) Explain why the two apparently different answers obtained in part (a) are really equivalent.

68. (a) Evaluate the integral $\int (5x - 1)^2 \, dx$ by two methods: first square and integrate, then let $u = 5x - 1$.

(b) Explain why the two apparently different answers obtained in part (a) are really equivalent.

The Definite Integral

READ:

Definite Integrals

Calculus I - Definition of the Definite Integral

Calculus I - Computing Definite Integrals

READ:

5.5.1 DEFINITION A function f is said to be ***integrable*** on a finite closed interval $[a, b]$ if the limit

$$\lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k$$

exists and does not depend on the choice of partitions or on the choice of the points x_k^* in the subintervals. When this is the case we denote the limit by the symbol

$$\int_a^b f(x)\,dx = \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k$$

which is called the ***definite integral*** of f from a to b. The numbers a and b are called the ***lower limit of integration*** and the ***upper limit of integration***, respectively, and $f(x)$ is called the ***integrand***.

The notation used for the definite integral deserves some comment. Historically, the expression "$f(x)\,dx$" was interpreted to be the "infinitesimal area" of a rectangle with height $f(x)$ and "infinitesimal" width dx. By "summing" these infinitesimal areas, the entire area under the curve was obtained. The integral symbol "$\int$" is an "elongated s" that was used to indicate this summation. For us, the integral symbol "$\int$" and the symbol "dx" can serve as reminders that the definite integral is actually a limit of a *summation* as $\Delta x_k \to 0$. The sum that appears in Definition 5.5.1 is called a ***Riemann sum***, and the definite integral is sometimes called the ***Riemann integral*** in honor of the German mathematician Bernhard Riemann who formulated many of the basic concepts of integral calculus. (The reason for the similarity in notation between the definite integral and the indefinite integral will become clear in the next section, where we will establish a link between the two types of "integration.")

5.5.2 THEOREM *If a function f is continuous on an interval $[a, b]$, then f is integrable on $[a, b]$, and the net signed area A between the graph of f and the interval $[a, b]$ is*

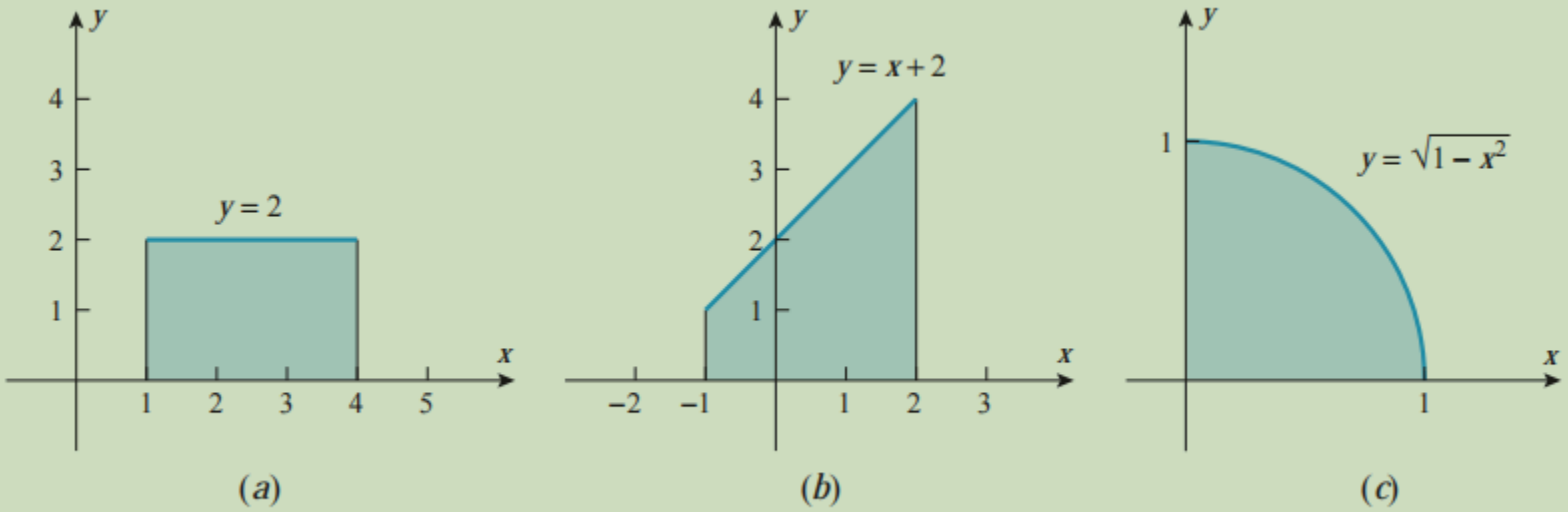
$$A = \int_a^b f(x)\,dx \tag{1}$$

In the simplest cases, definite integrals of continuous functions can be calculated using formulas from plane geometry to compute signed areas.

In Example 1, it is understood that the units of area are the squared units of length, even though we have not stated the units of length explicitly.

► **Example 1** Sketch the region whose area is represented by the definite integral, and evaluate the integral using an appropriate formula from geometry.

(a) $\int_1^4 2\,dx$ (b) $\int_{-1}^2 (x+2)\,dx$ (c) $\int_0^1 \sqrt{1-x^2}\,dx$



▲ **Figure 5.5.4**

Solution (a). The graph of the integrand is the horizontal line $y = 2$, so the region is a rectangle of height 2 extending over the interval from 1 to 4 (Figure 5.5.4*a*). Thus,

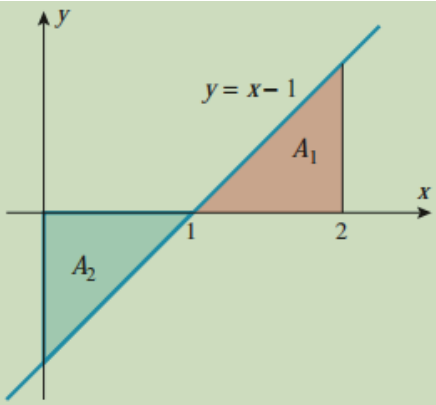
$$\int_1^4 2\,dx = (\text{area of rectangle}) = 2(3) = 6$$

Solution (b). The graph of the integrand is the line $y = x + 2$, so the region is a trapezoid whose base extends from $x = -1$ to $x = 2$ (Figure 5.5.4*b*). Thus,

$$\int_{-1}^2 (x+2)\,dx = (\text{area of trapezoid}) = \frac{1}{2}(1+4)(3) = \frac{15}{2}$$

Solution (c). The graph of $y = \sqrt{1-x^2}$ is the upper semicircle of radius 1, centered at the origin, so the region is the right quarter-circle extending from $x = 0$ to $x = 1$ (Figure 5.5.4*c*). Thus,

$$\int_0^1 \sqrt{1-x^2}\,dx = (\text{area of quarter-circle}) = \frac{1}{4}\pi(1^2) = \frac{\pi}{4} \quad \blacktriangleleft$$



▲ Figure 5.5.5

► **Example 2** Evaluate

$$\text{(a)} \int_0^2 (x-1)\,dx \qquad \text{(b)} \int_0^1 (x-1)\,dx$$

Solution. The graph of $y = x - 1$ is shown in Figure 5.5.5, and we leave it for you to verify that the shaded triangular regions both have area $\frac{1}{2}$. Over the interval $[0, 2]$ the net signed area is $A_1 - A_2 = \frac{1}{2} - \frac{1}{2} = 0$, and over the interval $[0, 1]$ the net signed area is $-A_2 = -\frac{1}{2}$. Thus,

$$\int_0^2 (x-1)\,dx = 0 \quad \text{and} \quad \int_0^1 (x-1)\,dx = -\tfrac{1}{2}$$

(Recall that in Example 7 of Section 5.4, we used Definition 5.4.5 to show that the net signed area between the graph of $y = x - 1$ and the interval $[0, 2]$ is zero.) ◄

PROPERTIES OF THE DEFINITE INTEGRAL

It is assumed in Definition 5.5.1 that $[a, b]$ is a finite closed interval with $a < b$, and hence the upper limit of integration in the definite integral is greater than the lower limit of integration. However, it will be convenient to extend this definition to allow for cases in which the upper and lower limits of integration are equal or the lower limit of integration is greater than the upper limit of integration. For this purpose we make the following special definitions.

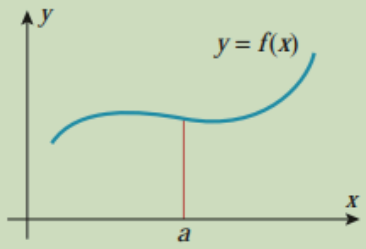
5.5.3 DEFINITION

(a) If a is in the domain of f, we define

$$\int_a^a f(x)\,dx = 0$$

(b) If f is integrable on $[a, b]$, then we define

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$$



The area between $y = f(x)$ and a is zero.

▲ **Figure 5.5.6**

Part (a) of this definition is consistent with the intuitive idea that the area between a point on the x-axis and a curve $y = f(x)$ should be zero (Figure 5.5.6). Part (b) of the definition is simply a useful convention; it states that interchanging the limits of integration reverses the sign of the integral.

► Example 3

(a) $\displaystyle\int_1^1 x^2\,dx = 0$

(b) $\displaystyle\int_1^0 \sqrt{1-x^2}\,dx = -\int_0^1 \sqrt{1-x^2}\,dx = -\frac{\pi}{4}$ ◄

Example 1(c)

Because definite integrals are defined as limits, they inherit many of the properties of limits. For example, we know that constants can be moved through limit signs and that the limit of a sum or difference is the sum or difference of the limits. Thus, you should not be surprised by the following theorem, which we state without formal proof.

5.5.4 THEOREM *If f and g are integrable on $[a, b]$ and if c is a constant, then cf, $f + g$, and $f - g$ are integrable on $[a, b]$ and*

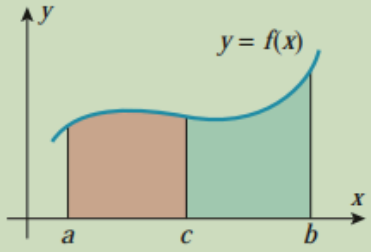
$$(a) \quad \int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$$

$$(b) \quad \int_a^b [f(x) + g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

$$(c) \quad \int_a^b [f(x) - g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

Part (b) of this theorem can be extended to more than two functions. More precisely,

$$\int_a^b [f_1(x) + f_2(x) + \cdots + f_n(x)]\,dx = \int_a^b f_1(x)\,dx + \int_a^b f_2(x)\,dx + \cdots + \int_a^b f_n(x)\,dx$$



▲ Figure 5.5.7

Some properties of definite integrals can be motivated by interpreting the integral as an area. For example, if f is continuous and nonnegative on the interval $[a, b]$, and if c is a point between a and b, then the area under $y = f(x)$ over the interval $[a, b]$ can be split into two parts and expressed as the area under the graph from a to c plus the area under the graph from c to b (Figure 5.5.7), that is,

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

This is a special case of the following theorem about definite integrals, which we state without proof.

5.5.5 THEOREM *If f is integrable on a closed interval containing the three points a, b, and c, then*

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

no matter how the points are ordered.

The following theorem, which we state without formal proof, can also be motivated by interpreting definite integrals as areas.

5.5.6 THEOREM

(a) *If f is integrable on $[a, b]$ and $f(x) \geq 0$ for all x in $[a, b]$, then*

$$\int_a^b f(x)\,dx \geq 0$$

(b) *If f and g are integrable on $[a, b]$ and $f(x) \geq g(x)$ for all x in $[a, b]$, then*

$$\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$$

WATCH:

- **7.11 Five properties of definite integrals**
- **8.2 Functions defined as integrals**
- **8.7 Summary: the three "notions of integral"**
- **the last question on a Harvard-MIT Math Tournament!**

Use the following online solver:

Integral Calculator

Integral Calculator - eMathHelp

Integral Calculator
Definite and Improper Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 5.5: The Definite Integral: 13-15, 22, 23, 25, 27, 37

13–16 Sketch the region whose signed area is represented by the definite integral, and evaluate the integral using an appropriate formula from geometry, where needed. ■

13. (a) $\int_0^3 x\,dx$ (b) $\int_{-2}^{-1} x\,dx$

(c) $\int_{-1}^4 x\,dx$ (d) $\int_{-5}^5 x\,dx$

14. (a) $\int_0^2 \left(1-\frac{1}{2}x\right)dx$ (b) $\int_{-1}^1 \left(1-\frac{1}{2}x\right)dx$

(c) $\int_2^3 \left(1-\frac{1}{2}x\right)dx$ (d) $\int_0^3 \left(1-\frac{1}{2}x\right)dx$

15. (a) $\int_0^5 2\,dx$ (b) $\int_0^\pi \cos x\,dx$

(c) $\int_{-1}^2 |2x-3|\,dx$ (d) $\int_{-1}^1 \sqrt{1-x^2}\,dx$

22. Find $\int_1^4 [3f(x) - g(x)]\,dx$ if

$$\int_1^4 f(x)\,dx = 2 \quad \text{and} \quad \int_1^4 g(x)\,dx = 10$$

23. Find $\int_1^5 f(x)\,dx$ if

$$\int_0^1 f(x)\,dx = -2 \quad \text{and} \quad \int_0^5 f(x)\,dx = 1$$

25–28 Use Theorem 5.5.4 and appropriate formulas from geometry to evaluate the integrals. ■

25. $\int_{-1}^3 (4 - 5x)\,dx$

27. $\int_0^1 (x + 2\sqrt{1 - x^2})\,dx$

37–38 Evaluate the integrals by completing the square and applying appropriate formulas from geometry. ■

37. $\int_0^{10} \sqrt{10x - x^2}\,dx$

Evaluating Definite Integral by Substitution

READ:
Calculus I - Substitution Rule for Definite Integrals

READ:

TWO METHODS FOR MAKING SUBSTITUTIONS IN DEFINITE INTEGRALS

Recall from Section 5.3 that indefinite integrals of the form

$$\int f(g(x))g'(x)\,dx$$

can sometimes be evaluated by making the u-substitution

$$u = g(x), \quad du = g'(x)\,dx \tag{1}$$

which converts the integral to the form

$$\int f(u)\,du$$

To apply this method to a definite integral of the form

$$\int_a^b f(g(x))g'(x)\,dx$$

we need to account for the effect that the substitution has on the x-limits of integration. There are two ways of doing this.

Method 1.

First evaluate the indefinite integral

$$\int f(g(x))g'(x)\,dx$$

by substitution, and then use the relationship

$$\int_a^b f(g(x))g'(x)\,dx = \left[\int f(g(x))g'(x)\,dx\right]_a^b$$

to evaluate the definite integral. This procedure does not require any modification of the x-limits of integration.

Method 2.

Make the substitution (1) directly in the definite integral, and then use the relationship $u = g(x)$ to replace the x-limits, $x = a$ and $x = b$, by corresponding u-limits, $u = g(a)$ and $u = g(b)$. This produces a new definite integral

$$\int_{g(a)}^{g(b)} f(u)\,du$$

that is expressed entirely in terms of u.

► **Example 1** Use the two methods above to evaluate $\int_0^2 x(x^2+1)^3\,dx$.

Solution by Method 1. If we let

$$u = x^2 + 1 \quad \text{so that} \quad du = 2x\,dx \tag{2}$$

then we obtain

$$\int x(x^2+1)^3\,dx = \frac{1}{2}\int u^3\,du = \frac{u^4}{8} + C = \frac{(x^2+1)^4}{8} + C$$

Thus,

$$\int_0^2 x(x^2+1)^3\,dx = \left[\int x(x^2+1)^3\,dx\right]_{x=0}^{2}$$

$$= \left.\frac{(x^2+1)^4}{8}\right]_{x=0}^{2} = \frac{625}{8} - \frac{1}{8} = 78$$

Solution by Method 2. If we make the substitution $u = x^2 + 1$ in (2), then

$$\text{if} \quad x = 0, \quad u = 1$$
$$\text{if} \quad x = 2, \quad u = 5$$

Thus,

$$\int_0^2 x(x^2+1)^3\,dx = \frac{1}{2}\int_1^5 u^3\,du$$
$$= \frac{u^4}{8}\Big]_{u=1}^{5} = \frac{625}{8} - \frac{1}{8} = 78$$

which agrees with the result obtained by Method 1. ◀

The following theorem states precise conditions under which Method 2 can be used.

5.9.1 THEOREM *If g' is continuous on $[a, b]$ and f is continuous on an interval containing the values of $g(x)$ for $a \le x \le b$, then*

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

▶ **Example 2** Evaluate

$$\text{(a)} \int_0^{\pi/8} \sin^5 2x \cos 2x \, dx \qquad \text{(b)} \int_2^5 (2x-5)(x-3)^9 \, dx$$

Solution (a). Let

$$u = \sin 2x \quad \text{so that} \quad du = 2\cos 2x \, dx \quad \left(\text{or } \tfrac{1}{2}\, du = \cos 2x \, dx\right)$$

With this substitution,

$$\text{if} \quad x = 0, \quad u = \sin(0) = 0$$
$$\text{if} \quad x = \pi/8, \quad u = \sin(\pi/4) = 1/\sqrt{2}$$

so

$$\int_0^{\pi/8} \sin^5 2x \cos 2x \, dx = \frac{1}{2} \int_0^{1/\sqrt{2}} u^5 \, du$$

$$= \frac{1}{2} \cdot \frac{u^6}{6} \Big]_{u=0}^{1/\sqrt{2}} = \frac{1}{2} \left[\frac{1}{6(\sqrt{2})^6} - 0 \right] = \frac{1}{96}$$

Solution (b). Let

$$u = x - 3 \quad \text{so that} \quad du = dx$$

This leaves a factor of $2x - 5$ unresolved in the integrand. However,

$$x = u + 3, \quad \text{so} \quad 2x - 5 = 2(u + 3) - 5 = 2u + 1$$

With this substitution,

$$\text{if} \quad x = 2, \quad u = 2 - 3 = -1$$
$$\text{if} \quad x = 5, \quad u = 5 - 3 = 2$$

so

$$\int_2^5 (2x - 5)(x - 3)^9 \, dx = \int_{-1}^2 (2u + 1)u^9 \, du = \int_{-1}^2 (2u^{10} + u^9) \, du$$

$$= \left[\frac{2u^{11}}{11} + \frac{u^{10}}{10}\right]_{u=-1}^2 = \left(\frac{2^{12}}{11} + \frac{2^{10}}{10}\right) - \left(-\frac{2}{11} + \frac{1}{10}\right)$$

$$= \frac{52,233}{110} \approx 474.8 \blacktriangleleft$$

▶ **Example 3** Evaluate

$$\text{(a)} \int_0^{3/4} \frac{dx}{1-x} \qquad \text{(b)} \int_0^{\ln 3} e^x (1+e^x)^{1/2}\, dx$$

Solution (a). Let

$$u = 1 - x \quad \text{so that} \quad du = -dx$$

With this substitution,

$$\text{if} \quad x = 0, \quad u = 1$$
$$\text{if} \quad x = \tfrac{3}{4}, \quad u = \tfrac{1}{4}$$

Thus,

$$\int_0^{3/4} \frac{dx}{1-x} = -\int_1^{1/4} \frac{du}{u}$$

$$= -\ln|u|\Big]_{u=1}^{1/4} = -\left[\ln\left(\frac{1}{4}\right) - \ln(1)\right] = \ln 4$$

The u-substitution in Example 3(a) produces an integral in which the upper u-limit is smaller than the lower u-limit. Use Definition 5.5.3(b) to convert this integral to one whose lower limit is smaller than the upper limit and verify that it produces an integral with the same value as that in the example.

Solution (b). Make the u-substitution

$$u = 1 + e^x, \quad du = e^x\,dx$$

and change the x-limits of integration ($x = 0, x = \ln 3$) to the u-limits

$$u = 1 + e^0 = 2, \quad u = 1 + e^{\ln 3} = 1 + 3 = 4$$

This yields

$$\int_0^{\ln 3} e^x(1 + e^x)^{1/2}\,dx = \int_2^4 u^{1/2}\,du$$

$$= \frac{2}{3}u^{3/2}\Big]_{u=2}^{4} = \frac{2}{3}[4^{3/2} - 2^{3/2}] = \frac{16 - 4\sqrt{2}}{3} \blacktriangleleft$$

WATCH:

9.3 Substitution for definite integrals

Substitution Method for Definite Integrals **careful!**

Use the following online solver:

Integral Calculator

Integral Calculator - eMathHelp

Integral Calculator

Definite and Improper Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 5.9: Evaluating Definite Integral by Substitution: 5-22, 32, 33, 35-40

5–18 Evaluate the definite integral two ways: first by a u-substitution in the definite integral and then by a u-substitution in the corresponding indefinite integral. ■

5. $\displaystyle\int_0^1 (2x+1)^3\,dx$

6. $\displaystyle\int_1^2 (4x-2)^3\,dx$

7. $\displaystyle\int_0^1 (2x-1)^3\,dx$

8. $\displaystyle\int_1^2 (4-3x)^8\,dx$

9. $\displaystyle\int_0^8 x\sqrt{1+x}\,dx$

10. $\displaystyle\int_{-3}^0 x\sqrt{1-x}\,dx$

11. $\displaystyle\int_0^{\pi/2} 4\sin(x/2)\,dx$

12. $\displaystyle\int_0^{\pi/6} 2\cos 3x\,dx$

13. $\displaystyle\int_{-2}^{-1} \frac{x}{(x^2+2)^3}\,dx$

14. $\displaystyle\int_{1-\pi}^{1+\pi} \sec^2\left(\tfrac{1}{4}x-\tfrac{1}{4}\right)\,dx$

15. $\displaystyle\int_{-\ln 3}^{\ln 3} \frac{e^x}{e^x+4}\,dx$

16. $\displaystyle\int_0^{\ln 5} e^x(3-4e^x)\,dx$

17. $\displaystyle\int_1^3 \frac{dx}{\sqrt{x}\,(x+1)}$

18. $\displaystyle\int_{\ln 2}^{\ln(2/\sqrt{3})} \frac{e^{-x}\,dx}{\sqrt{1-e^{-2x}}}$

19–22 Evaluate the definite integral by expressing it in terms of u and evaluating the resulting integral using a formula from geometry. ■

19. $\displaystyle\int_{-5/3}^{5/3} \sqrt{25 - 9x^2}\, dx;\ u = 3x$

20. $\displaystyle\int_{0}^{2} x\sqrt{16 - x^4}\, dx;\ u = x^2$

21. $\displaystyle\int_{\pi/3}^{\pi/2} \sin\theta\sqrt{1 - 4\cos^2\theta}\, d\theta;\ u = 2\cos\theta$

22. $\displaystyle\int_{e^{-3}}^{e^3} \frac{\sqrt{9 - (\ln x)^2}}{x}\, dx;\ u = \ln x$

31–50 Evaluate the integrals by any method. ■

31. $\displaystyle\int_1^5 \frac{dx}{\sqrt{2x-1}}$

32. $\displaystyle\int_1^2 \sqrt{5x-1}\,dx$

33. $\displaystyle\int_{-1}^1 \frac{x^2\,dx}{\sqrt{x^3+9}}$

34. $\displaystyle\int_{\pi/2}^{\pi} 6\sin x(\cos x+1)^5\,dx$

35. $\displaystyle\int_1^3 \frac{x+2}{\sqrt{x^2+4x+7}}\,dx$

36. $\displaystyle\int_1^2 \frac{dx}{x^2-6x+9}$

37. $\displaystyle\int_0^{\pi/4} 4\sin x\cos x\,dx$

38. $\displaystyle\int_0^{\pi/4} \sqrt{\tan x}\sec^2 x\,dx$

39. $\displaystyle\int_0^{\sqrt{\pi}} 5x\cos(x^2)\,dx$

40. $\displaystyle\int_{\pi^2}^{4\pi^2} \frac{1}{\sqrt{x}}\sin\sqrt{x}\,dx$

- **Improper Integrals, Integration by parts, Reduction formula, Integration by Trigonometric Substitution, Integration of Rational Function**

Integration by parts, Reduction formula, Improper Integrals, Infinite integrals

READ:

METHODS FOR APPROACHING INTEGRATION PROBLEMS

There are three basic approaches for evaluating unfamiliar integrals:

- **Technology**—CAS programs such as *Mathematica*, *Maple*, and the open source program *Sage* are capable of evaluating extremely complicated integrals, and such programs are increasingly available for both computers and handheld calculators.
- **Tables**—Prior to the development of CAS programs, scientists relied heavily on tables to evaluate difficult integrals arising in applications. Such tables were compiled over many years, incorporating the skills and experience of many people. One such table appears in the endpapers of this text, but more comprehensive tables appear in various reference books such as the *CRC Standard Mathematical Tables and Formulae*, CRC Press, Inc., 2002.
- **Transformation Methods**—Transformation methods are methods for converting unfamiliar integrals into familiar integrals. These include u-substitution, algebraic manipulation of the integrand, and other methods that we will discuss in this chapter.

None of the three methods is perfect; for example, CAS programs often encounter integrals that they cannot evaluate and they sometimes produce answers that are unnecessarily complicated, tables are not exhaustive and may not include a particular integral of interest, and transformation methods rely on human ingenuity that may prove to be inadequate in difficult problems.

In this chapter we will focus on transformation methods and tables, so it will *not be necessary* to have a CAS such as *Mathematica*, *Maple*, or *Sage*. However, if you have a CAS, then you can use it to confirm the results in the examples, and there are exercises that are designed to be solved with a CAS. If you have a CAS, keep in mind that many of the algorithms that it uses are based on the methods we will discuss here, so an understanding of these methods will help you to use your technology in a more informed way.

A REVIEW OF FAMILIAR INTEGRATION FORMULAS

The following is a list of basic integrals that we have encountered thus far:

CONSTANTS, POWERS, EXPONENTIALS

1. $\int du = u + C$
2. $\int a\,du = a\int du = au + C$
3. $\int u^r\,du = \dfrac{u^{r+1}}{r+1} + C,\ r \neq -1$
4. $\int \dfrac{du}{u} = \ln|u| + C$
5. $\int e^u\,du = e^u + C$
6. $\int b^u\,du = \dfrac{b^u}{\ln b} + C,\ b > 0, b \neq 1$

TRIGONOMETRIC FUNCTIONS

7. $\int \sin u\,du = -\cos u + C$
8. $\int \cos u\,du = \sin u + C$
9. $\int \sec^2 u\,du = \tan u + C$
10. $\int \csc^2 u\,du = -\cot u + C$
11. $\int \sec u \tan u\,du = \sec u + C$
12. $\int \csc u \cot u\,du = -\csc u + C$
13. $\int \tan u\,du = -\ln|\cos u| + C$
14. $\int \cot u\,du = \ln|\sin u| + C$

HYPERBOLIC FUNCTIONS

15. $\int \sinh u\,du = \cosh u + C$

16. $\int \cosh u\,du = \sinh u + C$

17. $\int \operatorname{sech}^2 u\,du = \tanh u + C$

18. $\int \operatorname{csch}^2 u\,du = -\coth u + C$

19. $\int \operatorname{sech} u \tanh u\,du = -\operatorname{sech} u + C$

20. $\int \operatorname{csch} u \coth u\,du = -\operatorname{csch} u + C$

ALGEBRAIC FUNCTIONS ($a > 0$)

21. $\displaystyle\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1}\frac{u}{a} + C \qquad (|u| < a)$

22. $\displaystyle\int \frac{du}{a^2 + u^2} = \frac{1}{a}\tan^{-1}\frac{u}{a} + C$

23. $\displaystyle\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{u}{a}\right| + C \qquad (0 < a < |u|)$

$$\textbf{24.}\ \int \frac{du}{\sqrt{a^2+u^2}} = \ln\left(u+\sqrt{u^2+a^2}\right)+C$$

$$\textbf{25.}\ \int \frac{du}{\sqrt{u^2-a^2}} = \ln\left|u+\sqrt{u^2-a^2}\right|+C \qquad (0<a<|u|)$$

$$\textbf{26.}\ \int \frac{du}{a^2-u^2} = \frac{1}{2a}\ln\left|\frac{a+u}{a-u}\right|+C$$

$$\textbf{27.}\ \int \frac{du}{u\sqrt{a^2-u^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{a^2-u^2}}{u}\right|+C \qquad (0<|u|<a)$$

$$\textbf{28.}\ \int \frac{du}{u\sqrt{a^2+u^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{a^2+u^2}}{u}\right|+C$$

REMARK Formula 25 is a generalization of a result in Theorem 6.9.6. Readers who did not cover Section 6.9 can ignore Formulas 24–28 for now, since we will develop other methods for obtaining them in this chapter.

WATCH:

READ: [Calculus I - Derivatives of Hyperbolic Functions]

OR WATCH: **Why hyperbolic functions are actually really nice**

Use the following online solver:

Integral Calculator

Integral Calculator - eMathHelp

Integral Calculator

Definite and Improper Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 7.1: Overview of Integration Methods: 1-30

1–30 Evaluate the integrals by making appropriate u-substitutions and applying the formulas reviewed in this section. ■

1. $\int (4-2x)^3\,dx$

2. $\int 3\sqrt{4+2x}\,dx$

3. $\int x\sec^2(x^2)\,dx$

4. $\int 4x\tan(x^2)\,dx$

5. $\int \frac{\sin 3x}{2+\cos 3x}\,dx$

6. $\int \frac{1}{9+4x^2}\,dx$

7. $\int e^x \sinh(e^x)\,dx$

8. $\int \frac{\sec(\ln x)\tan(\ln x)}{x}\,dx$

9. $\int e^{\tan x}\sec^2 x\,dx$

10. $\int \frac{x}{\sqrt{1-x^4}}\,dx$

11. $\int \cos^5 5x \sin 5x\,dx$

12. $\int \frac{\cos x}{\sin x\sqrt{\sin^2 x+1}}\,dx$

13. $\int \frac{e^x}{\sqrt{4+e^{2x}}}\,dx$

14. $\int \frac{e^{\tan^{-1}x}}{1+x^2}\,dx$

15. $\int \frac{e^{\sqrt{x-1}}}{\sqrt{x-1}}\,dx$

16. $\int (x+1)\cot(x^2+2x)\,dx$

17. $\int \frac{\cosh\sqrt{x}}{\sqrt{x}}\,dx$

18. $\int \frac{dx}{x(\ln x)^2}$

19. $\int \frac{dx}{\sqrt{x}\,3\sqrt{x}}$

20. $\int \sec(\sin\theta)\tan(\sin\theta)\cos\theta\, d\theta$

21. $\int \frac{\operatorname{csch}^2(2/x)}{x^2}\, dx$

22. $\int \frac{dx}{\sqrt{x^2-4}}$

23. $\int \frac{e^{-x}}{4-e^{-2x}}\, dx$

24. $\int \frac{\cos(\ln x)}{x}\, dx$

25. $\int \frac{e^x}{\sqrt{1-e^{2x}}}\, dx$

26. $\int \frac{\sinh(x^{-1/2})}{x^{3/2}}\, dx$

27. $\int \frac{x}{\csc(x^2)}\, dx$

28. $\int \frac{e^x}{\sqrt{4-e^{2x}}}\, dx$

29. $\int x4^{-x^2}\, dx$

30. $\int 2^{\pi x}\, dx$

Integration By Parts

READ:

Integration by Parts

READ:

THE PRODUCT RULE AND INTEGRATION BY PARTS

Our primary goal in this section is to develop a general method for attacking integrals of the form

$$\int f(x)g(x)\,dx$$

As a first step, let $G(x)$ be *any* antiderivative of $g(x)$. In this case $G'(x) = g(x)$, so the product rule for differentiating $f(x)G(x)$ can be expressed as

$$\frac{d}{dx}[f(x)G(x)] = f(x)G'(x) + f'(x)G(x) = f(x)g(x) + f'(x)G(x) \qquad (1)$$

This implies that $f(x)G(x)$ is an antiderivative of the function on the right side of (1), so we can express (1) in integral form as

$$\int [f(x)g(x) + f'(x)G(x)]\,dx = f(x)G(x)$$

or, equivalently, as

$$\int f(x)g(x)\,dx = f(x)G(x) - \int f'(x)G(x)\,dx \tag{2}$$

This formula allows us to integrate $f(x)g(x)$ by integrating $f'(x)G(x)$ instead, and in many cases the net effect is to replace a difficult integration with an easier one. The application of this formula is called ***integration by parts***.

In practice, we usually rewrite (2) by letting

$$u = f(x), \quad du = f'(x)\,dx$$
$$v = G(x), \quad dv = G'(x)\,dx = g(x)\,dx$$

This yields the following alternative form for (2):

$$\int u\,dv = uv - \int v\,du \tag{3}$$

Note that in Example 1 we omitted the constant of integration in calculating v from dv. Had we included a constant of integration, it would have eventually dropped out. This is always the case in integration by parts [Exercise 68(b)], so it is common to omit the constant at this stage of the computation. However, there are certain cases in which making a clever choice of a constant of integration to include with v can simplify the computation of $\int v\,du$ (Exercises 69–71).

► **Example 1** Use integration by parts to evaluate $\int x\cos x\,dx$.

Solution. We will apply Formula (3). The first step is to make a choice for u and dv to put the given integral in the form $\int u\,dv$. We will let

$$u = x \quad \text{and} \quad dv = \cos x\,dx$$

(Other possibilities will be considered later.) The second step is to compute du from u and v from dv. This yields

$$du = dx \quad \text{and} \quad v = \int dv = \int \cos x\,dx = \sin x$$

The third step is to apply Formula (3). This yields

$$\int \underbrace{x}_{u}\,\underbrace{\cos x\,dx}_{dv} = \underbrace{x}_{u}\,\underbrace{\sin x}_{v} - \int \underbrace{\sin x}_{v}\,\underbrace{dx}_{du}$$
$$= x\sin x - (-\cos x) + C = x\sin x + \cos x + C \quad \blacktriangleleft$$

GUIDELINES FOR INTEGRATION BY PARTS

The main goal in integration by parts is to choose u and dv to obtain a new integral that is easier to evaluate than the original. In general, there are no hard and fast rules for doing this; it is mainly a matter of experience that comes from lots of practice. A strategy that often works is to choose u and dv so that u becomes "simpler" when differentiated, while leaving a dv that can be readily integrated to obtain v. Thus, for the integral $\int x \cos x \, dx$ in Example 1, both goals were achieved by letting $u = x$ and $dv = \cos x \, dx$. In contrast, $u = \cos x$ would not have been a good first choice in that example, since $du/dx = -\sin x$ is no simpler than u. Indeed, had we chosen

$$u = \cos x \qquad dv = x \, dx$$

$$du = -\sin x \, dx \qquad v = \int x \, dx = \frac{x^2}{2}$$

then we would have obtained

$$\int x \cos x \, dx = \frac{x^2}{2} \cos x - \int \frac{x^2}{2}(-\sin x) \, dx = \frac{x^2}{2} \cos x + \frac{1}{2} \int x^2 \sin x \, dx$$

For this choice of u and dv, the new integral is actually more complicated than the original.

The LIATE method is discussed in the article "A Technique for Integration by Parts," *American Mathematical Monthly*, Vol. 90, 1983, pp. 210–211, by Herbert Kasube.

There is another useful strategy for choosing u and dv that can be applied when the integrand is a product of two functions from *different* categories in the list

Logarithmic, **I**nverse trigonometric, **A**lgebraic, **T**rigonometric, **E**xponential

In this case you will often be successful if you take u to be the function whose category occurs earlier in the list and take dv to be the rest of the integrand. The acronym LIATE will help you to remember the order. The method does not work all the time, but it works often enough to be useful.

Note, for example, that the integrand in Example 1 consists of the product of the *algebraic* function x and the *trigonometric* function $\cos x$. Thus, the LIATE method suggests that we should let $u = x$ and $dv = \cos x\, dx$, which proved to be a successful choice.

▶ **Example 2** Evaluate $\int xe^x\,dx$.

Solution. In this case the integrand is the product of the algebraic function x with the exponential function e^x. According to LIATE we should let

$$u = x \quad \text{and} \quad dv = e^x\,dx$$

so that

$$du = dx \quad \text{and} \quad v = \int e^x\,dx = e^x$$

Thus, from (3)

$$\int xe^x\,dx = \int u\,dv = uv - \int v\,du = xe^x - \int e^x\,dx = xe^x - e^x + C \quad \blacktriangleleft$$

▶ **Example 3** Evaluate $\int \ln x\,dx$.

Solution. One choice is to let $u = 1$ and $dv = \ln x\,dx$. But with this choice finding v is equivalent to evaluating $\int \ln x\,dx$ and we have gained nothing. Therefore, the only reasonable choice is to let

$$u = \ln x \qquad dv = dx$$
$$du = \frac{1}{x}\,dx \qquad v = \int dx = x$$

With this choice it follows from (3) that

$$\int \ln x\,dx = \int u\,dv = uv - \int v\,du = x\ln x - \int dx = x\ln x - x + C \quad \blacktriangleleft$$

REPEATED INTEGRATION BY PARTS

It is sometimes necessary to use integration by parts more than once in the same problem.

▶ Example 4 Evaluate $\int x^2 e^{-x}\,dx$.

Solution. Let

$$u = x^2, \quad dv = e^{-x}\,dx, \quad du = 2x\,dx, \quad v = \int e^{-x}\,dx = -e^{-x}$$

so that from (3)

$$\begin{aligned}\int x^2 e^{-x}\,dx &= \int u\,dv = uv - \int v\,du \\ &= x^2(-e^{-x}) - \int -e^{-x}(2x)\,dx \\ &= -x^2e^{-x} + 2\int xe^{-x}\,dx \end{aligned} \tag{4}$$

The last integral is similar to the original except that we have replaced x^2 by x. Another integration by parts applied to $\int xe^{-x}\,dx$ will complete the problem. We let

$$u = x, \quad dv = e^{-x}\,dx, \quad du = dx, \quad v = \int e^{-x}\,dx = -e^{-x}$$

so that

$$\int xe^{-x}\,dx = x(-e^{-x}) - \int -e^{-x}\,dx = -xe^{-x} + \int e^{-x}\,dx = -xe^{-x} - e^{-x} + C$$

Finally, substituting this into the last line of (4) yields

$$\begin{aligned}\int x^2e^{-x}\,dx &= -x^2e^{-x} + 2\int xe^{-x}\,dx = -x^2e^{-x} + 2(-xe^{-x} - e^{-x}) + C \\ &= -(x^2 + 2x + 2)e^{-x} + C \end{aligned}$$ ◀

▶ **Example 5** Evaluate $\int e^x \cos x \, dx$.

Solution. Let

$$u = \cos x, \quad dv = e^x \, dx, \quad du = -\sin x \, dx, \quad v = \int e^x \, dx = e^x$$

Thus,

$$\int e^x \cos x \, dx = \int u \, dv = uv - \int v \, du = e^x \cos x + \int e^x \sin x \, dx \tag{5}$$

Since the integral $\int e^x \sin x \, dx$ is similar in form to the original integral $\int e^x \cos x \, dx$, it seems that nothing has been accomplished. However, let us integrate this new integral by parts. We let

$$u = \sin x, \quad dv = e^x \, dx, \quad du = \cos x \, dx, \quad v = \int e^x \, dx = e^x$$

Thus,

$$\int e^x \sin x \, dx = \int u \, dv = uv - \int v \, du = e^x \sin x - \int e^x \cos x \, dx$$

Together with Equation (5) this yields

$$\int e^x \cos x \, dx = e^x \cos x + e^x \sin x - \int e^x \cos x \, dx \tag{6}$$

which is an equation we can solve for the unknown integral. We obtain

$$2\int e^x \cos x\, dx = e^x \cos x + e^x \sin x$$

and hence

$$\int e^x \cos x\, dx = \tfrac{1}{2}e^x \cos x + \tfrac{1}{2}e^x \sin x + C \quad \blacktriangleleft$$

A TABULAR METHOD FOR REPEATED INTEGRATION BY PARTS

Integrals of the form

$$\int p(x)\,f(x)\,dx$$

where $p(x)$ is a polynomial, can sometimes be evaluated using repeated integration by parts in which u is taken to be $p(x)$ or one of its derivatives at each stage. Since du is computed by differentiating u, the repeated differentiation of $p(x)$ will eventually produce 0, at which point you may be left with a simplified integration problem. A convenient method for organizing the computations into two columns is called ***tabular integration by parts***.

More information on tabular integration by parts can be found in the articles "Tabular Integration by Parts," *College Mathematics Journal*, Vol. 21, 1990, pp. 307–311, by David Horowitz and "More on Tabular Integration by Parts," *College Mathematics Journal*, Vol. 22, 1991, pp. 407–410, by Leonard Gillman.

Tabular Integration by Parts

Step 1. Differentiate $p(x)$ repeatedly until you obtain 0, and list the results in the first column.

Step 2. Integrate $f(x)$ repeatedly and list the results in the second column.

Step 3. Draw an arrow from each entry in the first column to the entry that is one row down in the second column.

Step 4. Label the arrows with alternating $+$ and $-$ signs, starting with a $+$.

Step 5. For each arrow, form the product of the expressions at its tip and tail and then multiply that product by $+1$ or -1 in accordance with the sign on the arrow. Add the results to obtain the value of the integral.

This process is illustrated in Figure 7.2.1 for the integral $\int (x^2 - x)\cos x\,dx$.

REPEATED DIFFERENTIATION		REPEATED INTEGRATION
$x^2 - x$	+	$\cos x$
$2x - 1$	−	$\sin x$
2	+	$-\cos x$
0		$-\sin x$

$$\int (x^2 - x)\cos x\,dx = (x^2 - x)\sin x + (2x - 1)\cos x - 2\sin x + C$$
$$= (x^2 - x - 2)\sin x + (2x - 1)\cos x + C$$

▶ **Figure 7.2.1**

▶ **Example 6** In Example 11 of Section 5.3 we evaluated $\int x^2\sqrt{x-1}\,dx$ using u-substitution. Evaluate this integral using tabular integration by parts.

Solution.

REPEATED DIFFERENTIATION		REPEATED INTEGRATION
x^2	$+$	$(x-1)^{1/2}$
$2x$	$-$	$\frac{2}{3}(x-1)^{3/2}$
2	$+$	$\frac{4}{15}(x-1)^{5/2}$
0		$\frac{8}{105}(x-1)^{7/2}$

The result obtained in Example 6 looks quite different from that obtained in Example 11 of Section 5.3. Show that the two answers are equivalent.

Thus, it follows that

$$\int x^2\sqrt{x-1}\,dx = \tfrac{2}{3}x^2(x-1)^{3/2} - \tfrac{8}{15}x(x-1)^{5/2} + \tfrac{16}{105}(x-1)^{7/2} + C \quad \blacktriangleleft$$

INTEGRATION BY PARTS FOR DEFINITE INTEGRALS

For definite integrals the formula corresponding to (3) is

$$\int_a^b u\,dv = uv\Big]_a^b - \int_a^b v\,du \tag{7}$$

REMARK It is important to keep in mind that the variables u and v in this formula are functions of x and that the limits of integration in (7) are limits on the variable x. Sometimes it is helpful to emphasize this by writing (7) as

$$\int_{x=a}^b u\,dv = uv\Big]_{x=a}^b - \int_{x=a}^b v\,du \tag{8}$$

The next example illustrates how integration by parts can be used to integrate the inverse trigonometric functions.

▶ Example 7 Evaluate $\int_0^1 \tan^{-1} x\,dx$.

Solution. Let

$$u = \tan^{-1} x, \quad dv = dx, \quad du = \frac{1}{1+x^2}\,dx, \quad v = x$$

Thus,

$$\int_0^1 \tan^{-1} x\,dx = \int_0^1 u\,dv = uv\Big]_0^1 - \int_0^1 v\,du$$

The limits of integration refer to x; that is, $x = 0$ and $x = 1$.

$$= x\tan^{-1} x\,\Big]_0^1 - \int_0^1 \frac{x}{1+x^2}\,dx$$

But

$$\int_0^1 \frac{x}{1+x^2}\,dx = \frac{1}{2}\int_0^1 \frac{2x}{1+x^2}\,dx = \frac{1}{2}\ln(1+x^2)\Big]_0^1 = \frac{1}{2}\ln 2$$

so

$$\int_0^1 \tan^{-1} x\,dx = x\tan^{-1} x\,\Big]_0^1 - \frac{1}{2}\ln 2 = \left(\frac{\pi}{4} - 0\right) - \frac{1}{2}\ln 2 = \frac{\pi}{4} - \ln\sqrt{2}$$ ◀

WATCH:

9.4 Integration by parts - The theory

9.5 Integration by parts - Examples

9.6 Integration by parts - One more example

Intro to Integration By Parts --- Examples: ∫xsinxdx & ∫arctan(x)dx

Two Tricky Integration By Parts Examples

life changing integration by parts trick

Use the following online solver:

Integral Calculator

Integral Calculator - eMathHelp

Integral Calculator

Definite and Improper Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 7.2 : Integration By Parts: 10-35

9. $\int x \ln x \, dx$

10. $\int \sqrt{x} \ln x \, dx$

11. $\int (\ln x)^2 \, dx$

12. $\int \frac{\ln x}{\sqrt{x}} \, dx$

13. $\int \ln(3x - 2) \, dx$

14. $\int \ln(x^2 + 4) \, dx$

15. $\int \sin^{-1} x \, dx$

16. $\int \cos^{-1}(2x) \, dx$

17. $\int \tan^{-1}(3x) \, dx$

18. $\int x \tan^{-1} x \, dx$

19. $\int e^x \sin x \, dx$

20. $\int e^{3x} \cos 2x \, dx$

21. $\int \sin(\ln x) \, dx$

22. $\int \cos(\ln x) \, dx$

23. $\int x \sec^2 x \, dx$

24. $\int x \tan^2 x \, dx$

25. $\int x^3 e^{x^2} \, dx$

26. $\int \frac{xe^x}{(x+1)^2} \, dx$

27. $\int_0^2 xe^{2x} \, dx$

28. $\int_0^1 xe^{-5x} \, dx$

29. $\int_1^e x^2 \ln x \, dx$

30. $\int_{\sqrt{e}}^e \frac{\ln x}{x^2} \, dx$

31. $\displaystyle\int_{-1}^{1} \ln(x+2)\,dx$

32. $\displaystyle\int_{0}^{\sqrt{3}/2} \sin^{-1} x\,dx$

33. $\displaystyle\int_{2}^{4} \sec^{-1}\sqrt{\theta}\,d\theta$

34. $\displaystyle\int_{1}^{2} x\sec^{-1} x\,dx$

35. $\displaystyle\int_{0}^{\pi} x\sin 2x\,dx$

36. $\displaystyle\int_{0}^{\pi} (x + x\cos x)\,dx$

Integrating Trigonometric functions (Reduction Formula)

READ:

REDUCTION FORMULAS

Integration by parts can be used to derive ***reduction formulas*** for integrals. These are formulas that express an integral involving a power of a function in terms of an integral that involves a *lower* power of that function. For example, if n is a positive integer and $n \geq 2$, then integration by parts can be used to obtain the reduction formulas

$$\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1} x\cos x + \frac{n-1}{n}\int \sin^{n-2} x\,dx \tag{9}$$

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1} x\sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx \tag{10}$$

To illustrate how such formulas can be obtained, let us derive (10). We begin by writing $\cos^n x$ as $\cos^{n-1} x \cdot \cos x$ and letting

$$u = \cos^{n-1} x \qquad dv = \cos x\,dx$$

$$du = (n-1)\cos^{n-2} x(-\sin x)\,dx \qquad v = \sin x$$

$$= -(n-1)\cos^{n-2} x\sin x\,dx$$

so that

$$\int \cos^n x\,dx = \int \cos^{n-1} x \cos x\,dx = \int u\,dv = uv - \int v\,du$$

$$= \cos^{n-1} x \sin x + (n-1) \int \sin^2 x \cos^{n-2} x\,dx$$

$$= \cos^{n-1} x \sin x + (n-1) \int (1 - \cos^2 x) \cos^{n-2} x\,dx$$

$$= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x\,dx - (n-1) \int \cos^n x\,dx$$

Moving the last term on the right to the left side yields

$$n \int \cos^n x\,dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x\,dx$$

from which (10) follows. The derivation of reduction formula (9) is similar (Exercise 63).

Reduction formulas (9) and (10) reduce the exponent of sine (or cosine) by 2. Thus, if the formulas are applied repeatedly, the exponent can eventually be reduced to 0 if n is even or 1 if n is odd, at which point the integration can be completed. We will discuss this method in more detail in the next section, but for now, here is an example that illustrates how reduction formulas work.

► **Example 8** Evaluate $\int \cos^4 x\, dx$.

Solution. From (10) with $n = 4$

$$\int \cos^4 x\, dx = \tfrac{1}{4}\cos^3 x \sin x + \tfrac{3}{4}\int \cos^2 x\, dx$$

Now apply (10) with $n = 2$.

$$= \tfrac{1}{4}\cos^3 x \sin x + \tfrac{3}{4}\left(\tfrac{1}{2}\cos x \sin x + \tfrac{1}{2}\int dx\right)$$

$$= \tfrac{1}{4}\cos^3 x \sin x + \tfrac{3}{8}\cos x \sin x + \tfrac{3}{8}x + C \quad \blacktriangleleft$$

INTEGRATING POWERS OF SINE AND COSINE

We begin by recalling two reduction formulas from the preceding section.

$$\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1} x\cos x + \frac{n-1}{n}\int \sin^{n-2} x\,dx \tag{1}$$

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1} x\sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx \tag{2}$$

In the case where $n = 2$, these formulas yield

$$\int \sin^2 x\,dx = -\tfrac{1}{2}\sin x\cos x + \tfrac{1}{2}\int dx = \tfrac{1}{2}x - \tfrac{1}{2}\sin x\cos x + C \tag{3}$$

$$\int \cos^2 x\,dx = \tfrac{1}{2}\cos x\sin x + \tfrac{1}{2}\int dx = \tfrac{1}{2}x + \tfrac{1}{2}\sin x\cos x + C \tag{4}$$

Alternative forms of these integration formulas can be derived from the trigonometric identities

$$\sin^2 x = \tfrac{1}{2}(1-\cos 2x) \quad \text{and} \quad \cos^2 x = \tfrac{1}{2}(1+\cos 2x) \tag{5–6}$$

which follow from the double-angle formulas

$$\cos 2x = 1 - 2\sin^2 x \quad \text{and} \quad \cos 2x = 2\cos^2 x - 1$$

These identities yield

$$\int \sin^2 x\,dx = \tfrac{1}{2}\int (1-\cos 2x)\,dx = \tfrac{1}{2}x - \tfrac{1}{4}\sin 2x + C \tag{7}$$

$$\int \cos^2 x\,dx = \tfrac{1}{2}\int (1+\cos 2x)\,dx = \tfrac{1}{2}x + \tfrac{1}{4}\sin 2x + C \tag{8}$$

Observe that the antiderivatives in Formulas (3) and (4) involve both sines and cosines, whereas those in (7) and (8) involve sines alone. However, the apparent discrepancy is easy to resolve by using the identity

$$\sin 2x = 2 \sin x \cos x$$

to rewrite (7) and (8) in forms (3) and (4), or conversely.

In the case where $n = 3$, the reduction formulas for integrating $\sin^3 x$ and $\cos^3 x$ yield

$$\int \sin^3 x\, dx = -\tfrac{1}{3} \sin^2 x \cos x + \tfrac{2}{3} \int \sin x\, dx = -\tfrac{1}{3} \sin^2 x \cos x - \tfrac{2}{3} \cos x + C \qquad (9)$$

$$\int \cos^3 x\, dx = \tfrac{1}{3} \cos^2 x \sin x + \tfrac{2}{3} \int \cos x\, dx = \tfrac{1}{3} \cos^2 x \sin x + \tfrac{2}{3} \sin x + C \qquad (10)$$

If desired, Formula (9) can be expressed in terms of cosines alone by using the identity $\sin^2 x = 1 - \cos^2 x$, and Formula (10) can be expressed in terms of sines alone by using the identity $\cos^2 x = 1 - \sin^2 x$. We leave it for you to do this and confirm that

$$\int \sin^3 x\, dx = \tfrac{1}{3} \cos^3 x - \cos x + C \qquad (11)$$

$$\int \cos^3 x\, dx = \sin x - \tfrac{1}{3} \sin^3 x + C \qquad (12)$$

We leave it as an exercise to obtain the following formulas by first applying the reduction formulas, and then using appropriate trigonometric identities.

$$\int \sin^4 x\, dx = \tfrac{3}{8}x - \tfrac{1}{4} \sin 2x + \tfrac{1}{32} \sin 4x + C \qquad (13)$$

$$\int \cos^4 x\, dx = \tfrac{3}{8}x + \tfrac{1}{4} \sin 2x + \tfrac{1}{32} \sin 4x + C \qquad (14)$$

Table 7.3.1

INTEGRATING PRODUCTS OF SINES AND COSINES

$\int \sin^m x \cos^n x \, dx$	PROCEDURE	RELEVANT IDENTITIES
n odd	• Split off a factor of $\cos x$. • Apply the relevant identity. • Make the substitution $u = \sin x$.	$\cos^2 x = 1 - \sin^2 x$
m odd	• Split off a factor of $\sin x$. • Apply the relevant identity. • Make the substitution $u = \cos x$.	$\sin^2 x = 1 - \cos^2 x$
$\begin{cases} m \text{ even} \\ n \text{ even} \end{cases}$	• Use the relevant identities to reduce the powers on $\sin x$ and $\cos x$.	$\begin{cases} \sin^2 x = \frac{1}{2}(1 - \cos 2x) \\ \cos^2 x = \frac{1}{2}(1 + \cos 2x) \end{cases}$

INTEGRATING PRODUCTS OF SINES AND COSINES

If m and n are positive integers, then the integral

$$\int \sin^m x \cos^n x \, dx$$

can be evaluated by one of the three procedures stated in Table 7.3.1, depending on whether m and n are odd or even.

► Example 2 Evaluate

(a) $\int \sin^4 x \cos^5 x \, dx$ (b) $\int \sin^4 x \cos^4 x \, dx$

Solution (a). Since $n = 5$ is odd, we will follow the first procedure in Table 7.3.1:

$$\begin{aligned}
\int \sin^4 x \cos^5 x \, dx &= \int \sin^4 x \cos^4 x \cos x \, dx \\
&= \int \sin^4 x (1 - \sin^2 x)^2 \cos x \, dx \\
&= \int u^4 (1 - u^2)^2 \, du \\
&= \int (u^4 - 2u^6 + u^8) \, du \\
&= \tfrac{1}{5}u^5 - \tfrac{2}{7}u^7 + \tfrac{1}{9}u^9 + C \\
&= \tfrac{1}{5}\sin^5 x - \tfrac{2}{7}\sin^7 x + \tfrac{1}{9}\sin^9 x + C
\end{aligned}$$

Solution (b). Since $m = n = 4$, both exponents are even, so we will follow the third procedure in Table 7.3.1:

$$\int \sin^4 x \cos^4 x \, dx = \int (\sin^2 x)^2 (\cos^2 x)^2 \, dx$$

$$= \int \left(\tfrac{1}{2}[1 - \cos 2x]\right)^2 \left(\tfrac{1}{2}[1 + \cos 2x]\right)^2 \, dx$$

$$= \tfrac{1}{16} \int (1 - \cos^2 2x)^2 \, dx$$

$$= \tfrac{1}{16} \int \sin^4 2x \, dx$$

Note that this can be obtained more directly from the original integral using the identity $\sin x \cos x = \frac{1}{2} \sin 2x$.

$$= \tfrac{1}{32} \int \sin^4 u \, du$$

$u = 2x$
$du = 2\,dx$ or $dx = \frac{1}{2}\,du$

$$= \tfrac{1}{32} \left(\tfrac{3}{8}u - \tfrac{1}{4} \sin 2u + \tfrac{1}{32} \sin 4u\right) + C$$

Formula (13)

$$= \tfrac{3}{128}x - \tfrac{1}{128} \sin 4x + \tfrac{1}{1024} \sin 8x + C \quad \blacktriangleleft$$

Integrals of the form

$$\int \sin mx \cos nx \, dx, \quad \int \sin mx \sin nx \, dx, \quad \int \cos mx \cos nx \, dx \tag{15}$$

can be found by using the trigonometric identities

$$\sin\alpha \cos\beta = \tfrac{1}{2}[\sin(\alpha - \beta) + \sin(\alpha + \beta)] \tag{16}$$

$$\sin\alpha \sin\beta = \tfrac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)] \tag{17}$$

$$\cos\alpha \cos\beta = \tfrac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)] \tag{18}$$

to express the integrand as a sum or difference of sines and cosines.

► **Example 3** Evaluate $\int \sin 7x \cos 3x \, dx$.

Solution. Using (16) yields

$$\int \sin 7x \cos 3x \, dx = \tfrac{1}{2} \int (\sin 4x + \sin 10x) \, dx = -\tfrac{1}{8} \cos 4x - \tfrac{1}{20} \cos 10x + C \quad \blacktriangleleft$$

INTEGRATING POWERS OF TANGENT AND SECANT

The procedures for integrating powers of tangent and secant closely parallel those for sine and cosine. The idea is to use the following reduction formulas (which were derived in Exercise 64 of Section 7.2) to reduce the exponent in the integrand until the resulting integral can be evaluated:

$$\int \tan^n x\,dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x\,dx \tag{19}$$

$$\int \sec^n x\,dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2} x\,dx \tag{20}$$

In the case where n is odd, the exponent can be reduced to 1, leaving us with the problem of integrating $\tan x$ or $\sec x$. These integrals are given by

$$\int \tan x\,dx = \ln|\sec x| + C \tag{21}$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C \tag{22}$$

Formula (21) can be obtained by writing

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$$

$$= -\ln|\cos x| + C \qquad \begin{array}{l} u = \cos x \\ du = -\sin x\,dx \end{array}$$

$$= \ln|\sec x| + C \qquad \ln|\cos x| = -\ln\frac{1}{|\cos x|}$$

To obtain Formula (22) we write

$$\int \sec x\,dx = \int \sec x\left(\frac{\sec x + \tan x}{\sec x + \tan x}\right)dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\,dx$$

$$= \ln|\sec x + \tan x| + C \qquad \begin{array}{l} u = \sec x + \tan x \\ du = (\sec^2 x + \sec x \tan x)\,dx \end{array}$$

The following basic integrals occur frequently and are worth noting:

$$\int \tan^2 x\,dx = \tan x - x + C \tag{23}$$

$$\int \sec^2 x\,dx = \tan x + C \tag{24}$$

Formula (24) is already known to us, since the derivative of $\tan x$ is $\sec^2 x$. Formula (23) can be obtained by applying reduction formula (19) with $n = 2$ (verify) or, alternatively, by using the identity

$$1 + \tan^2 x = \sec^2 x$$

to write

$$\int \tan^2 x\,dx = \int (\sec^2 x - 1)\,dx = \tan x - x + C$$

The formulas

$$\int \tan^3 x\,dx = \tfrac{1}{2}\tan^2 x - \ln|\sec x| + C \tag{25}$$

$$\int \sec^3 x\,dx = \tfrac{1}{2}\sec x \tan x + \tfrac{1}{2}\ln|\sec x + \tan x| + C \tag{26}$$

can be deduced from (21), (22), and reduction formulas (19) and (20) as follows:

$$\int \tan^3 x\,dx = \tfrac{1}{2}\tan^2 x - \int \tan x\,dx = \tfrac{1}{2}\tan^2 x - \ln|\sec x| + C$$

$$\int \sec^3 x\,dx = \tfrac{1}{2}\sec x \tan x + \tfrac{1}{2}\int \sec x\,dx = \tfrac{1}{2}\sec x \tan x + \tfrac{1}{2}\ln|\sec x + \tan x| + C$$

INTEGRATING PRODUCTS OF TANGENTS AND SECANTS

If m and n are positive integers, then the integral

$$\int \tan^m x \sec^n x \, dx$$

can be evaluated by one of the three procedures stated in Table 7.3.2, depending on whether m and n are odd or even.

Table 7.3.2

INTEGRATING PRODUCTS OF TANGENTS AND SECANTS

$\int \tan^m x \sec^n x \, dx$	PROCEDURE	RELEVANT IDENTITIES
n even	• Split off a factor of $\sec^2 x$. • Apply the relevant identity. • Make the substitution $u = \tan x$.	$\sec^2 x = \tan^2 x + 1$
m odd	• Split off a factor of $\sec x \tan x$. • Apply the relevant identity. • Make the substitution $u = \sec x$.	$\tan^2 x = \sec^2 x - 1$
$\begin{cases} m \text{ even} \\ n \text{ odd} \end{cases}$	• Use the relevant identities to reduce the integrand to powers of $\sec x$ alone. • Then use the reduction formula for powers of $\sec x$.	$\tan^2 x = \sec^2 x - 1$

▶ **Example 4** Evaluate

$$\text{(a)} \int \tan^2 x \sec^4 x \, dx \qquad \text{(b)} \int \tan^3 x \sec^3 x \, dx \qquad \text{(c)} \int \tan^2 x \sec x \, dx$$

Solution (a). Since $n = 4$ is even, we will follow the first procedure in Table 7.3.2:

$$\begin{aligned}
\int \tan^2 x \sec^4 x \, dx &= \int \tan^2 x \sec^2 x \sec^2 x \, dx \\
&= \int \tan^2 x (\tan^2 x + 1) \sec^2 x \, dx \\
&= \int u^2 (u^2 + 1) \, du \\
&= \tfrac{1}{5}u^5 + \tfrac{1}{3}u^3 + C = \tfrac{1}{5}\tan^5 x + \tfrac{1}{3}\tan^3 x + C
\end{aligned}$$

Solution (b). Since $m = 3$ is odd, we will follow the second procedure in Table 7.3.2:

$$\begin{aligned}
\int \tan^3 x \sec^3 x \, dx &= \int \tan^2 x \sec^2 x (\sec x \tan x) \, dx \\
&= \int (\sec^2 x - 1) \sec^2 x (\sec x \tan x) \, dx \\
&= \int (u^2 - 1) u^2 \, du \\
&= \tfrac{1}{5}u^5 - \tfrac{1}{3}u^3 + C = \tfrac{1}{5}\sec^5 x - \tfrac{1}{3}\sec^3 x + C
\end{aligned}$$

Solution (c). Since $m = 2$ is even and $n = 1$ is odd, we will follow the third procedure in Table 7.3.2:

$$\begin{aligned}\int \tan^2 x \sec x \, dx &= \int (\sec^2 x - 1) \sec x \, dx \\ &= \int \sec^3 x \, dx - \int \sec x \, dx \qquad \text{See (26) and (22)} \\ &= \tfrac{1}{2} \sec x \tan x + \tfrac{1}{2} \ln |\sec x + \tan x| - \ln |\sec x + \tan x| + C \\ &= \tfrac{1}{2} \sec x \tan x - \tfrac{1}{2} \ln |\sec x + \tan x| + C \quad \blacktriangleleft\end{aligned}$$

AN ALTERNATIVE METHOD FOR INTEGRATING POWERS OF SINE, COSINE, TANGENT, AND SECANT

The methods in Tables 7.3.1 and 7.3.2 can sometimes be applied if $m = 0$ or $n = 0$ to integrate positive integer powers of sine, cosine, tangent, and secant without reduction formulas. For example, instead of using the reduction formula to integrate $\sin^3 x$, we can apply the second procedure in Table 7.3.1:

$$\begin{aligned}\int \sin^3 x \, dx &= \int (\sin^2 x) \sin x \, dx \\ &= \int (1 - \cos^2 x) \sin x \, dx \qquad \begin{array}{c} u = \cos x \\ du = -\sin x \, dx \end{array} \\ &= -\int (1 - u^2) \, du \\ &= \tfrac{1}{3}u^3 - u + C = \tfrac{1}{3} \cos^3 x - \cos x + C\end{aligned}$$

which agrees with (11).

With the aid of the identity

$$1 + \cot^2 x = \csc^2 x$$

the techniques in Table 7.3.2 can be adapted to evaluate integrals of the form

$$\int \cot^m x \, \csc^n x \, dx$$

It is also possible to derive reduction formulas for powers of cot and csc that are analogous to Formulas (19) and (20).

WATCH:

9.7 Integration of products of trigonometric functions (Part 1)

9.8 Integration of products of trigonometric functions (Part 2)

9.9 The integral of secant

Trigonometric Integrals --- ∫sin^n(x)cos^m(x)dx via Pythagorean or Half-Angle Identities

Use the following online solver:

Integral Calculator

Integral Calculator - eMathHelp

Integral Calculator

Definite and Improper Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 7.3: Integrating Trigonometric functions (Reduction Formula): 5, 7, 13, 19, 43, 44, 47

5. $\int \sin^3 a\theta \, d\theta$ **7.** $\int \sin ax \cos ax \, dx$ **13.** $\int \sin 2x \cos 3x \, dx$ **19.** $\int_0^{\pi/3} \sin^4 3x \cos^3 3x \, dx$

43. $\int \sqrt{\tan x} \sec^4 x \, dx$ **44.** $\int \tan x \sec^{3/2} x \, dx$ **47.** $\int_0^{\pi/2} \tan^5 \frac{x}{2} \, dx$

Improper Integrals

READ:

IMPROPER INTEGRALS

It is assumed in the definition of the definite integral

$$\int_a^b f(x)\,dx$$

that $[a, b]$ is a finite interval and that the limit that defines the integral exists; that is, the function f is integrable. We observed in Theorems 5.5.2 and 5.5.8 that continuous functions are integrable, as are bounded functions with finitely many points of discontinuity. We also observed in Theorem 5.5.8 that functions that are not bounded on the interval of integration are not integrable. Thus, for example, a function with a vertical asymptote within the interval of integration would not be integrable.

Our main objective in this section is to extend the concept of a definite integral to allow for infinite intervals of integration and integrands with vertical asymptotes within the interval of integration. We will call the vertical asymptotes ***infinite discontinuities***, and we will call

integrals with infinite intervals of integration or infinite discontinuities within the interval of integration ***improper integrals***. Here are some examples:

- Improper integrals with infinite intervals of integration:

$$\int_1^{+\infty} \frac{dx}{x^2}, \quad \int_{-\infty}^{0} e^x\,dx, \quad \int_{-\infty}^{+\infty} \frac{dx}{1+x^2}$$

- Improper integrals with infinite discontinuities in the interval of integration:

$$\int_{-3}^{3} \frac{dx}{x^2}, \quad \int_1^2 \frac{dx}{x-1}, \quad \int_0^{\pi} \tan x\,dx$$

- Improper integrals with infinite discontinuities and infinite intervals of integration:

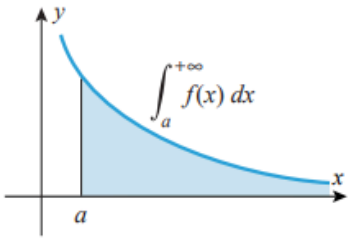
$$\int_0^{+\infty} \frac{dx}{\sqrt{x}}, \quad \int_{-\infty}^{+\infty} \frac{dx}{x^2-9}, \quad \int_1^{+\infty} \sec x\,dx$$

INTEGRALS OVER INFINITE INTERVALS

To motivate a reasonable definition for improper integrals of the form

$$\int_a^{+\infty} f(x)\,dx$$

let us begin with the case where f is continuous and nonnegative on $[a, +\infty)$, so we can think of the integral as the area under the curve $y = f(x)$ over the interval $[a, +\infty)$ (Figure 7.8.1). At first, you might be inclined to argue that this area is infinite because the region has infinite extent. However, such an argument would be based on vague intuition rather than precise mathematical logic, since the concept of area has only been defined over intervals of *finite extent*. Thus, before we can make any reasonable statements about the area of the region in Figure 7.8.1, we need to begin by defining what we mean by the area of this region. For that purpose, it will help to focus on a specific example.



▲ **Figure 7.8.1**

Suppose we are interested in the area A of the region that lies below the curve $y = 1/x^2$ and above the interval $[1, +\infty)$ on the x-axis. Instead of trying to find the entire area at once, let us begin by calculating the portion of the area that lies above a finite interval $[1, b]$, where $b > 1$ is arbitrary. That area is

$$\int_1^b \frac{dx}{x^2} = -\frac{1}{x}\Big]_1^b = 1 - \frac{1}{b}$$

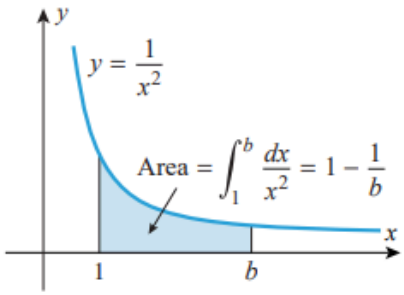


Figure 7.8.2

(Figure 7.8.2). If we now allow b to increase so that $b \to +\infty$, then the portion of the area over the interval $[1, b]$ will begin to fill out the area over the entire interval $[1, +\infty)$ (Figure 7.8.3), and hence we can reasonably define the area A under $y = 1/x^2$ over the interval $[1, +\infty)$ to be

$$A = \int_1^{+\infty} \frac{dx}{x^2} = \lim_{b \to +\infty} \int_1^b \frac{dx}{x^2} = \lim_{b \to +\infty} \left(1 - \frac{1}{b}\right) = 1 \qquad (1)$$

Thus, the area has a finite value of 1 and is not infinite as we first conjectured.

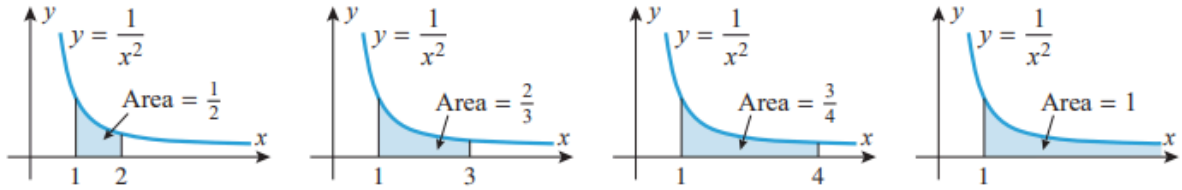


Figure 7.8.3

With the preceding discussion as our guide, we make the following definition (which is applicable to functions with both positive and negative values).

If f is nonnegative over the interval $[a, +\infty)$, then the improper integral in Definition 7.8.1 can be interpreted to be the area under the graph of f over the interval $[a, +\infty)$. If the integral converges, then the area is finite and equal to the value of the integral, and if the integral diverges, then the area is regarded to be infinite.

7.8.1 DEFINITION The ***improper integral of f over the interval*** $[a, +\infty)$ is defined to be

$$\int_a^{+\infty} f(x)\,dx = \lim_{b \to +\infty} \int_a^b f(x)\,dx$$

In the case where the limit exists, the improper integral is said to ***converge***, and the limit is defined to be the value of the integral. In the case where the limit does not exist, the improper integral is said to ***diverge***, and it is not assigned a value.

▶ **Example 1** Evaluate

$$\text{(a)} \int_1^{+\infty} \frac{dx}{x^3} \qquad \text{(b)} \int_1^{+\infty} \frac{dx}{x}$$

Solution (a). Following the definition, we replace the infinite upper limit by a finite upper limit b, and then take the limit of the resulting integral. This yields

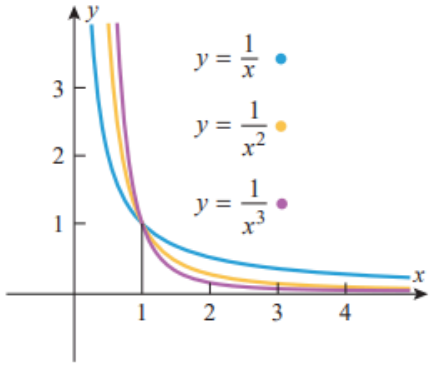
$$\int_1^{+\infty} \frac{dx}{x^3} = \lim_{b\to+\infty} \int_1^b \frac{dx}{x^3} = \lim_{b\to+\infty} \left[-\frac{1}{2x^2}\right]_1^b = \lim_{b\to+\infty} \left(\frac{1}{2} - \frac{1}{2b^2}\right) = \frac{1}{2}$$

Since the limit is finite, the integral converges and its value is $1/2$.

Solution (b).

$$\int_1^{+\infty} \frac{dx}{x} = \lim_{b\to+\infty} \int_1^b \frac{dx}{x} = \lim_{b\to+\infty} \left[\ln x\right]_1^b = \lim_{b\to+\infty} \ln b = +\infty$$

In this case the integral diverges and hence has no value. ◀



▲ **Figure 7.8.4**

Because the functions $1/x^3$, $1/x^2$, and $1/x$ are nonnegative over the interval $[1, +\infty)$, it follows from (1) and the last example that over this interval the area under $y = 1/x^3$ is $\frac{1}{2}$, the area under $y = 1/x^2$ is 1, and the area under $y = 1/x$ is infinite. However, on the surface the graphs of the three functions seem very much alike (Figure 7.8.4), and there is nothing to suggest why one of the areas should be infinite and the other two finite. One explanation is that $1/x^3$ and $1/x^2$ approach zero more rapidly than $1/x$ as $x \to +\infty$, so that the area over the interval $[1, b]$ accumulates less rapidly under the curves $y = 1/x^3$ and $y = 1/x^2$ than under $y = 1/x$ as $b \to +\infty$, and the difference is just enough that the first two areas are finite and the third is infinite.

► **Example 2** For what values of p does the integral $\displaystyle\int_1^{+\infty} \frac{dx}{x^p}$ converge?

Solution. We know from the preceding example that the integral diverges if $p = 1$, so let us assume that $p \neq 1$. In this case we have

$$\int_1^{+\infty} \frac{dx}{x^p} = \lim_{b\to+\infty} \int_1^b x^{-p}\,dx = \lim_{b\to+\infty} \frac{x^{1-p}}{1-p}\bigg]_1^b = \lim_{b\to+\infty} \left[\frac{b^{1-p}}{1-p} - \frac{1}{1-p}\right]$$

If $p > 1$, then the exponent $1 - p$ is negative and $b^{1-p} \to 0$ as $b \to +\infty$; and if $p < 1$, then the exponent $1 - p$ is positive and $b^{1-p} \to +\infty$ as $b \to +\infty$. Thus, the integral converges if $p > 1$ and diverges otherwise. In the convergent case the value of the integral is

$$\int_1^{+\infty} \frac{dx}{x^p} = \left[0 - \frac{1}{1-p}\right] = \frac{1}{p-1} \quad (p > 1) \;\blacktriangleleft$$

The following theorem summarizes this result.

7.8.2 THEOREM

$$\int_1^{+\infty} \frac{dx}{x^p} = \begin{cases} \dfrac{1}{p-1} & \text{if} \quad p > 1 \\ \text{diverges} & \text{if} \quad p \le 1 \end{cases}$$

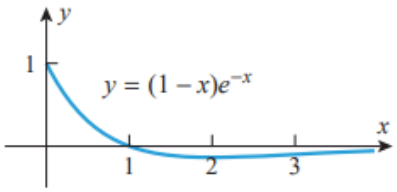
▶ **Example 3** Evaluate $\displaystyle\int_0^{+\infty} (1-x)e^{-x}\,dx$.

Solution. We begin by evaluating the indefinite integral using integration by parts. Setting $u = 1 - x$ and $dv = e^{-x}\,dx$ yields

$$\int (1-x)e^{-x}\,dx = -e^{-x}(1-x) - \int e^{-x}\,dx = -e^{-x} + xe^{-x} + e^{-x} + C = xe^{-x} + C$$

Thus,

$$\int_0^{+\infty} (1-x)e^{-x}\,dx = \lim_{b\to+\infty} \int_0^b (1-x)e^{-x}\,dx = \lim_{b\to+\infty} \left[xe^{-x}\right]_0^b = \lim_{b\to+\infty} \frac{b}{e^b}$$



The net signed area between the graph and the interval $[0, +\infty)$ is zero.

▲ **Figure 7.8.5**

If f is nonnegative over the interval $(-\infty, +\infty)$, then the improper integral

$$\int_{-\infty}^{+\infty} f(x)\,dx$$

can be interpreted to be the area under the graph of f over the interval $(-\infty, +\infty)$. The area is finite and equal to the value of the integral if the integral converges and is infinite if it diverges.

The limit is an indeterminate form of type ∞/∞, so we will apply L'Hôpital's rule by differentiating the numerator and denominator with respect to b. This yields

$$\int_0^{+\infty} (1-x)e^{-x}\,dx = \lim_{b\to+\infty} \frac{1}{e^b} = 0$$

We can interpret this to mean that the net signed area between the graph of $y = (1-x)e^{-x}$ and the interval $[0, +\infty)$ is 0 (Figure 7.8.5). ◀

7.8.3 DEFINITION The ***improper integral of f over the interval*** $(-\infty, b]$ is defined to be

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to-\infty} \int_a^b f(x)\,dx \tag{2}$$

The integral is said to ***converge*** if the limit exists and ***diverge*** if it does not.

The ***improper integral of f over the interval*** $(-\infty, +\infty)$ is defined as

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_c^{+\infty} f(x)\,dx \tag{3}$$

where c is any real number. The improper integral is said to ***converge*** if *both* terms converge and ***diverge*** if *either* term diverges.

▶ Example 4 Evaluate $\displaystyle\int_{-\infty}^{+\infty} \frac{dx}{1+x^2}$.

Solution. We will evaluate the integral by choosing $c = 0$ in (3). With this value for c we obtain

$$\int_0^{+\infty} \frac{dx}{1+x^2} = \lim_{b\to+\infty} \int_0^b \frac{dx}{1+x^2} = \lim_{b\to+\infty} \Big[\tan^{-1} x\Big]_0^b = \lim_{b\to+\infty} (\tan^{-1} b) = \frac{\pi}{2}$$

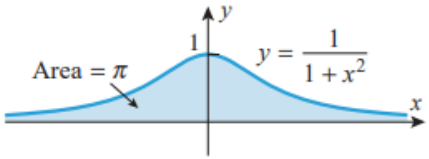
$$\int_{-\infty}^0 \frac{dx}{1+x^2} = \lim_{a\to-\infty} \int_a^0 \frac{dx}{1+x^2} = \lim_{a\to-\infty} \Big[\tan^{-1} x\Big]_a^0 = \lim_{a\to-\infty} (-\tan^{-1} a) = \frac{\pi}{2}$$

Although we usually choose $c = 0$ in (3), the choice does not matter because it can be proved that neither the convergence nor the value of the integral is affected by the choice of c.

Thus, the integral converges and its value is

$$\int_{-\infty}^{+\infty} \frac{dx}{1+x^2} = \int_{-\infty}^0 \frac{dx}{1+x^2} + \int_0^{+\infty} \frac{dx}{1+x^2} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

Since the integrand is nonnegative on the interval $(-\infty, +\infty)$, the integral represents the area of the region shown in Figure 7.8.6. ◀

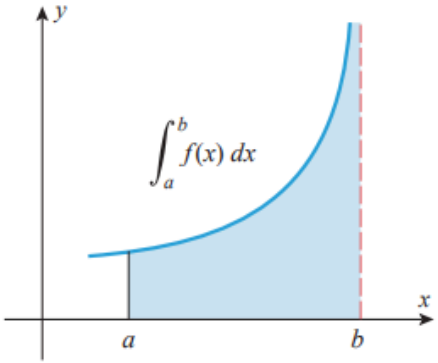


▲ **Figure 7.8.6**

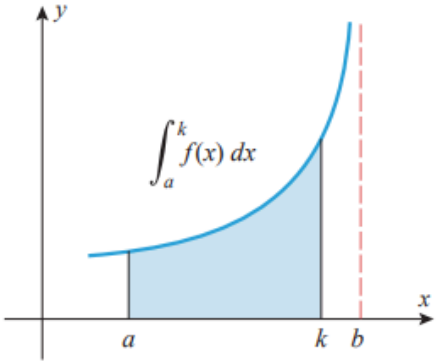
INTEGRALS WHOSE INTEGRANDS HAVE INFINITE DISCONTINUITIES

Next we will consider improper integrals whose integrands have infinite discontinuities. We will start with the case where the interval of integration is a finite interval $[a, b]$ and the infinite discontinuity occurs at the right-hand endpoint.

To motivate an appropriate definition for such an integral let us consider the case where f is nonnegative on $[a, b]$, so we can interpret the improper integral $\int_a^b f(x)\,dx$ as the area of the region in Figure 7.8.7*a*. The problem of finding the area of this region is complicated by the fact that it extends indefinitely in the positive y-direction. However, instead of trying to find the entire area at once, we can proceed indirectly by calculating the portion of the area over the interval $[a, k]$, where $a \le k < b$, and then letting k approach b to fill out the area of the entire region (Figure 7.8.7*b*). Motivated by this idea, we make the following definition.



(*a*)



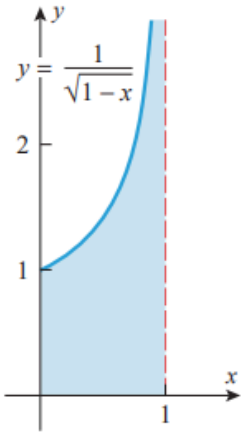
(*b*)

▲ **Figure 7.8.7**

7.8.4 DEFINITION If f is continuous on the interval $[a, b]$, except for an infinite discontinuity at b, then the ***improper integral of f over the interval [a, b]*** is defined as

$$\int_a^b f(x)\,dx = \lim_{k \to b^-} \int_a^k f(x)\,dx \tag{4}$$

In the case where the indicated limit exists, the improper integral is said to ***converge***, and the limit is defined to be the value of the integral. In the case where the limit does not exist, the improper integral is said to ***diverge***, and it is not assigned a value.



▲ **Figure 7.8.8**

► **Example 5** Evaluate $\displaystyle\int_0^1 \frac{dx}{\sqrt{1-x}}$.

Solution. The integral is improper because the integrand approaches $+\infty$ as x approaches the upper limit 1 from the left (Figure 7.8.8). From (4),

$$\int_0^1 \frac{dx}{\sqrt{1-x}} = \lim_{k\to 1^-} \int_0^k \frac{dx}{\sqrt{1-x}} = \lim_{k\to 1^-} \left[-2\sqrt{1-x}\right]_0^k$$

$$= \lim_{k\to 1^-} \left[-2\sqrt{1-k}+2\right] = 2 \quad \blacktriangleleft$$

Improper integrals with an infinite discontinuity at the left-hand endpoint or inside the interval of integration are defined as follows.

7.8.5 DEFINITION If f is continuous on the interval $[a, b]$, except for an infinite discontinuity at a, then the ***improper integral of f over the interval [a, b]*** is defined as

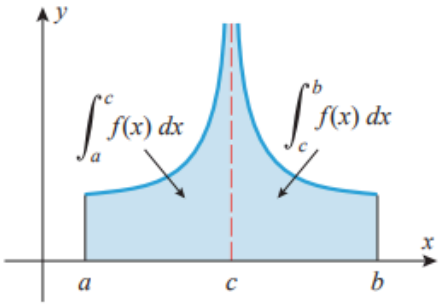
$$\int_a^b f(x)\,dx = \lim_{k \to a^+} \int_k^b f(x)\,dx \tag{5}$$

The integral is said to ***converge*** if the indicated limit exists and ***diverge*** if it does not.

If f is continuous on the interval $[a, b]$, except for an infinite discontinuity at a point c in (a, b), then the ***improper integral of f over the interval [a, b]*** is defined as

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx \tag{6}$$

where the two integrals on the right side are themselves improper. The improper integral on the left side is said to ***converge*** if *both* terms on the right side converge and ***diverge*** if *either* term on the right side diverges (Figure 7.8.9).

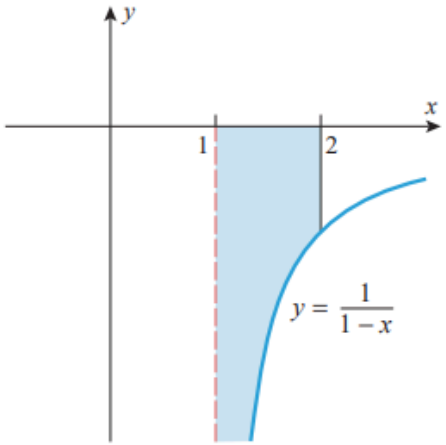


$\int_a^b f(x)\,dx$ is improper.

▲ **Figure 7.8.9**

► **Example 6** Evaluate

(a) $\displaystyle\int_1^2 \frac{dx}{1-x}$ (b) $\displaystyle\int_1^4 \frac{dx}{(x-2)^{2/3}}$



▲ **Figure 7.8.10**

Solution (a). The integral is improper because the integrand approaches $-\infty$ as x approaches the lower limit 1 from the right (Figure 7.8.10). From Definition 7.8.5 we obtain

$$\int_1^2 \frac{dx}{1-x} = \lim_{k\to 1^+} \int_k^2 \frac{dx}{1-x} = \lim_{k\to 1^+} \Big[-\ln|1-x|\Big]_k^2$$

$$= \lim_{k\to 1^+} \big[-\ln|-1| + \ln|1-k|\big] = \lim_{k\to 1^+} \ln|1-k| = -\infty$$

so the integral diverges.

Solution (b). The integral is improper because the integrand approaches $+\infty$ at $x = 2$, which is inside the interval of integration. From Definition 7.8.5 we obtain

$$\int_1^4 \frac{dx}{(x-2)^{2/3}} = \int_1^2 \frac{dx}{(x-2)^{2/3}} + \int_2^4 \frac{dx}{(x-2)^{2/3}} \tag{7}$$

and we must investigate the convergence of both improper integrals on the right. Since

$$\int_1^2 \frac{dx}{(x-2)^{2/3}} = \lim_{k\to 2^-} \int_1^k \frac{dx}{(x-2)^{2/3}} = \lim_{k\to 2^-} \Big[3(k-2)^{1/3} - 3(1-2)^{1/3}\Big] = 3$$

$$\int_2^4 \frac{dx}{(x-2)^{2/3}} = \lim_{k\to 2^+} \int_k^4 \frac{dx}{(x-2)^{2/3}} = \lim_{k\to 2^+} \Big[3(4-2)^{1/3} - 3(k-2)^{1/3}\Big] = 3\sqrt[3]{2}$$

we have from (7) that

$$\int_1^4 \frac{dx}{(x-2)^{2/3}} = 3 + 3\sqrt[3]{2} \quad \blacktriangleleft$$

WARNING It is sometimes tempting to apply the Fundamental Theorem of Calculus directly to an improper integral without taking the appropriate limits. To illustrate what can go wrong with this procedure, suppose we fail to recognize that the integral

$$\int_0^2 \frac{dx}{(x-1)^2} \tag{8}$$

is improper and mistakenly evaluate this integral as

$$-\frac{1}{x-1}\Big]_0^2 = -1 - (1) = -2$$

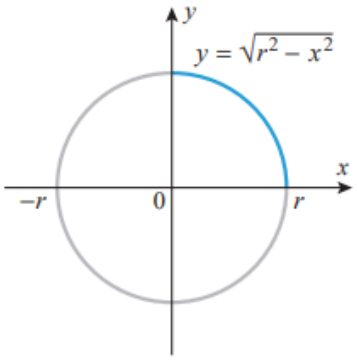
This result is clearly incorrect because the integrand is never negative and hence the integral cannot be negative! To evaluate (8) correctly we should first write

$$\int_0^2 \frac{dx}{(x-1)^2} = \int_0^1 \frac{dx}{(x-1)^2} + \int_1^2 \frac{dx}{(x-1)^2}$$

and then treat each term as an improper integral. For the first term,

$$\int_0^1 \frac{dx}{(x-1)^2} = \lim_{k\to 1^-} \int_0^k \frac{dx}{(x-1)^2} = \lim_{k\to 1^-} \left[-\frac{1}{k-1} - 1\right] = +\infty$$

so (8) diverges.



▲ **Figure 7.8.11**

▶ **Example 7** Derive the formula for the circumference of a circle of radius r.

Solution. For convenience, let us assume that the circle is centered at the origin, in which case its equation is $x^2 + y^2 = r^2$. We will find the arc length of the portion of the circle that lies in the first quadrant and then multiply by 4 to obtain the total circumference (Figure 7.8.11).

Since the equation of the upper semicircle is $y = \sqrt{r^2 - x^2}$, it follows from Formula (4) of Section 6.4 that the circumference C is

$$C = 4\int_0^r \sqrt{1 + (dy/dx)^2}\,dx = 4\int_0^r \sqrt{1 + \left(-\frac{x}{\sqrt{r^2 - x^2}}\right)^2}\,dx$$

$$= 4r\int_0^r \frac{dx}{\sqrt{r^2 - x^2}}$$

This integral is improper because of the infinite discontinuity at $x = r$, and hence we evaluate it by writing

$$C = 4r \lim_{k \to r^-} \int_0^k \frac{dx}{\sqrt{r^2 - x^2}}$$

$$= 4r \lim_{k \to r^-} \left[\sin^{-1}\left(\frac{x}{r}\right)\right]_0^k$$

Formula (77) in the Endpaper Integral Table

$$= 4r \lim_{k \to r^-} \left[\sin^{-1}\left(\frac{k}{r}\right) - \sin^{-1} 0\right]$$

$$= 4r[\sin^{-1} 1 - \sin^{-1} 0] = 4r\left(\frac{\pi}{2} - 0\right) = 2\pi r \quad ◀$$

WATCH:

12.1 Improper integrals: Definition and Example 1

12.2 Improper integrals: Example 2

12.3 Improper integrals: Example 3

12.4 Improper integral - Example 4 ("p functions")

12.5 Improper integral: Example 5 (vertical asymptote)

12.6 Doubly improper integrals

Improper Integrals: How to Integrate with Infinities, 2 ways!

Use the following online solver:

Integral Calculator

Integral Calculator - eMathHelp

Integral Calculator

Definite and Improper Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 7.8: Improper Integrals: 3-32

1. In each part, determine whether the integral is improper, and if so, explain why.

(d) $\int_1^{+\infty} e^{-x}\,dx$

3–32 Evaluate the integrals that converge. ■

3. $\displaystyle\int_0^{+\infty} e^{-2x}\,dx$

4. $\displaystyle\int_{-1}^{+\infty} \frac{x}{1+x^2}\,dx$

5. $\displaystyle\int_3^{+\infty} \frac{2}{x^2-1}\,dx$

6. $\displaystyle\int_0^{+\infty} xe^{-x^2}\,dx$

7. $\displaystyle\int_e^{+\infty} \frac{1}{x\ln^3 x}\,dx$

8. $\displaystyle\int_2^{+\infty} \frac{1}{x\sqrt{\ln x}}\,dx$

9. $\displaystyle\int_{-\infty}^0 \frac{dx}{(2x-1)^3}$

10. $\displaystyle\int_{-\infty}^3 \frac{dx}{x^2+9}$

11. $\displaystyle\int_{-\infty}^0 e^{3x}\,dx$

12. $\displaystyle\int_{-\infty}^0 \frac{e^x\,dx}{3-2e^x}$

13. $\displaystyle\int_{-\infty}^{+\infty} x\,dx$

14. $\displaystyle\int_{-\infty}^{+\infty} \frac{x}{\sqrt{x^2+2}}\,dx$

15. $\displaystyle\int_{-\infty}^{+\infty} \frac{x}{(x^2+3)^2}\,dx$

16. $\displaystyle\int_{-\infty}^{+\infty} \frac{e^{-t}}{1+e^{-2t}}\,dt$

17. $\displaystyle\int_0^4 \frac{dx}{(x-4)^2}$

18. $\displaystyle\int_0^8 \frac{dx}{\sqrt[3]{x}}$

19. $\displaystyle\int_0^{\pi/2} \tan x\,dx$

20. $\displaystyle\int_0^4 \frac{dx}{\sqrt{4-x}}$

21. $\displaystyle\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

22. $\displaystyle\int_{-3}^1 \frac{x\,dx}{\sqrt{9-x^2}}$

23. $\displaystyle\int_{\pi/3}^{\pi/2} \frac{\sin x}{\sqrt{1-2\cos x}}\,dx$

24. $\displaystyle\int_{0}^{\pi/4} \frac{\sec^2 x}{1-\tan x}\,dx$

25. $\displaystyle\int_{0}^{3} \frac{dx}{x-2}$

26. $\displaystyle\int_{-2}^{2} \frac{dx}{x^2}$

27. $\displaystyle\int_{-1}^{8} x^{-1/3}\,dx$

28. $\displaystyle\int_{0}^{1} \frac{dx}{(x-1)^{2/3}}$

29. $\displaystyle\int_{0}^{+\infty} \frac{1}{x^2}\,dx$

30. $\displaystyle\int_{1}^{+\infty} \frac{dx}{x\sqrt{x^2-1}}$

31. $\displaystyle\int_{0}^{1} \frac{dx}{\sqrt{x}(x+1)}$

32. $\displaystyle\int_{0}^{+\infty} \frac{dx}{\sqrt{x}(x+1)}$

$$\int_{-\infty}^{\infty} x^3\,\mathrm{d}x$$

$$\int_{3}^{4} \frac{1}{(x-3)^2}\,\mathrm{d}x$$

$$\int_{0}^{\frac{\pi}{6}} \frac{\cos(x)}{\sqrt{1-2\sin(x)}}\,\mathrm{d}x$$

Integration by Trigonometric Substitution, Integration of Rational Function

Integration by Trigonometric Substitution

READ:

THE METHOD OF TRIGONOMETRIC SUBSTITUTION

To start, we will be concerned with integrals that contain expressions of the form

$$\sqrt{a^2 - x^2}, \quad \sqrt{x^2 + a^2}, \quad \sqrt{x^2 - a^2}$$

in which a is a positive constant. The basic idea for evaluating such integrals is to make a substitution for x that will eliminate the radical. For example, to eliminate the radical in the expression $\sqrt{a^2 - x^2}$, we can make the substitution

$$x = a \sin\theta, \quad -\pi/2 \le \theta \le \pi/2 \tag{1}$$

which yields

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2\theta} = \sqrt{a^2(1 - \sin^2\theta)}$$

$$= a\sqrt{\cos^2\theta} = a|\cos\theta| = a\cos\theta \qquad \cos\theta \ge 0 \text{ since } -\pi/2 \le \theta \le \pi/2$$

The restriction on θ in (1) serves two purposes—it enables us to replace $|\cos\theta|$ by $\cos\theta$ to simplify the calculations, and it also ensures that the substitutions can be rewritten as $\theta = \sin^{-1}(x/a)$, if needed.

Table 7.4.1

TRIGONOMETRIC SUBSTITUTIONS

EXPRESSION IN THE INTEGRAND	SUBSTITUTION	RESTRICTION ON θ	SIMPLIFICATION
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$-\pi/2 \le \theta \le \pi/2$	$a^2 - x^2 = a^2 - a^2\sin^2\theta = a^2\cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$-\pi/2 < \theta < \pi/2$	$a^2 + x^2 = a^2 + a^2\tan^2\theta = a^2\sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$\begin{cases} 0 \le \theta < \pi/2 & (\text{if } x \ge a) \\ \pi/2 < \theta \le \pi & (\text{if } x \le -a) \end{cases}$	$x^2 - a^2 = a^2\sec^2\theta - a^2 = a^2\tan^2\theta$

▶ Example 1 Evaluate $\displaystyle\int \frac{dx}{x^2\sqrt{4-x^2}}$.

Solution. To eliminate the radical we make the substitution

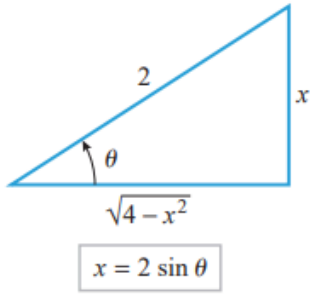
$$x = 2\sin\theta, \quad dx = 2\cos\theta\, d\theta$$

This yields

$$\int \frac{dx}{x^2\sqrt{4-x^2}} = \int \frac{2\cos\theta\, d\theta}{(2\sin\theta)^2\sqrt{4-4\sin^2\theta}}$$

$$= \int \frac{2\cos\theta\, d\theta}{(2\sin\theta)^2(2\cos\theta)} = \frac{1}{4}\int \frac{d\theta}{\sin^2\theta}$$

$$= \frac{1}{4}\int \csc^2\theta\, d\theta = -\frac{1}{4}\cot\theta + C \tag{2}$$



▲ **Figure 7.4.1**

At this point we have completed the integration; however, because the original integral was expressed in terms of x, it is desirable to express $\cot\theta$ in terms of x as well. This can be done using trigonometric identities, but the expression can also be obtained by writing the substitution $x = 2\sin\theta$ as $\sin\theta = x/2$ and representing it geometrically as in Figure 7.4.1. From that figure we obtain

$$\cot\theta = \frac{\sqrt{4-x^2}}{x}$$

Substituting this in (2) yields

$$\int \frac{dx}{x^2\sqrt{4-x^2}} = -\frac{1}{4}\frac{\sqrt{4-x^2}}{x} + C \quad ◀$$

► Example 2 Evaluate $\displaystyle\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}}$.

Solution. There are two possible approaches: we can make the substitution in the indefinite integral (as in Example 1) and then evaluate the definite integral using the x-limits of integration, or we can make the substitution in the definite integral and convert the x-limits to the corresponding θ-limits.

Method 1.

Using the result from Example 1 with the x-limits of integration yields

$$\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}} = -\frac{1}{4}\left[\frac{\sqrt{4-x^2}}{x}\right]_1^{\sqrt{2}} = -\frac{1}{4}\left[1-\sqrt{3}\right] = \frac{\sqrt{3}-1}{4}$$

Method 2.

The substitution $x = 2\sin\theta$ can be expressed as $x/2 = \sin\theta$ or $\theta = \sin^{-1}(x/2)$, so the θ-limits that correspond to $x = 1$ and $x = \sqrt{2}$ are

$$x = 1: \quad \theta = \sin^{-1}(1/2) = \pi/6$$
$$x = \sqrt{2}: \quad \theta = \sin^{-1}(\sqrt{2}/2) = \pi/4$$

Thus, from (2) in Example 1 we obtain

$$\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}} = \frac{1}{4}\int_{\pi/6}^{\pi/4} \csc^2\theta\, d\theta$$

Convert x-limits to θ-limits.

$$= -\frac{1}{4}\left[\cot\theta\right]_{\pi/6}^{\pi/4} = -\frac{1}{4}\left[1-\sqrt{3}\right] = \frac{\sqrt{3}-1}{4} \quad \blacktriangleleft$$

► Example 5 Evaluate $\displaystyle\int \frac{\sqrt{x^2-25}}{x}\,dx$, assuming that $x \geq 5$.

Solution. The integrand involves a radical of the form $\sqrt{x^2 - a^2}$ with $a = 5$, so from Table 7.4.1 we make the substitution

$$x = 5\sec\theta, \quad 0 \le \theta < \pi/2$$

$$\frac{dx}{d\theta} = 5\sec\theta\tan\theta \quad \text{or} \quad dx = 5\sec\theta\tan\theta\, d\theta$$

Thus,

$$\int \frac{\sqrt{x^2 - 25}}{x}\, dx = \int \frac{\sqrt{25\sec^2\theta - 25}}{5\sec\theta}(5\sec\theta\tan\theta)\, d\theta$$

$$= \int \frac{5|\tan\theta|}{5\sec\theta}(5\sec\theta\tan\theta)\, d\theta$$

$$= 5\int \tan^2\theta\, d\theta \qquad \tan\theta \ge 0 \text{ since } 0 \le \theta < \pi/2$$

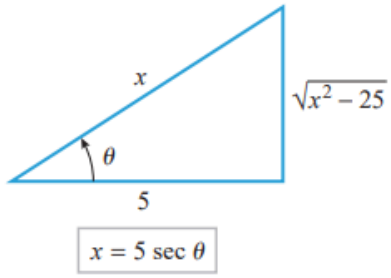
$$= 5\int (\sec^2\theta - 1)\, d\theta = 5\tan\theta - 5\theta + C$$

To express the solution in terms of x, we will represent the substitution $x = 5\sec\theta$ geometrically by the triangle in Figure 7.4.5, from which we obtain

$$\tan\theta = \frac{\sqrt{x^2 - 25}}{5}$$

From this and the fact that the substitution can be expressed as $\theta = \sec^{-1}(x/5)$, we obtain

$$\int \frac{\sqrt{x^2 - 25}}{x}\, dx = \sqrt{x^2 - 25} - 5\sec^{-1}\left(\frac{x}{5}\right) + C \quad \blacktriangleleft$$



▲ **Figure 7.4.5**

INTEGRALS INVOLVING $ax^2 + bx + c$

Integrals that involve a quadratic expression $ax^2 + bx + c$, where $a \neq 0$ and $b \neq 0$, can often be evaluated by first completing the square, then making an appropriate substitution. The following example illustrates this idea.

▶ Example 6 Evaluate $\displaystyle\int \frac{x}{x^2 - 4x + 8}\,dx$.

Solution. Completing the square yields

$$x^2 - 4x + 8 = (x^2 - 4x + 4) + 8 - 4 = (x - 2)^2 + 4$$

Thus, the substitution

$$u = x - 2, \quad du = dx$$

yields

$$\begin{aligned}
\int \frac{x}{x^2 - 4x + 8}\,dx &= \int \frac{x}{(x-2)^2 + 4}\,dx = \int \frac{u+2}{u^2+4}\,du \\
&= \int \frac{u}{u^2+4}\,du + 2\int \frac{du}{u^2+4} \\
&= \frac{1}{2}\int \frac{2u}{u^2+4}\,du + 2\int \frac{du}{u^2+4} \\
&= \frac{1}{2}\ln(u^2+4) + 2\left(\frac{1}{2}\right)\tan^{-1}\frac{u}{2} + C \\
&= \frac{1}{2}\ln[(x-2)^2+4] + \tan^{-1}\left(\frac{x-2}{2}\right) + C \quad \blacktriangleleft
\end{aligned}$$

WATCH:

All the TRIGONOMETRY you need for calculus actually explained

Intro to Trigonometric Substitution --- Ex: Deriving Area of Circle Formula

Trig Subs | How To Choose The Substitution & Deal With Indefinite Integrals

Deconstructing a messy integral | Trig subs & u-subs combined

Integrating by Weierstrass Substitution (visual proof)

Use EULER'S SUBSTITUTION not TRIG SUBSTITUTION!

Use the following online solver:

Integral Calculator

Integral Calculator - eMathHelp

Integral Calculator

Definite and Improper Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 7.4: Trigonometric Substitution: 3, 5, 7, 19, 23, 27

3. $\int \frac{x^2}{\sqrt{16-x^2}}\,dx$ **5.** $\int \frac{dx}{(4+x^2)^2}$ **7.** $\int \frac{\sqrt{x^2-9}}{x}\,dx$

19. $\int e^x\sqrt{1-e^{2x}}\,dx$ **23.** $\int_{\sqrt{2}}^{2} \frac{dx}{x^2\sqrt{x^2-1}}$

27–30 True–False Determine whether the statement is true or false. Explain your answer. ■

27. An integrand involving a radical of the form $\sqrt{a^2-x^2}$ suggests the substitution $x = a\sin\theta$.

28. The trigonometric substitution $x = a\sin\theta$ is made with the restriction $0 \le \theta \le \pi$.

29. An integrand involving a radical of the form $\sqrt{x^2-a^2}$ suggests the substitution $x = a\cos\theta$.

Integration of Rational Function by partial fractions

READ:

Partial Fractions

Polynomials - Long Division

READ:

PARTIAL FRACTIONS

In algebra, one learns to combine two or more fractions into a single fraction by finding a common denominator. For example,

$$\frac{2}{x-4}+\frac{3}{x+1}=\frac{2(x+1)+3(x-4)}{(x-4)(x+1)}=\frac{5x-10}{x^2-3x-4} \qquad (1)$$

However, for purposes of integration, the left side of (1) is preferable to the right side since each of the terms is easy to integrate:

$$\int \frac{5x-10}{x^2-3x-4}\,dx = \int \frac{2}{x-4}\,dx + \int \frac{3}{x+1}\,dx = 2\ln|x-4| + 3\ln|x+1| + C$$

Thus, it is desirable to have some method that will enable us to obtain the left side of (1), starting with the right side. To illustrate how this can be done, we begin by noting that on the left side the numerators are constants and the denominators are the factors of the denominator on the right side. Thus, to find the left side of (1), starting from the right side, we could factor the denominator of the right side and look for constants A and B such that

$$\frac{5x-10}{(x-4)(x+1)} = \frac{A}{x-4} + \frac{B}{x+1} \tag{2}$$

One way to find the constants A and B is to multiply (2) through by $(x-4)(x+1)$ to clear fractions. This yields

$$5x - 10 = A(x+1) + B(x-4) \tag{3}$$

This relationship holds for all x, so it holds in particular if $x = 4$ or $x = -1$. Substituting $x = 4$ in (3) makes the second term on the right drop out and yields the equation $10 = 5A$ or $A = 2$; and substituting $x = -1$ in (3) makes the first term on the right drop out and yields the equation $-15 = -5B$ or $B = 3$. Substituting these values in (2) we obtain

$$\frac{5x-10}{(x-4)(x+1)} = \frac{2}{x-4} + \frac{3}{x+1} \tag{4}$$

which agrees with (1).

A second method for finding the constants A and B is to multiply out the right side of (3) and collect like powers of x to obtain

$$5x - 10 = (A + B)x + (A - 4B)$$

Since the polynomials on the two sides are identical, their corresponding coefficients must be the same. Equating the corresponding coefficients on the two sides yields the following system of equations in the unknowns A and B:

$$\begin{aligned} A + B &= 5 \\ A - 4B &= -10 \end{aligned}$$

Solving this system yields $A = 2$ and $B = 3$ as before (verify).

The terms on the right side of (4) are called ***partial fractions*** of the expression on the left side because they each constitute *part* of that expression. To find those partial fractions we first had to make a guess about their form, and then we had to find the unknown constants. Our next objective is to extend this idea to general rational functions. For this purpose, suppose that $P(x)/Q(x)$ is a ***proper rational function***, by which we mean that the degree of the numerator is less than the degree of the denominator. There is a theorem in advanced algebra which states that every proper rational function can be expressed as a sum

$$\frac{P(x)}{Q(x)} = F_1(x) + F_2(x) + \cdots + F_n(x)$$

where $F_1(x), F_2(x), \ldots, F_n(x)$ are rational functions of the form

$$\frac{A}{(ax+b)^k} \quad \text{or} \quad \frac{Ax+B}{(ax^2+bx+c)^k}$$

in which the denominators are factors of $Q(x)$. The sum is called the ***partial fraction decomposition*** of $P(x)/Q(x)$, and the terms are called ***partial fractions***. As in our opening example, there are two parts to finding a partial fraction decomposition: determining the exact form of the decomposition and finding the unknown constants.

FINDING THE FORM OF A PARTIAL FRACTION DECOMPOSITION

The first step in finding the form of the partial fraction decomposition of a proper rational function $P(x)/Q(x)$ is to factor $Q(x)$ completely into linear and irreducible quadratic factors, and then collect all repeated factors so that $Q(x)$ is expressed as a product of *distinct* factors of the form

$$(ax+b)^m \quad \text{and} \quad (ax^2+bx+c)^m$$

From these factors we can determine the form of the partial fraction decomposition using two rules that we will now discuss.

LINEAR FACTORS

If all of the factors of $Q(x)$ are linear, then the partial fraction decomposition of $P(x)/Q(x)$ can be determined by using the following rule:

LINEAR FACTOR RULE For each factor of the form $(ax+b)^m$, the partial fraction decomposition contains the following sum of m partial fractions:

$$\frac{A_1}{ax+b}+\frac{A_2}{(ax+b)^2}+\cdots+\frac{A_m}{(ax+b)^m}$$

where $A_1, A_2, \ldots, A_m$ are constants to be determined. In the case where $m=1$, only the first term in the sum appears.

▶ Example 1 Evaluate $\displaystyle\int \frac{dx}{x^2+x-2}$.

Solution. The integrand is a proper rational function that can be written as

$$\frac{1}{x^2+x-2} = \frac{1}{(x-1)(x+2)}$$

The factors $x-1$ and $x+2$ are both linear and appear to the first power, so each contributes one term to the partial fraction decomposition by the linear factor rule. Thus, the decomposition has the form

$$\frac{1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2} \tag{5}$$

where A and B are constants to be determined. Multiplying this expression through by $(x-1)(x+2)$ yields

$$1 = A(x+2) + B(x-1) \tag{6}$$

As discussed earlier, there are two methods for finding A and B: we can substitute values of x that are chosen to make terms on the right drop out, or we can multiply out on the right and equate corresponding coefficients on the two sides to obtain a system of equations that can be solved for A and B. We will use the first approach.

Setting $x = 1$ makes the second term in (6) drop out and yields $1 = 3A$ or $A = \frac{1}{3}$; and setting $x = -2$ makes the first term in (6) drop out and yields $1 = -3B$ or $B = -\frac{1}{3}$. Substituting these values in (5) yields the partial fraction decomposition

$$\frac{1}{(x-1)(x+2)} = \frac{\frac{1}{3}}{x-1} + \frac{-\frac{1}{3}}{x+2}$$

The integration can now be completed as follows:

$$\int \frac{dx}{(x-1)(x+2)} = \frac{1}{3}\int \frac{dx}{x-1} - \frac{1}{3}\int \frac{dx}{x+2}$$

$$= \frac{1}{3}\ln|x-1| - \frac{1}{3}\ln|x+2| + C = \frac{1}{3}\ln\left|\frac{x-1}{x+2}\right| + C \quad \blacktriangleleft$$

If the factors of $Q(x)$ are linear and none are repeated, as in the last example, then the recommended method for finding the constants in the partial fraction decomposition is to substitute appropriate values of x to make terms drop out. However, if some of the linear factors are repeated, then it will not be possible to find all of the constants in this way. In this case the recommended procedure is to find as many constants as possible by substitution and then find the rest by equating coefficients. This is illustrated in the next example.

▶ **Example 2** Evaluate $\int \frac{2x+4}{x^3-2x^2}\,dx$.

Solution. The integrand can be rewritten as

$$\frac{2x+4}{x^3-2x^2} = \frac{2x+4}{x^2(x-2)}$$

Although x^2 is a quadratic factor, it is *not* irreducible since $x^2 = xx$. Thus, by the linear factor rule, x^2 introduces two terms (since $m = 2$) of the form

$$\frac{A}{x} + \frac{B}{x^2}$$

and the factor $x - 2$ introduces one term (since $m = 1$) of the form

$$\frac{C}{x-2}$$

so the partial fraction decomposition is

$$\frac{2x+4}{x^2(x-2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2} \tag{7}$$

Multiplying by $x^2(x-2)$ yields

$$2x+4 = Ax(x-2) + B(x-2) + Cx^2 \tag{8}$$

which, after multiplying out and collecting like powers of x, becomes

$$2x+4 = (A+C)x^2 + (-2A+B)x - 2B \tag{9}$$

Setting $x = 0$ in (8) makes the first and third terms drop out and yields $B = -2$, and setting $x = 2$ in (8) makes the first and second terms drop out and yields $C = 2$ (verify). However, there is no substitution in (8) that produces A directly, so we look to Equation (9) to find this value. This can be done by equating the coefficients of x^2 on the two sides to obtain

$$A + C = 0 \quad \text{or} \quad A = -C = -2$$

Substituting the values $A = -2$, $B = -2$, and $C = 2$ in (7) yields the partial fraction decomposition

$$\frac{2x+4}{x^2(x-2)} = \frac{-2}{x} + \frac{-2}{x^2} + \frac{2}{x-2}$$

Thus,

$$\int \frac{2x+4}{x^2(x-2)}\,dx = -2\int \frac{dx}{x} - 2\int \frac{dx}{x^2} + 2\int \frac{dx}{x-2}$$

$$= -2\ln|x| + \frac{2}{x} + 2\ln|x-2| + C = 2\ln\left|\frac{x-2}{x}\right| + \frac{2}{x} + C \quad \blacktriangleleft$$

QUADRATIC FACTORS

If some of the factors of $Q(x)$ are irreducible quadratics, then the contribution of those factors to the partial fraction decomposition of $P(x)/Q(x)$ can be determined from the following rule:

QUADRATIC FACTOR RULE For each factor of the form $(ax^2 + bx + c)^m$, the partial fraction decomposition contains the following sum of m partial fractions:

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_mx + B_m}{(ax^2 + bx + c)^m}$$

where $A_1, A_2, \ldots, A_m, B_1, B_2, \ldots, B_m$ are constants to be determined. In the case where $m = 1$, only the first term in the sum appears.

▶ **Example 3** Evaluate $\displaystyle\int \frac{x^2+x-2}{3x^3-x^2+3x-1}\,dx$.

Solution. The denominator in the integrand can be factored by grouping:

$$3x^3-x^2+3x-1 = x^2(3x-1)+(3x-1) = (3x-1)(x^2+1)$$

By the linear factor rule, the factor $3x-1$ introduces one term, namely,

$$\frac{A}{3x-1}$$

and by the quadratic factor rule, the factor x^2+1 introduces one term, namely,

$$\frac{Bx+C}{x^2+1}$$

Thus, the partial fraction decomposition is

$$\frac{x^2+x-2}{(3x-1)(x^2+1)} = \frac{A}{3x-1}+\frac{Bx+C}{x^2+1} \tag{10}$$

Multiplying by $(3x-1)(x^2+1)$ yields

$$x^2+x-2 = A(x^2+1)+(Bx+C)(3x-1) \tag{11}$$

We could find A by substituting $x = \frac{1}{3}$ to make the last term drop out, and then find the rest of the constants by equating corresponding coefficients. However, in this case it is just as easy to find *all* of the constants by equating coefficients and solving the resulting system. For this purpose we multiply out the right side of (11) and collect like terms:

$$x^2 + x - 2 = (A + 3B)x^2 + (-B + 3C)x + (A - C)$$

Equating corresponding coefficients gives

$$\begin{aligned} A + 3B \qquad &= 1 \\ -B + 3C &= 1 \\ A \qquad - C &= -2 \end{aligned}$$

To solve this system, subtract the third equation from the first to eliminate A. Then use the resulting equation together with the second equation to solve for B and C. Finally, determine A from the first or third equation. This yields (verify)

$$A = -\frac{7}{5}, \quad B = \frac{4}{5}, \quad C = \frac{3}{5}$$

Thus, (10) becomes

$$\frac{x^2+x-2}{(3x-1)(x^2+1)} = \frac{-\frac{7}{5}}{3x-1} + \frac{\frac{4}{5}x+\frac{3}{5}}{x^2+1}$$

and

$$\begin{aligned}\int \frac{x^2+x-2}{(3x-1)(x^2+1)}\,dx &= -\frac{7}{5}\int \frac{dx}{3x-1} + \frac{4}{5}\int \frac{x}{x^2+1}\,dx + \frac{3}{5}\int \frac{dx}{x^2+1} \\ &= -\frac{7}{15}\ln|3x-1| + \frac{2}{5}\ln(x^2+1) + \frac{3}{5}\tan^{-1}x + C \quad \blacktriangleleft\end{aligned}$$

▶ Example 4 Evaluate $\displaystyle\int \frac{3x^4+4x^3+16x^2+20x+9}{(x+2)(x^2+3)^2}\,dx.$

Solution. Observe that the integrand is a proper rational function since the numerator has degree 4 and the denominator has degree 5. Thus, the method of partial fractions is applicable. By the linear factor rule, the factor $x+2$ introduces the single term

$$\frac{A}{x+2}$$

and by the quadratic factor rule, the factor $(x^2+3)^2$ introduces two terms (since $m=2$):

$$\frac{Bx+C}{x^2+3}+\frac{Dx+E}{(x^2+3)^2}$$

Thus, the partial fraction decomposition of the integrand is

$$\frac{3x^4+4x^3+16x^2+20x+9}{(x+2)(x^2+3)^2}=\frac{A}{x+2}+\frac{Bx+C}{x^2+3}+\frac{Dx+E}{(x^2+3)^2} \tag{12}$$

Multiplying by $(x+2)(x^2+3)^2$ yields

$$\begin{aligned}3x^4+4x^3+16x^2+20x+9&\\ =A(x^2+3)^2+(Bx+C)(x^2+3)(x+2)+(Dx+E)(x+2)&\end{aligned} \tag{13}$$

which, after multiplying out and collecting like powers of x, becomes

$$\begin{aligned}3x^4+4x^3+16x^2+20x+9&\\ =(A+B)x^4+(2B+C)x^3+(6A+3B+2C+D)x^2&\\ +(6B+3C+2D+E)x+(9A+6C+2E)&\end{aligned} \tag{14}$$

Equating corresponding coefficients in (14) yields the following system of five linear equations in five unknowns:

$$\begin{aligned} A + B &= 3 \\ 2B + C &= 4 \\ 6A + 3B + 2C + D &= 16 \\ 6B + 3C + 2D + E &= 20 \\ 9A + 6C + 2E &= 9 \end{aligned} \tag{15}$$

Efficient methods for solving systems of linear equations such as this are studied in a branch of mathematics called ***linear algebra***; those methods are outside the scope of this text. However, as a practical matter most linear systems of any size are solved by computer, and most computer algebra systems have commands that in many cases can solve linear systems exactly. In this particular case we can simplify the work by first substituting $x = -2$

in (13), which yields $A = 1$. Substituting this known value of A in (15) yields the simpler system

$$\begin{aligned} B &= 2 \\ 2B + C &= 4 \\ 3B + 2C + D &= 10 \\ 6B + 3C + 2D + E &= 20 \\ 6C + 2E &= 0 \end{aligned} \tag{16}$$

This system can be solved by starting at the top and working down, first substituting $B = 2$ in the second equation to get $C = 0$, then substituting the known values of B and C in the third equation to get $D = 4$, and so forth. This yields

$$A = 1, \quad B = 2, \quad C = 0, \quad D = 4, \quad E = 0$$

Thus, (12) becomes

$$\frac{3x^4 + 4x^3 + 16x^2 + 20x + 9}{(x+2)(x^2+3)^2} = \frac{1}{x+2} + \frac{2x}{x^2+3} + \frac{4x}{(x^2+3)^2}$$

and so

$$\begin{aligned} \int \frac{3x^4 + 4x^3 + 16x^2 + 20x + 9}{(x+2)(x^2+3)^2}\,dx &= \int \frac{dx}{x+2} + \int \frac{2x}{x^2+3}\,dx + 4\int \frac{x}{(x^2+3)^2}\,dx \\ &= \ln|x+2| + \ln(x^2+3) - \frac{2}{x^2+3} + C \end{aligned}$$

INTEGRATING IMPROPER RATIONAL FUNCTIONS

Although the method of partial fractions only applies to proper rational functions, an improper rational function can be integrated by performing a long division and expressing the function as the quotient plus the remainder over the divisor. The remainder over the divisor will be a proper rational function, which can then be decomposed into partial fractions. This idea is illustrated in the following example.

▶ Example 5 Evaluate $\displaystyle\int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2}\,dx.$

Solution. The integrand is an improper rational function since the numerator has degree 4 and the denominator has degree 2. Thus, we first perform the long division

$$
\begin{array}{r}
3x^2 \quad +1 \\
x^2+x-2 \;\overline{)\; 3x^4+3x^3-5x^2+x-1} \\
\underline{3x^4+3x^3-6x^2} \\
x^2+x-1 \\
\underline{x^2+x-2} \\
1
\end{array}
$$

It follows that the integrand can be expressed as

$$\frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2} = (3x^2+1) + \frac{1}{x^2+x-2}$$

and hence

$$\int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2}\,dx = \int (3x^2+1)\,dx + \int \frac{dx}{x^2+x-2}$$

The second integral on the right now involves a proper rational function and can thus be evaluated by a partial fraction decomposition. Using the result of Example 1 we obtain

$$\int \frac{3x^4 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2}\,dx = x^3 + x + \frac{1}{3}\ln\left|\frac{x-1}{x+2}\right| + C \quad \blacktriangleleft$$

CONCLUDING REMARKS

There are some cases in which the method of partial fractions is inappropriate. For example, it would be inefficient to use partial fractions to perform the integration

$$\int \frac{3x^2 + 2}{x^3 + 2x - 8}\,dx = \ln|x^3 + 2x - 8| + C$$

since the substitution $u = x^3 + 2x - 8$ is more direct. Similarly, the integration

$$\int \frac{2x - 1}{x^2 + 1}\,dx = \int \frac{2x}{x^2 + 1}\,dx - \int \frac{dx}{x^2 + 1} = \ln(x^2 + 1) - \tan^{-1} x + C$$

requires only a little algebra since the integrand is already in partial fraction form.

WATCH:

9.10 Integration of rational functions - Example 1

9.11 Integration of rational functions - Example 2

9.12 Integration of rational functions - Example 3

Integration by Partial Fractions | Big Idea + First Example

Partial Fractions | Repeating and irreducible Quadratic Terms

Use the following online solver:

Integral Calculator

Integral Calculator - eMathHelp

Integral Calculator

Definite and Improper Integral Calculator - eMathHelp

Polynomial Long Division Calculator - eMathHelp

Partial Fraction Decomposition Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 7.5: Integrating rational functions by partial fractions: 1-34

1–8 Write out the form of the partial fraction decomposition. (Do not find the numerical values of the coefficients.) ■

1. $\dfrac{3x-1}{(x-3)(x+4)}$

2. $\dfrac{5}{x(x^2-4)}$

3. $\dfrac{2x-3}{x^3-x^2}$

4. $\dfrac{x^2}{(x+2)^3}$

5. $\dfrac{1-x^2}{x^3(x^2+2)}$

6. $\dfrac{3x}{(x-1)(x^2+6)}$

7. $\dfrac{4x^3-x}{(x^2+5)^2}$

8. $\dfrac{1-3x^4}{(x-2)(x^2+1)^2}$

9–34 Evaluate the integral. ■

9. $\displaystyle\int \frac{dx}{x^2-3x-4}$

10. $\displaystyle\int \frac{dx}{x^2-6x-7}$

11. $\int \frac{11x+17}{2x^2+7x-4}\,dx$

12. $\int \frac{5x-5}{3x^2-8x-3}\,dx$

13. $\int \frac{2x^2-9x-9}{x^3-9x}\,dx$

14. $\int \frac{dx}{x(x^2-1)}$

15. $\int \frac{x^2-8}{x+3}\,dx$

16. $\int \frac{x^2+1}{x-1}\,dx$

17. $\int \frac{3x^2-10}{x^2-4x+4}\,dx$

18. $\int \frac{x^2}{x^2-3x+2}\,dx$

19. $\int \frac{2x-3}{x^2-3x-10}\,dx$

20. $\int \frac{3x+1}{3x^2+2x-1}\,dx$

21. $\int \frac{x^5+x^2+2}{x^3-x}\,dx$

22. $\int \frac{x^5-4x^3+1}{x^3-4x}\,dx$

23. $\int \frac{2x^2+3}{x(x-1)^2}\,dx$

24. $\int \frac{3x^2-x+1}{x^3-x^2}\,dx$

25. $\int \frac{2x^2 - 10x + 4}{(x+1)(x-3)^2}\,dx$

26. $\int \frac{2x^2 - 2x - 1}{x^3 - x^2}\,dx$

27. $\int \frac{x^2}{(x+1)^3}\,dx$

28. $\int \frac{2x^2 + 3x + 3}{(x+1)^3}\,dx$

29. $\int \frac{2x^2 - 1}{(4x-1)(x^2+1)}\,dx$

30. $\int \frac{dx}{x^3 + 2x}$

31. $\int \frac{x^3 + 3x^2 + x + 9}{(x^2+1)(x^2+3)}\,dx$

32. $\int \frac{x^3 + x^2 + x + 2}{(x^2+1)(x^2+2)}\,dx$

33. $\int \frac{x^3 - 2x^2 + 2x - 2}{x^2 + 1}\,dx$

34. $\int \frac{x^4 + 6x^3 + 10x^2 + x}{x^2 + 6x + 10}\,dx$

- **Beta and Gamma Function**

READ:

Gamma Function

How to Integrate Using the Gamma Function - wikiHow

Beta Function | Brilliant Math & Science Wiki

How to Integrate Using the Beta Function - wikiHow

WATCH:

Gamma Function | Integral Functions - YouTube

Beta Function | Integral Functions - YouTube

Use the following online solver:

PRACTISE:

1. **Evaluate in terms of gamma function: (WEEK 2)**

$$(i)\int_0^4 x^{3/2}(4-x)^{5/2}dx,\quad (ii)\int_0^b y^5\sqrt{b^2-y^2}\,dy,\quad (iii)\int_0^\infty e^{-t^2}dt,$$

$$(iv)\int_0^\infty x^5e^{-4x}dx,\quad (v)\int_0^\infty e^{-y^2}y^5dy,\quad (vi)\int_0^\infty e^{-x^2}dx,\quad (vii)\int_0^\infty x^6e^{-3x}dx,$$

$$(viii)\int_0^\infty e^{-x^2}x^9dx,\quad (ix)\int_0^\infty \sqrt{x}e^{-x^2}dx,\quad (x)\int_0^1 \frac{x^3}{\sqrt{1-x^3}}dx,$$

$$(xi)\int_0^1 \frac{1}{\sqrt{x\ln(1/x)}}dx,\quad (xii)\int_0^1\left(1-\frac{1}{x}\right)^{1/3}dx.$$

2. **Evaluate in terms of beta function: (WEEK 3)**

$$(i)\int_0^1 \frac{x^2}{\sqrt{1-x}}dx,\quad (ii)\int_0^1 x^7(1-x)^3dx,\quad (iii)\int_0^1 \frac{1}{\sqrt{1-x^3}}dx,$$

$$(iv)\int_0^1 (1-x)^{1/2}x^3dx,\quad (v)\int_0^1 x^{5/2}(1-x)^{3/2}dx,$$

$$(vi)\int_0^a y^7\sqrt{a^4-y^4}\,dy,\quad (vii)\int_0^4 y^3\sqrt{64-y^3}\,dy,$$

$$(viii)\int_0^1 x^2(1-x^3)^{3/2}dx,\quad (ix)\int_0^\infty \frac{1}{1+x^4}dx.$$

3. **Evaluate the following integrals: (WEEK 3)**

$$(i)\int_0^\pi \sin^5\theta\cos^4\theta d\theta,\quad (ii)\int_0^\pi \sin^6\theta\cos^7\theta d\theta,$$

$$(iii)\int_0^{\pi/6} \sin^2 6x\cos^4 3xdx,\quad (iv)\int_0^{\pi/4} \sin^2 4\theta\cos^3 2\theta d\theta,$$

$$(v)\int_0^{\pi/2} \sin^4\theta\cos^2\theta d\theta,\quad (vi)\int_0^{\pi/8} \sin^2 8x\cos^4 4xdx.$$

Formula

1. $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx, \quad where \quad n > 0$
2. $\beta(m,n) = \int_0^{1} x^{m-1}(1-x)^{n-1} dx, \quad where \quad m > 0, n > 0.$
3. $\int_0^{\pi/2} \sin^p x \cos^q x dx = \frac{\Gamma(\frac{p+1}{2})\,\Gamma(\frac{q+1}{2})}{2\,\Gamma(\frac{p+q+2}{2})}$
4. $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$
5. $\Gamma(n) = (n-1)!$
6. $\Gamma(n+1) = n\Gamma(n) = n!$
7. $\Gamma(\frac{1}{2}) = \sqrt{\pi}$
8. $\Gamma(1) = 1$
9. $\Gamma(\frac{p}{2}) = (\frac{p}{2}-1)(\frac{p}{2}-2)(\frac{p}{2}-3).....\frac{1}{2}.\Gamma(\frac{1}{2})$
10. $\Gamma(n) = (n-1)(n-2).....3.2.1$
11. $\int_0^{\pi/2} 2sin^{2x-1}(t)cos^{2y-1}(t)dt = \beta(x,y)$

Gamma Beta Distribution Solution

1. (i) $8.4.32\frac{\Gamma\left(\frac{5}{2}\right)\Gamma\left(\frac{7}{2}\right)}{\Gamma(6)}$

   (ii) $\frac{b^7}{2}\frac{\Gamma(3)\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{9}{2}\right)}$

   (iii) ½ (1/2)

   (iv) $\frac{1}{4^6}(6)$

   (v) ½ (3)

   (vii) $\frac{1}{3^7}(7)$

   (viii) ½ (5)

   (ix) ½ (3/4)

   (x) $\frac{1}{3}\frac{\Gamma\left(\frac{4}{3}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{4}{3}+\frac{1}{2}\right)}$

   (xi) $\sqrt{2}\,(1/2)$

   (xii) $-\frac{\Gamma\left(\frac{2}{3}\right)\Gamma\left(\frac{4}{3}\right)}{\Gamma(2)}$

2. (i) β(3, ½)

   (ii) β(8, 4)

   (iii) $\frac{1}{3}$β(1/3, ½)

   (iv) β(4, 3/2)

   (v) β(7/2, 5/2)

   (vi) $\frac{a^{10}}{4}$β(2, 3/2)

   (vii) $4^4.8.\frac{1}{3}$β(4/3, 3/2)

   (viii) $\frac{1}{3}$β(1, 5/2)

   (ix) ½ $\frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)}{2\Gamma(1)}=\frac{1}{4}\,\beta\left(\frac{1}{4},\frac{3}{4}\right)$

3. (i) $\frac{\Gamma(3)\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{11}{2}\right)}$

   (ii) $\frac{\Gamma\left(\frac{7}{2}\right)\Gamma(4)}{\Gamma\left(\frac{15}{2}\right)}$

   (iii) $\frac{4}{3}\frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{7}{2}\right)}{2\Gamma(5)}$

   (iv) $\frac{\Gamma\left(\frac{3}{2}\right)\Gamma(3)}{\Gamma\left(\frac{9}{2}\right)}$

   (v) $\frac{\Gamma\left(\frac{5}{2}\right)\Gamma\left(\frac{3}{2}\right)}{2\Gamma(4)}$

   (vi) $\frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{7}{2}\right)}{2\Gamma(5)}$

- **Length of a curve, Surface area of revolution**

Length of a curve

READ: Arc Length

READ:

6.4.2 DEFINITION If $y = f(x)$ is a smooth curve on the interval $[a, b]$, then the arc length L of this curve over $[a, b]$ is defined as

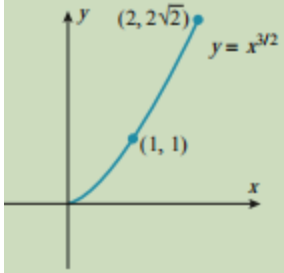
$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx \tag{3}$$

This result provides both a definition and a formula for computing arc lengths. Where convenient, (3) can also be expressed as

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \tag{4}$$

Moreover, for a curve expressed in the form $x = g(y)$, where g' is continuous on $[c, d]$, the arc length L from $y = c$ to $y = d$ can be expressed as

$$L = \int_c^d \sqrt{1 + [g'(y)]^2}\, dy = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \tag{5}$$



gure 6.4.4

► **Example 1** Find the arc length of the curve $y = x^{3/2}$ from (1, 1) to $(2, 2\sqrt{2})$ (Figure 6.4.4) in two ways: (a) using Formula (4) and (b) using Formula (5).

Solution (a).

$$\frac{dy}{dx} = \frac{3}{2}x^{1/2}$$

and since the curve extends from $x = 1$ to $x = 2$, it follows from (4) that

$$L = \int_1^2 \sqrt{1 + \left(\tfrac{3}{2}x^{1/2}\right)^2}\,dx = \int_1^2 \sqrt{1 + \tfrac{9}{4}x}\,dx$$

To evaluate this integral we make the u-substitution

$$u = 1 + \tfrac{9}{4}x, \quad du = \tfrac{9}{4}dx$$

and then change the x-limits of integration ($x = 1, x = 2$) to the corresponding u-limits $\left(u = \frac{13}{4}, u = \frac{22}{4}\right)$:

$$L = \frac{4}{9}\int_{13/4}^{22/4} u^{1/2}\,du = \frac{8}{27}u^{3/2}\Big]_{13/4}^{22/4} = \frac{8}{27}\left[\left(\frac{22}{4}\right)^{3/2} - \left(\frac{13}{4}\right)^{3/2}\right]$$

$$= \frac{22\sqrt{22} - 13\sqrt{13}}{27} \approx 2.09$$

Solution (b). To apply Formula (5) we must first rewrite the equation $y = x^{3/2}$ so that x is expressed as a function of y. This yields $x = y^{2/3}$ and

$$\frac{dx}{dy} = \frac{2}{3}y^{-1/3}$$

Since the curve extends from $y = 1$ to $y = 2\sqrt{2}$, it follows from (5) that

$$L = \int_1^{2\sqrt{2}} \sqrt{1 + \tfrac{4}{9}y^{-2/3}}\,dy = \frac{1}{3}\int_1^{2\sqrt{2}} y^{-1/3}\sqrt{9y^{2/3} + 4}\,dy$$

The arc from the point $(1, 1)$ to the point $(2, 2\sqrt{2})$ in Figure 6.4.4 is nearly a straight line, so the arc length should be only slightly larger than the straight-line distance between these points. Show that this is so.

To evaluate this integral we make the u-substitution

$$u = 9y^{2/3} + 4, \quad du = 6y^{-1/3}\,dy$$

and change the y-limits of integration ($y = 1$, $y = 2\sqrt{2}$) to the corresponding u-limits ($u = 13$, $u = 22$). This gives

$$L = \frac{1}{18}\int_{13}^{22} u^{1/2}\,du = \frac{1}{27}u^{3/2}\Big]_{13}^{22} = \frac{1}{27}[(22)^{3/2} - (13)^{3/2}] = \frac{22\sqrt{22} - 13\sqrt{13}}{27}$$

The answer in part (b) agrees with that in part (a); however, the integration in part (b) is more tedious. In problems where there is a choice between using (4) or (5), it is often the case that one of the formulas leads to a simpler integral than the other. ◀

WATCH:

Arclength Formula | Derivation & Ex: Circumference of a Circle

Use the following online solver:

Curve Arc Length Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 6.4: Length of a Plane Curve: 3-8, 27-32

3–8 Find the exact arc length of the curve over the interval.

3. $y = 3x^{3/2} - 1$ from $x = 0$ to $x = 1$

4. $x = \frac{1}{3}(y^2 + 2)^{3/2}$ from $y = 0$ to $y = 1$

5. $y = x^{2/3}$ from $x = 1$ to $x = 8$

6. $y = (x^6 + 8)/(16x^2)$ from $x = 2$ to $x = 3$

7. $24xy = y^4 + 48$ from $y = 2$ to $y = 4$

8. $x = \frac{1}{8}y^4 + \frac{1}{4}y^{-2}$ from $y = 1$ to $y = 4$

C **26.** Assume that no segment of the curve

$$x = x(t), \quad y = y(t), \qquad (a \le t \le b)$$

is traced more than once as t increases from a to b. Divide the interval $[a, b]$ into n subintervals by inserting points $t_1, t_2, \ldots, t_{n-1}$ between $a = t_0$ and $b = t_n$. Let L denote the arc length of the curve. Give an informal argument for the approximation

$$L \approx \sum_{k=1}^{n} \sqrt{[x(t_k) - x(t_{k-1})]^2 + [y(t_k) - y(t_{k-1})]^2}$$

If dx/dt and dy/dt are continuous functions for $a \le t \le b$, then it can be shown that as $\max \Delta t_k \to 0$, this sum converges to

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

27–32 Use the arc length formula from Exercise 26 to find the arc length of the curve. ■

27. $x = \frac{1}{3}t^3,\ y = \frac{1}{2}t^2 \quad (0 \le t \le 1)$

28. $x = (1+t)^2,\ y = (1+t)^3 \quad (0 \le t \le 1)$

29. $x = \cos 2t,\ y = \sin 2t \quad (0 \le t \le \pi/2)$

30. $x = \cos t + t \sin t,\ y = \sin t - t \cos t \quad (0 \le t \le \pi)$

31. $x = e^t \cos t,\ y = e^t \sin t \quad (0 \le t \le \pi/2)$

32. $x = e^t(\sin t + \cos t),\ y = e^t(\cos t - \sin t) \quad (1 \le t \le 4)$

Surface Area of revolution

READ:

6.5.2 DEFINITION If f is a smooth, nonnegative function on $[a, b]$, then the surface area S of the surface of revolution that is generated by revolving the portion of the curve $y = f(x)$ between $x = a$ and $x = b$ about the x-axis is defined as

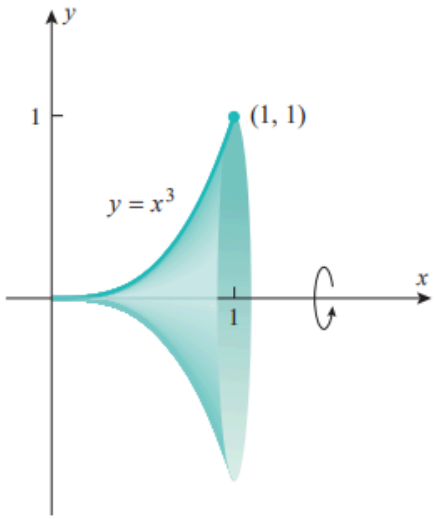
$$S = \int_a^b 2\pi f(x)\sqrt{1 + [f'(x)]^2}\, dx$$

This result provides both a definition and a formula for computing surface areas. Where convenient, this formula can also be expressed as

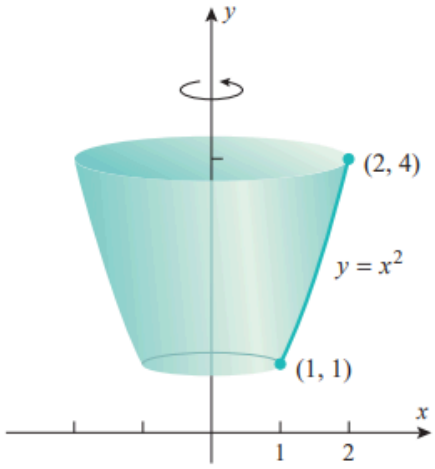
$$S = \int_a^b 2\pi f(x)\sqrt{1 + [f'(x)]^2}\, dx = \int_a^b 2\pi y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \tag{4}$$

Moreover, if g is nonnegative and $x = g(y)$ is a smooth curve on the interval $[c, d]$, then the area of the surface that is generated by revolving the portion of a curve $x = g(y)$ between $y = c$ and $y = d$ about the y-axis can be expressed as

$$S = \int_c^d 2\pi g(y)\sqrt{1 + [g'(y)]^2}\, dy = \int_c^d 2\pi x\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \tag{5}$$



▲ Figure 6.5.6



▲ Figure 6.5.7

► **Example 1** Find the area of the surface that is generated by revolving the portion of the curve $y = x^3$ between $x = 0$ and $x = 1$ about the x-axis.

Solution. First sketch the curve; then imagine revolving it about the x-axis (Figure 6.5.6). Since $y = x^3$, we have $dy/dx = 3x^2$, and hence from (4) the surface area S is

$$S = \int_0^1 2\pi y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$$

$$= \int_0^1 2\pi x^3\sqrt{1 + (3x^2)^2}\,dx$$

$$= 2\pi\int_0^1 x^3(1 + 9x^4)^{1/2}\,dx$$

$$= \frac{2\pi}{36}\int_1^{10} u^{1/2}\,du \qquad \begin{array}{|c|}\hline u = 1 + 9x^4 \\ du = 36x^3\,dx \\ \hline \end{array}$$

$$= \frac{2\pi}{36}\cdot\frac{2}{3}u^{3/2}\Big]_{u=1}^{10} = \frac{\pi}{27}(10^{3/2} - 1) \approx 3.56 \quad ◄$$

► **Example 2** Find the area of the surface that is generated by revolving the portion of the curve $y = x^2$ between $x = 1$ and $x = 2$ about the y-axis.

Solution. First sketch the curve; then imagine revolving it about the y-axis (Figure 6.5.7). Because the curve is revolved about the y-axis we will apply Formula (5). Toward this end, we rewrite $y = x^2$ as $x = \sqrt{y}$ and observe that the y-values corresponding to $x = 1$ and

$x = 2$ are $y = 1$ and $y = 4$. Since $x = \sqrt{y}$, we have $dx/dy = 1/(2\sqrt{y})$, and hence from (5) the surface area S is

$$S = \int_1^4 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

$$= \int_1^4 2\pi\sqrt{y}\sqrt{1 + \left(\frac{1}{2\sqrt{y}}\right)^2}\, dy$$

$$= \pi \int_1^4 \sqrt{4y + 1}\, dy$$

$$= \frac{\pi}{4} \int_5^{17} u^{1/2}\, du \qquad \begin{array}{l} u = 4y + 1 \\ du = 4\, dy \end{array}$$

$$= \frac{\pi}{4} \cdot \frac{2}{3} u^{3/2} \Big]_{u=5}^{17} = \frac{\pi}{6}(17^{3/2} - 5^{3/2}) \approx 30.85 \quad \blacktriangleleft$$

WATCH:

Area of Surfaces of Revolution | Derivation & Example

Use the following online solver:

Area of Surface of Revolution Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 6.5: Area of Surface of Revolution: 1-4, 5-8

1–4 Find the area of the surface generated by revolving the given curve about the x-axis. ■

1. $y = 7x,\ 0 \le x \le 1$
2. $y = \sqrt{x},\ 1 \le x \le 4$
3. $y = \sqrt{4 - x^2},\ -1 \le x \le 1$
4. $x = \sqrt[3]{y},\ 1 \le y \le 8$

5–8 Find the area of the surface generated by revolving the given curve about the y-axis. ■

5. $x = 9y + 1,\ 0 \le y \le 2$
6. $x = y^3,\ 0 \le y \le 1$
7. $x = \sqrt{9 - y^2},\ -2 \le y \le 2$
8. $x = 2\sqrt{1 - y},\ -1 \le y \le 0$

- **Basic concept of Double integral, Double integral in Cartesian and Polar Coordinate system**

Prerequisite

READ:
HOWARD ANTON - 11.8: Cylindrical and Spherical Coordinates
Read only the selected parts from:
https://shorturl.at/sySX8

WATCH:
Pixar: The math behind the movies - Tony DeRose
Coordinate Systems: Animated
READ: Polar and Cartesian Coordinates or **WATCH:** **Intro to Polar Coordinates**
Sketching Polar Curves

CalcBLUE 3 : Ch. 13.5 : Cylindrical Coordinates
CalcBLUE 3 : Ch. 14.1 : Intro to Spherical Coordinates
CalcBLUE 3 : Ch. 14.2 : Visualizing Spherical Coordinates
Spherical Coordinate System (With 3D Animation)
Twos-Day Tuesday: 22 Slices of Pizza with Six Cuts

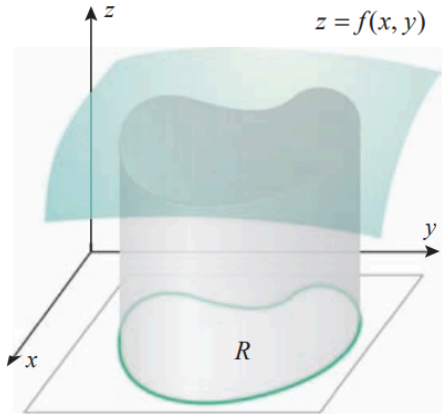
READ: HOWARD ANTON: 14.1: Double Integrals

VOLUME

Recall that the definite integral of a function of one variable

$$\int_a^b f(x)\,dx = \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} f(x_k^*)\Delta x_k = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x_k \qquad (1)$$

arose from the problem of finding areas under curves. [In the rightmost expression in (1), we use the "limit as $n \to +\infty$" to encapsulate the process by which we increase the number of subintervals of $[a, b]$ in such a way that the lengths of the subintervals approach zero.] Integrals of functions of two variables arise from the problem of finding volumes under surfaces.



▲ **Figure 14.1.1**

14.1.1 THE VOLUME PROBLEM Given a function f of two variables that is continuous and nonnegative on a region R in the xy-plane, find the volume of the solid enclosed between the surface $z = f(x, y)$ and the region R (Figure 14.1.1).

Later, we will place more restrictions on the region R, but for now we will just assume that the entire region can be enclosed within some suitably large rectangle with sides parallel to the coordinate axes. This ensures that R does not extend indefinitely in any direction.

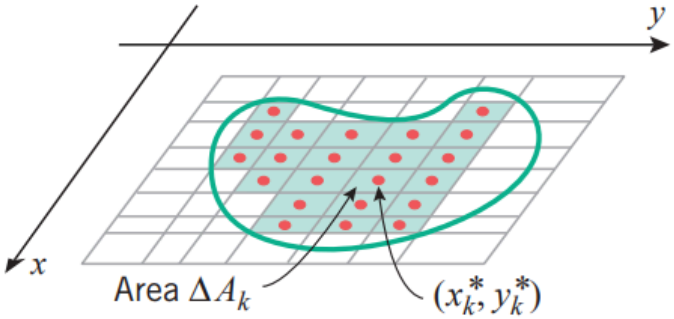
The procedure for finding the volume V of the solid in Figure 14.1.1 will be similar to the limiting process used for finding areas, except that now the approximating elements will be rectangular parallelepipeds rather than rectangles. We proceed as follows:

- Using lines parallel to the coordinate axes, divide the rectangle enclosing the region R into subrectangles, and exclude from consideration all those subrectangles that contain any points outside of R. This leaves only rectangles that are subsets of R (Figure 14.1.2). Assume that there are n such rectangles, and denote the area of the kth such rectangle by ΔA_k.
- Choose any arbitrary point in each subrectangle, and denote the point in the kth subrectangle by (x_k^*, y_k^*). As shown in Figure 14.1.3, the product $f(x_k^*, y_k^*)\Delta A_k$ is the volume of a rectangular parallelepiped with base area ΔA_k and height $f(x_k^*, y_k^*)$, so the sum

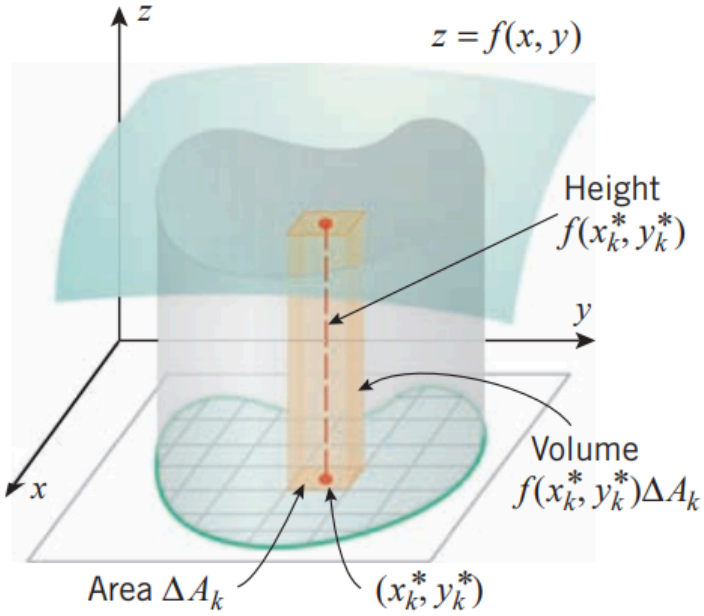
$$\sum_{k=1}^{n} f(x_k^*, y_k^*)\Delta A_k$$

  can be viewed as an approximation to the volume V of the entire solid.
- There are two sources of error in the approximation: first, the parallelepipeds have flat tops, whereas the surface $z = f(x, y)$ may be curved; second, the rectangles that form the bases of the parallelepipeds may not completely cover the region R. However, if we repeat the above process with more and more subdivisions in such a way that both the lengths and the widths of the subrectangles approach zero, then it is plausible that the errors of both types approach zero, and the exact volume of the solid will be

$$V = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*, y_k^*)\Delta A_k$$



▲ **Figure 14.1.2**



▲ **Figure 14.1.3**

This suggests the following definition.

Definition 14.1.2 is satisfactory for our present purposes, but some issues would have to be resolved before it could be regarded as rigorous. For example, we would have to prove that the limit actually exists and that its value does not depend on how the points $(x_1^*, y_1^*), (x_2^*, y_2^*), \ldots, (x_n^*, y_n^*)$ are chosen. These facts are true if the region R is not too "complicated" and if f is continuous on R. The details are beyond the scope of this text.

14.1.2 DEFINITION (*Volume Under a Surface*) If f is a function of two variables that is continuous and nonnegative on a region R in the xy-plane, then the volume of the solid enclosed between the surface $z = f(x, y)$ and the region R is defined by

$$V = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*, y_k^*) \Delta A_k \tag{2}$$

Here, $n \to +\infty$ indicates the process of increasing the number of subrectangles of the rectangle enclosing R in such a way that both the lengths and the widths of the subrectangles approach zero.

DEFINITION OF A DOUBLE INTEGRAL

As in Definition 14.1.2, the notation $n \to +\infty$ in (3) encapsulates a process in which the enclosing rectangle for R is repeatedly subdivided in such a way that both the lengths and the widths of the subrectangles approach zero. Note that subdividing so that the subrectangle lengths approach zero forces the mesh of the partition of the length of the enclosing rectangle for R to approach zero. Similarly, subdividing so that the subrectangle widths approach zero forces the mesh of the partition of the width of the enclosing rectangle for R to approach zero. Thus, we have extended the notion conveyed by Formula (1) where the definite integral of a one-variable function is expressed as a limit of Riemann sums. By extension, the sums in (3) are also called ***Riemann sums***, and the limit of the Riemann sums is denoted by

$$\iint_R f(x, y)\, dA = \lim_{n \to +\infty} \sum_{k=1}^{n} f(x_k^*, y_k^*) \Delta A_k \tag{4}$$

which is called the ***double integral*** of $f(x, y)$ over R.

If f is continuous and nonnegative on the region R, then the volume formula in (2) can be expressed as

$$V = \iint_R f(x, y)\, dA \tag{5}$$

If f has both positive and negative values on R, then a positive value for the double integral of f over R means that there is more volume above R than below, a negative value for the double integral means that there is more volume below R than above, and a value of zero means that the volume above R is the same as the volume below R.

EVALUATING DOUBLE INTEGRALS

Except in the simplest cases, it is impractical to obtain the value of a double integral from the limit in (4). However, we will now show how to evaluate double integrals by calculating

two successive single integrals. For the rest of this section we will limit our discussion to the case where R is a rectangle; in the next section we will consider double integrals over more complicated regions.

The partial derivatives of a function $f(x, y)$ are calculated by holding one of the variables fixed and differentiating with respect to the other variable. Let us consider the reverse of this process, ***partial integration***. The symbols

$$\int_a^b f(x, y)\,dx \quad \text{and} \quad \int_c^d f(x, y)\,dy$$

denote ***partial definite integrals***; the first integral, called the ***partial definite integral with respect to x***, is evaluated by holding y fixed and integrating with respect to x, and the second integral, called the ***partial definite integral with respect to y***, is evaluated by holding x fixed and integrating with respect to y. As the following example shows, the partial definite integral with respect to x is a function of y, and the partial definite integral with respect to y is a function of x.

▶ Example 1

$$\int_0^1 xy^2\,dx = y^2\int_0^1 x\,dx = \frac{y^2x^2}{2}\Big]_{x=0}^{1} = \frac{y^2}{2}$$

$$\int_0^1 xy^2\,dy = x\int_0^1 y^2\,dy = \frac{xy^3}{3}\Big]_{y=0}^{1} = \frac{x}{3}$$ ◀

A partial definite integral with respect to x is a function of y and hence can be integrated with respect to y; similarly, a partial definite integral with respect to y can be integrated with respect to x. This two-stage integration process is called ***iterated*** (or ***repeated***) ***integration***. We introduce the following notation:

$$\int_c^d \int_a^b f(x, y)\,dx\,dy = \int_c^d \left[\int_a^b f(x, y)\,dx\right] dy \tag{6}$$

$$\int_a^b \int_c^d f(x, y)\,dy\,dx = \int_a^b \left[\int_c^d f(x, y)\,dy\right] dx \tag{7}$$

These integrals are called ***iterated integrals***.

► **Example 2** Evaluate

$$\text{(a)}\ \int_1^3 \int_2^4 (40 - 2xy)\, dy\, dx \qquad \text{(b)}\ \int_2^4 \int_1^3 (40 - 2xy)\, dx\, dy$$

Solution (a).

$$\begin{aligned}
\int_1^3 \int_2^4 (40 - 2xy)\, dy\, dx &= \int_1^3 \left[\int_2^4 (40 - 2xy)\, dy \right] dx \\
&= \int_1^3 (40y - xy^2)\Big]_{y=2}^4 dx \\
&= \int_1^3 [(160 - 16x) - (80 - 4x)]\, dx \\
&= \int_1^3 (80 - 12x)\, dx \\
&= (80x - 6x^2)\Big]_1^3 = 112
\end{aligned}$$

Solution (b).

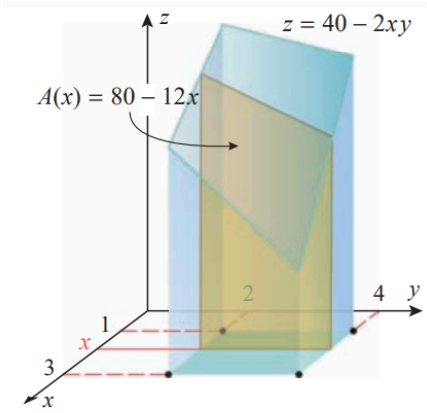
$$\int_2^4 \int_1^3 (40 - 2xy)\,dx\,dy = \int_2^4 \left[\int_1^3 (40 - 2xy)\,dx\right] dy$$

$$= \int_2^4 (40x - x^2y)\Big]_{x=1}^{3}\,dy$$

$$= \int_2^4 [(120 - 9y) - (40 - y)]\,dy$$

$$= \int_2^4 (80 - 8y)\,dy$$

$$= (80y - 4y^2)\Big]_2^4 = 112 \quad \blacktriangleleft$$



▲ **Figure 14.1.4**

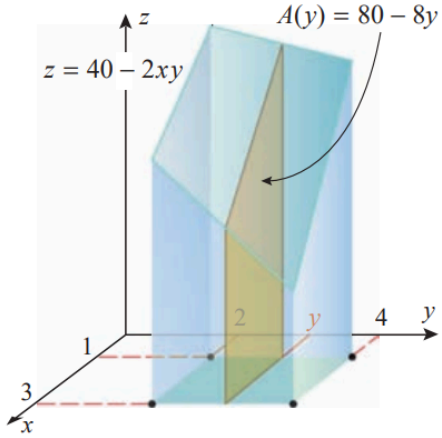
It is no accident that both parts of Example 2 produced the same answer. Consider the solid S bounded above by the surface $z = 40 - 2xy$ and below by the rectangle R defined by $1 \le x \le 3$ and $2 \le y \le 4$. By the method of slicing discussed in Section 6.2, the volume of S is given by

$$V = \int_1^3 A(x)\,dx$$

where $A(x)$ is the area of a vertical cross section of S taken perpendicular to the x-axis (Figure 14.1.4). For a fixed value of x, $1 \le x \le 3$, $z = 40 - 2xy$ is a function of y, so the integral

$$A(x) = \int_2^4 (40 - 2xy)\,dy$$

represents the area under the graph of this function of y. Thus,



▲ **Figure 14.1.5**

represents the area under the graph of this function of y. Thus,

$$V = \int_1^3 \left[\int_2^4 (40 - 2xy)\, dy \right] dx = \int_1^3 \int_2^4 (40 - 2xy)\, dy\, dx$$

is the volume of S. Similarly, by the method of slicing with cross sections of S taken perpendicular to the y-axis, the volume of S is given by

$$V = \int_2^4 A(y)\, dy = \int_2^4 \left[\int_1^3 (40 - 2xy)\, dx \right] dy = \int_2^4 \int_1^3 (40 - 2xy)\, dx\, dy$$

(Figure 14.1.5). Thus, the iterated integrals in parts (a) and (b) of Example 2 both measure the volume of S, which by Formula (5) is the double integral of $z = 40 - 2xy$ over R. That is,

$$\int_1^3 \int_2^4 (40 - 2xy)\, dy\, dx = \iint_R (40 - 2xy)\, dA = \int_2^4 \int_1^3 (40 - 2xy)\, dx\, dy$$

We will often denote the rectangle

$$\{(x, y) : a \le x \le b, c \le y \le d\}$$

as $[a, b] \times [c, d]$ for simplicity.

The geometric argument above applies to any continuous function $f(x, y)$ that is nonnegative on a rectangle $R = [a, b] \times [c, d]$, as is the case for $f(x, y) = 40 - 2xy$ on $[1, 3] \times [2, 4]$. The conclusion that the double integral of $f(x, y)$ over R has the same value as either of the two possible iterated integrals is true even when f is negative at some points in R. We state this result in the following theorem and omit a formal proof.

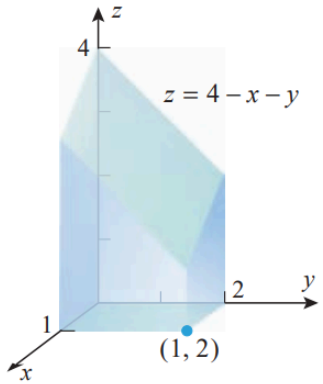
14.1.3 THEOREM (*Fubini's Theorem*) *Let R be the rectangle defined by the inequalities*

$$a \le x \le b, \quad c \le y \le d$$

If $f(x, y)$ is continuous on this rectangle, then

$$\iint_R f(x, y)\, dA = \int_c^d \int_a^b f(x, y)\, dx\, dy = \int_a^b \int_c^d f(x, y)\, dy\, dx$$

Theorem 14.1.3 allows us to evaluate a double integral over a rectangle by converting it to an iterated integral. This can be done in two ways, both of which produce the value of the double integral.



▲ **Figure 14.1.6**

TECHNOLOGY MASTERY

If you have a CAS with a built-in capability for computing iterated double integrals, use it to check Example 3.

► **Example 3** Use a double integral to find the volume of the solid that is bounded above by the plane $z = 4 - x - y$ and below by the rectangle $R = [0, 1] \times [0, 2]$ (Figure 14.1.6).

Solution. The volume is the double integral of $z = 4 - x - y$ over R. Using Theorem 14.1.3, this can be obtained from either of the iterated integrals

$$\int_0^2 \int_0^1 (4 - x - y)\, dx\, dy \quad \text{or} \quad \int_0^1 \int_0^2 (4 - x - y)\, dy\, dx \tag{8}$$

Using the first of these, we obtain

$$V = \iint_R (4 - x - y)\, dA = \int_0^2 \int_0^1 (4 - x - y)\, dx\, dy$$
$$= \int_0^2 \left[4x - \frac{x^2}{2} - xy\right]_{x=0}^{1} dy = \int_0^2 \left(\frac{7}{2} - y\right) dy$$
$$= \left[\frac{7}{2}y - \frac{y^2}{2}\right]_0^2 = 5$$

You can check this result by evaluating the second integral in (8). ◄

Theorem 14.1.3 guarantees that the double integral in Example 4 can also be evaluated by integrating first with respect to y and then with respect to x. Verify this.

► **Example 4** Evaluate the double integral

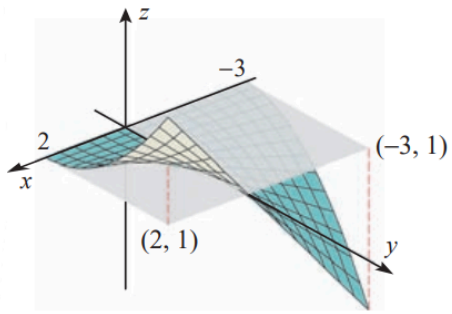
$$\iint_R y^2 x \, dA$$

over the rectangle $R = \{(x, y) : -3 \le x \le 2, 0 \le y \le 1\}$.

Solution. In view of Theorem 14.1.3, the value of the double integral can be obtained by evaluating one of two possible iterated double integrals. We choose to integrate first with respect to x and then with respect to y.

$$\iint_R y^2 x \, dA = \int_0^1 \int_{-3}^2 y^2 x \, dx \, dy = \int_0^1 \left[\frac{1}{2} y^2 x^2\right]_{x=-3}^2 dy$$

$$= \int_0^1 \left(-\frac{5}{2} y^2\right) dy = -\frac{5}{6} y^3 \Big]_0^1 = -\frac{5}{6} \quad \blacktriangleleft$$



$z = y^2 x$ on $[-3, 2] \times [0, 1]$

▲ **Figure 14.1.7**

The integral in Example 4 can be interpreted as the net signed volume between the rectangle $[-3, 2] \times [0, 1]$ and the surface $z = y^2 x$. That is, it is the volume below $z = y^2 x$ and above $[0, 2] \times [0, 1]$ minus the volume above $z = y^2 x$ and below $[-3, 0] \times [0, 1]$ (Figure 14.1.7).

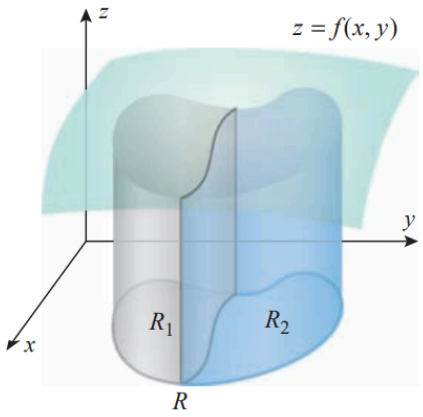
PROPERTIES OF DOUBLE INTEGRALS

To distinguish between double integrals of functions of two variables and definite integrals of functions of one variable, we will refer to the latter as ***single integrals***. Because double integrals, like single integrals, are defined as limits, they inherit many of the properties of limits. The following results, which we state without proof, are analogs of those in Theorem 5.5.4.

$$\iint_R cf(x, y)\,dA = c\iint_R f(x, y)\,dA \quad (c \text{ a constant}) \tag{9}$$

$$\iint_R [f(x, y) + g(x, y)]\,dA = \iint_R f(x, y)\,dA + \iint_R g(x, y)\,dA \tag{10}$$

$$\iint_R [f(x, y) - g(x, y)]\,dA = \iint_R f(x, y)\,dA - \iint_R g(x, y)\,dA \tag{11}$$



The volume of the entire solid is the sum of the volumes of the solids above R_1 and R_2.

▲ **Figure 14.1.8**

It is evident intuitively that if $f(x, y)$ is nonnegative on a region R, then subdividing R into two regions R_1 and R_2 has the effect of subdividing the solid between R and $z = f(x, y)$ into two solids, the sum of whose volumes is the volume of the entire solid (Figure 14.1.8). This suggests the following result, which holds even if f has negative values:

$$\iint_R f(x, y)\,dA = \iint_{R_1} f(x, y)\,dA + \iint_{R_2} f(x, y)\,dA \tag{12}$$

The proof of this result will be omitted.

WATCH:

- **CalcBLUE 3 : Ch. 1.1 : Integrals, Definite & Indefinite**
- **CalcBLUE 3 : Ch. 1.2 : Recalling Riemann Sums**
- **CalcBLUE 3 : Ch. 1.3 : The Definite Integral**
- **CalcBLUE 3 : Ch. 1.4 : Multiple Integrals - What & Why?**
- **CalcBLUE 3 : Ch. 1.5 : Integrating Constants**
- **CalcBLUE 3 : Ch. 1.6 : BONUS! Integration Theory**

- **CalcBLUE 3 : Ch. 2.1 : What Do We Do Now?**
- **CalcBLUE 3 : Ch. 2.2 : The Fubini Theorem**
- **CalcBLUE 3 : Ch. 2.3 : Iterated Integrals**
- **CalcBLUE 3 : Ch. 2.4 : Fubini & Areas**

- **CalcBLUE 3 : Ch. 3.1 : Limits & Double Integrals**
- **CalcBLUE 3 : Ch. 3.2 : Examples - Simple Double Integrals**
- **CalcBLUE 3 : Ch. 3.3 : Example - Order of Integration**
- **CalcBLUE 3 : Ch. 3.4 : Example - Additivity of Integrals**
- **CalcBLUE 3 : Ch. 3.5 : Example - Improper, Unbounded Integrals**
- **Double integrals and Polar integrals: Explained with 3D visualizations**
- **How to Set Up Double Integrals**
- **Defining Double Integration with Riemann Sums | Volume under a Surface**
- **What are double integrals? What are they for? | #SoME1 #3b1b**

Use the following online solver:

Double Integral Calculator

Double Integral Calculator

Multiple (Double, Triple) Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 14.1: Double Integrals: 1-16

1–12 Evaluate the iterated integrals. ■

1. $\int_0^1 \int_0^2 (x+3)\,dy\,dx$

2. $\int_1^3 \int_{-1}^1 (2x-4y)\,dy\,dx$

3. $\int_2^4 \int_0^1 x^2 y\,dx\,dy$

4. $\int_{-2}^0 \int_{-1}^2 (x^2+y^2)\,dx\,dy$

5. $\int_0^{\ln 3} \int_0^{\ln 2} e^{x+y}\,dy\,dx$

6. $\int_0^2 \int_0^1 y \sin x\,dy\,dx$

7. $\int_{-1}^0 \int_2^5 dx\,dy$

8. $\int_4^6 \int_{-3}^7 dy\,dx$

9. $\int_0^1 \int_0^1 \frac{x}{(xy+1)^2}\,dy\,dx$

10. $\int_{\pi/2}^{\pi} \int_1^2 x \cos xy\,dy\,dx$

11. $\int_0^{\ln 2} \int_0^1 xye^{y^2x}\,dy\,dx$

12. $\int_3^4 \int_1^2 \frac{1}{(x+y)^2}\,dy\,dx$

13–16 Evaluate the double integral over the rectangular region R. ■

13. $\iint_R 4xy^3\,dA;\ R = \{(x, y) : -1 \le x \le 1, -2 \le y \le 2\}$

14. $\iint_R \dfrac{xy}{\sqrt{x^2 + y^2 + 1}}\,dA;$

$R = \{(x, y) : 0 \le x \le 1, 0 \le y \le 1\}$

15. $\iint_R x\sqrt{1 - x^2}\,dA;\ R = \{(x, y) : 0 \le x \le 1, 2 \le y \le 3\}$

16. $\iint_R (x \sin y - y \sin x)\,dA;$

$R = \{(x, y) : 0 \le x \le \pi/2, 0 \le y \le \pi/3\}$

Double Integrals over non rectangular regions

READ:
READ: HOWARD ANTON: 14.2: Double Integrals over non rectangular regions

ITERATED INTEGRALS WITH NONCONSTANT LIMITS OF INTEGRATION

Later in this section we will see that double integrals over nonrectangular regions can often be evaluated as iterated integrals of the following types:

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y)\, dy \right] dx \tag{1}$$

$$\int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\, dy = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x, y)\, dx \right] dy \tag{2}$$

We begin with an example that illustrates how to evaluate such integrals.

Note that in (1) and (2) the limits of integration in the outer integral are constants. This is consistent with the fact that the value of each iterated integral is a number that represents a net signed volume.

► **Example 1** Evaluate

$$\text{(a)} \int_0^1 \int_{-x}^{x^2} y^2 x \, dy \, dx \qquad \text{(b)} \int_0^{\pi/3} \int_0^{\cos y} x \sin y \, dx \, dy$$

Solution (a).

$$\int_0^1 \int_{-x}^{x^2} y^2 x \, dy \, dx = \int_0^1 \left[\int_{-x}^{x^2} y^2 x \, dy \right] dx = \int_0^1 \left. \frac{y^3 x}{3} \right]_{y=-x}^{x^2} dx$$

$$= \int_0^1 \left[\frac{x^7}{3} + \frac{x^4}{3} \right] dx = \left(\frac{x^8}{24} + \frac{x^5}{15} \right) \Big]_0^1 = \frac{13}{120}$$

Solution (b).

$$\int_0^{\pi/3} \int_0^{\cos y} x \sin y \, dx \, dy = \int_0^{\pi/3} \left[\int_0^{\cos y} x \sin y \, dx \right] dy = \int_0^{\pi/3} \left. \frac{x^2}{2} \sin y \right]_{x=0}^{\cos y} dy$$

$$= \int_0^{\pi/3} \left[\frac{1}{2} \cos^2 y \sin y \right] dy = -\frac{1}{6} \cos^3 y \Big]_0^{\pi/3} = \frac{7}{48} \quad ◄$$

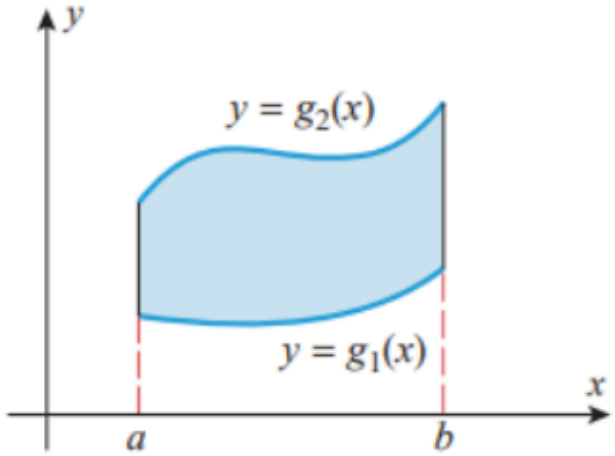
DOUBLE INTEGRALS OVER NONRECTANGULAR REGIONS

Plane regions can be extremely complex, and the theory of double integrals over very general regions is a topic for advanced courses in mathematics. We will limit our study of double integrals to two basic types of regions, which we will call *type I* and *type II*; they are defined as follows.

14.2.1 DEFINITION

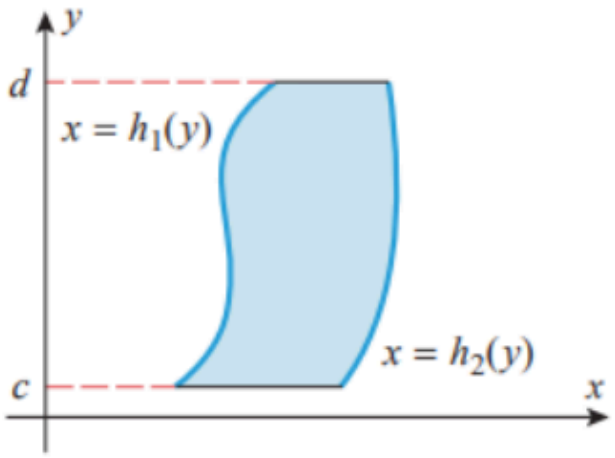
(a) A ***type I region*** is bounded on the left and right by vertical lines $x = a$ and $x = b$ and is bounded below and above by continuous curves $y = g_1(x)$ and $y = g_2(x)$, where $g_1(x) \leq g_2(x)$ for $a \leq x \leq b$ (Figure 14.2.1*a*).

(b) A ***type II region*** is bounded below and above by horizontal lines $y = c$ and $y = d$ and is bounded on the left and right by continuous curves $x = h_1(y)$ and $x = h_2(y)$ satisfying $h_1(y) \leq h_2(y)$ for $c \leq y \leq d$ (Figure 14.2.1*b*).



A type I region

(a)



A type II region

(b)

The following theorem will enable us to evaluate double integrals over type I and type II regions using iterated integrals.

14.2.2 THEOREM

(*a*) *If R is a type I region on which $f(x, y)$ is continuous, then*

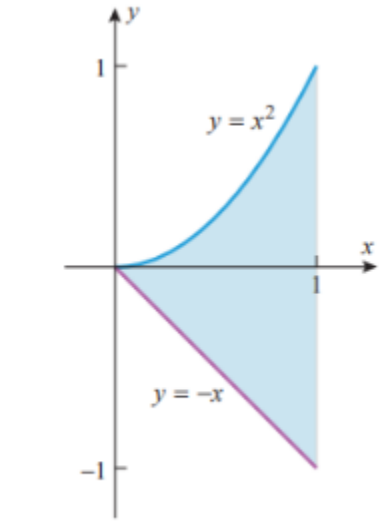
$$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx \tag{3}$$

(*b*) *If R is a type II region on which $f(x, y)$ is continuous, then*

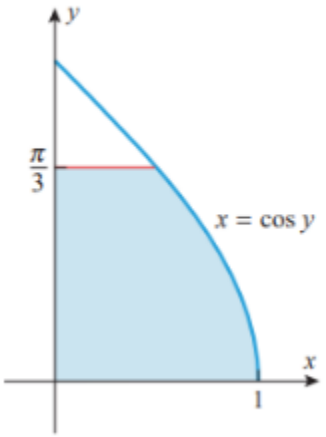
$$\iint_R f(x, y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\, dy \tag{4}$$

► **Example 2** Each of the iterated integrals in Example 1 is equal to a double integral over a region R. Identify the region R in each case.

Solution. Using Theorem 14.2.2, the integral in Example 1(a) is the double integral of the function $f(x, y) = y^2x$ over the type I region R bounded on the left and right by the vertical lines $x = 0$ and $x = 1$ and bounded below and above by the curves $y = -x$ and $y = x^2$ (Figure 14.2.2). The integral in Example 1(b) is the double integral of the function $f(x, y) = x \sin y$ over the type II region R bounded below and above by the horizontal lines $y = 0$ and $y = \pi/3$ and bounded on the left and right by the curves $x = 0$ and $x = \cos y$ (Figure 14.2.3). ◄



▲ Figure 14.2.2



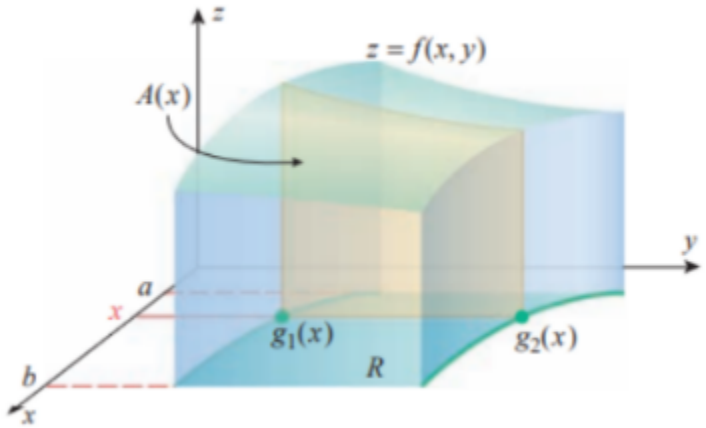
▲ Figure 14.2.3

We will not prove Theorem 14.2.2, but for the case where $f(x, y)$ is nonnegative on the region R, it can be made plausible by a geometric argument that is similar to that given for Theorem 14.1.3. Since $f(x, y)$ is nonnegative, the double integral can be interpreted as the volume of the solid S that is bounded above by the surface $z = f(x, y)$ and below by the region R, so it suffices to show that the iterated integrals also represent this volume. Consider the iterated integral in (3), for example. For a fixed value of x, the function $f(x, y)$ is a function of y, and hence the integral

$$A(x) = \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy$$

represents the area under the graph of this function of y between $y = g_1(x)$ and $y = g_2(x)$. This area, shown in yellow in Figure 14.2.4, is the cross-sectional area at x of the solid S, and hence by the method of slicing, the volume V of the solid S is

$$V = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$$

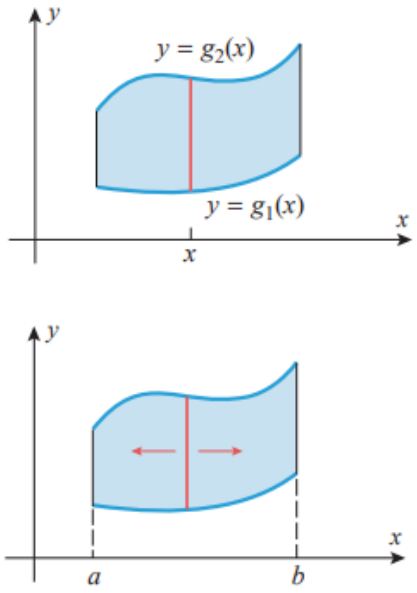


► **Figure 14.2.4**

which shows that in (3) the iterated integral is equal to the double integral. Similarly, the iterated integral in (4) is equal to the corresponding double integral.

SETTING UP LIMITS OF INTEGRATION FOR EVALUATING DOUBLE INTEGRALS

To apply Theorem 14.2.2, it is helpful to start with a two-dimensional sketch of the region R. [It is not necessary to graph $f(x, y)$.] For a type I region, the limits of integration in Formula (3) can be obtained as follows:

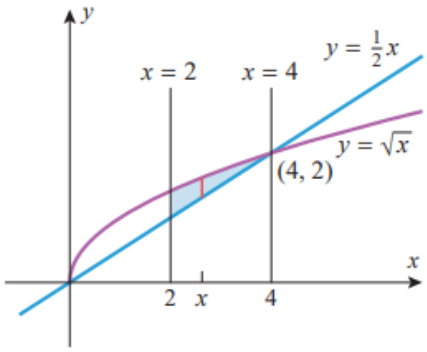


▲ **Figure 14.2.5**

Determining Limits of Integration: Type I Region

Step 1. Since x is held fixed for the first integration, we draw a vertical line through the region R at an arbitrary fixed value x (Figure 14.2.5). This line crosses the boundary of R twice. The lower point of intersection is on the curve $y = g_1(x)$ and the higher point is on the curve $y = g_2(x)$. These two intersections determine the lower and upper y-limits of integration in Formula (3).

Step 2. Imagine moving the line drawn in Step 1 first to the left and then to the right (Figure 14.2.5). The leftmost position where the line intersects the region R is $x = a$, and the rightmost position where the line intersects the region R is $x = b$. This yields the limits for the x-integration in Formula (3).



▲ **Figure 14.2.6**

► **Example 3** Evaluate

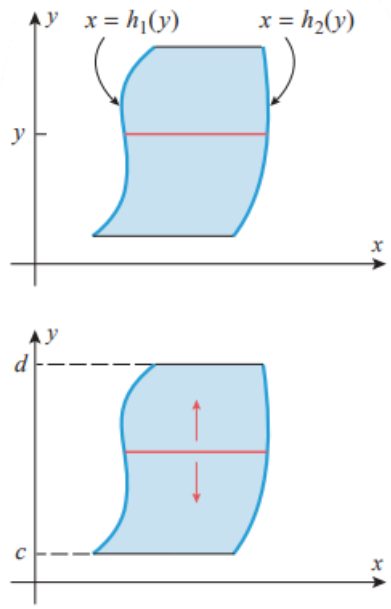
$$\iint_R xy\,dA$$

over the region R enclosed between $y = \frac{1}{2}x$, $y = \sqrt{x}$, $x = 2$, and $x = 4$.

Solution. We view R as a type I region. The region R and a vertical line corresponding to a fixed x are shown in Figure 14.2.6. This line meets the region R at the lower boundary $y = \frac{1}{2}x$ and the upper boundary $y = \sqrt{x}$. These are the y-limits of integration. Moving this line first left and then right yields the x-limits of integration, $x = 2$ and $x = 4$. Thus,

$$\iint_R xy\,dA = \int_2^4 \int_{x/2}^{\sqrt{x}} xy\,dy\,dx = \int_2^4 \left[\frac{xy^2}{2}\right]_{y=x/2}^{\sqrt{x}} dx = \int_2^4 \left(\frac{x^2}{2} - \frac{x^3}{8}\right) dx$$

$$= \left[\frac{x^3}{6} - \frac{x^4}{32}\right]_2^4 = \left(\frac{64}{6} - \frac{256}{32}\right) - \left(\frac{8}{6} - \frac{16}{32}\right) = \frac{11}{6} \quad ◄$$



▲ **Figure 14.2.7**

If R is a type II region, then the limits of integration in Formula (4) can be obtained as follows:

Determining Limits of Integration: Type II Region

Step 1. Since y is held fixed for the first integration, we draw a horizontal line through the region R at a fixed value y (Figure 14.2.7). This line crosses the boundary of R twice. The leftmost point of intersection is on the curve $x = h_1(y)$ and the rightmost point is on the curve $x = h_2(y)$. These intersections determine the x-limits of integration in (4).

Step 2. Imagine moving the line drawn in Step 1 first down and then up (Figure 14.2.7). The lowest position where the line intersects the region R is $y = c$, and the highest position where the line intersects the region R is $y = d$. This yields the y-limits of integration in (4).

▶ **Example 4** Evaluate

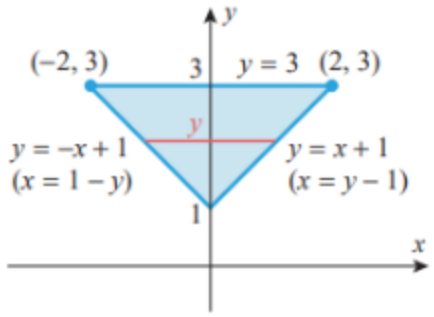
$$\iint_R (2x - y^2)\, dA$$

over the triangular region R enclosed between the lines $y = -x + 1$, $y = x + 1$, and $y = 3$.

Solution. We view R as a type II region. The region R and a horizontal line corresponding to a fixed y are shown in Figure 14.2.8. This line meets the region R at its left-hand boundary $x = 1 - y$ and its right-hand boundary $x = y - 1$. These are the x-limits of integration. Moving this line first down and then up yields the y-limits, $y = 1$ and $y = 3$. Thus,

$$\begin{aligned}
\iint_R (2x - y^2)\, dA &= \int_1^3 \int_{1-y}^{y-1} (2x - y^2)\, dx\, dy = \int_1^3 \left[x^2 - y^2 x\right]_{x=1-y}^{y-1} dy \\
&= \int_1^3 [(1 - 2y + 2y^2 - y^3) - (1 - 2y + y^3)]\, dy \\
&= \int_1^3 (2y^2 - 2y^3)\, dy = \left[\frac{2y^3}{3} - \frac{y^4}{2}\right]_1^3 = -\frac{68}{3}
\end{aligned}$$

◀



▲ Figure 14.2.8

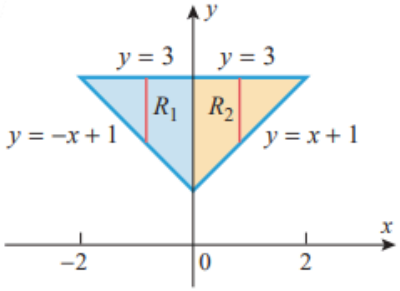
To integrate over a type II region, the left- and right-hand boundaries must be expressed in the form $x = h_1(y)$ and $x = h_2(y)$. This is why we rewrote the boundary equations

$$y = -x + 1 \quad \text{and} \quad y = x + 1$$

as

$$x = 1 - y \quad \text{and} \quad x = y - 1$$

in Example 4.



▲ Figure 14.2.9

In Example 4 we could have treated R as a type I region, but with an added complication. Viewed as a type I region, the upper boundary of R is the line $y = 3$ (Figure 14.2.9) and the lower boundary consists of two parts, the line $y = -x + 1$ to the left of the y-axis and the line $y = x + 1$ to the right of the y-axis. To carry out the integration it is necessary to decompose the region R into two parts, R_1 and R_2, as shown in Figure 14.2.9, and write

$$\iint_R (2x - y^2)\,dA = \iint_{R_1} (2x - y^2)\,dA + \iint_{R_2} (2x - y^2)\,dA$$

$$= \int_{-2}^{0}\int_{-x+1}^{3} (2x - y^2)\,dy\,dx + \int_{0}^{2}\int_{x+1}^{3} (2x - y^2)\,dy\,dx$$

This will yield the same result that was obtained in Example 4. (Verify.)

▶ **Example 5** Use a double integral to find the volume of the tetrahedron bounded by the coordinate planes and the plane $z = 4 - 4x - 2y$.

Solution. The tetrahedron in question is bounded above by the plane

$$z = 4 - 4x - 2y \tag{5}$$

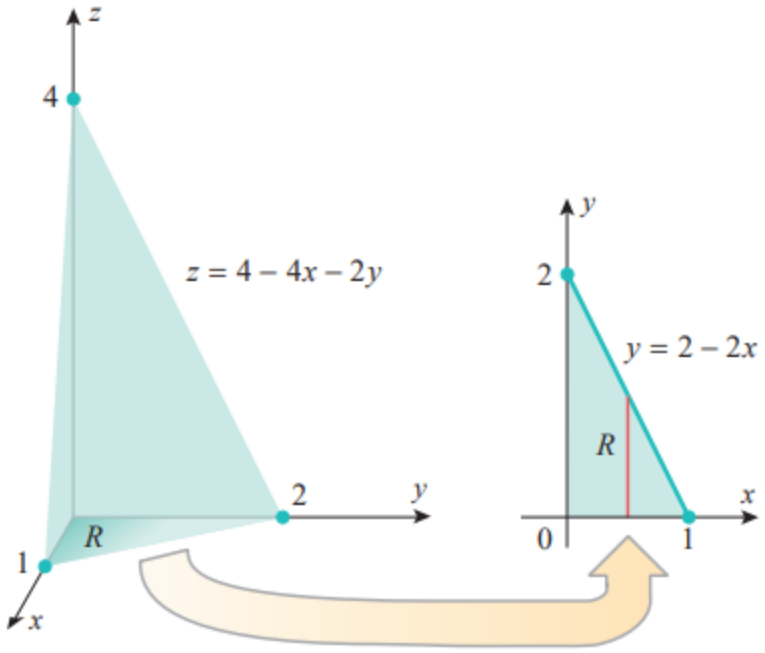
and below by the triangular region R shown in Figure 14.2.10. Thus, the volume is given by

$$V = \iint_R (4 - 4x - 2y)\, dA$$

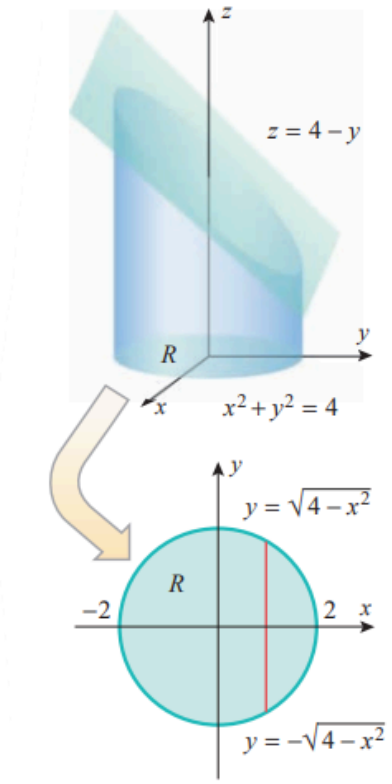
The region R is bounded by the x-axis, the y-axis, and the line $y = 2 - 2x$ [set $z = 0$ in (5)], so that treating R as a type I region yields

$$V = \iint_R (4 - 4x - 2y)\, dA = \int_0^1 \int_0^{2-2x} (4 - 4x - 2y)\, dy\, dx$$

$$= \int_0^1 \left[4y - 4xy - y^2\right]_{y=0}^{2-2x} dx = \int_0^1 (4 - 8x + 4x^2)\, dx = \frac{4}{3} \quad \blacktriangleleft$$



▶ **Figure 14.2.10**



▲ **Figure 14.2.11**

► **Example 6** Find the volume of the solid bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$.

Solution. The solid shown in Figure 14.2.11 is bounded above by the plane $z = 4 - y$ and below by the region R within the circle $x^2 + y^2 = 4$. The volume is given by

$$V = \iint_R (4 - y)\, dA$$

Treating R as a type I region we obtain

$$V = \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4 - y)\, dy\, dx = \int_{-2}^{2} \left[4y - \frac{1}{2}y^2 \right]_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx$$

$$= \int_{-2}^{2} 8\sqrt{4 - x^2}\, dx = 8(2\pi) = 16\pi$$

See Formula (3) of Section 7.4. ◄

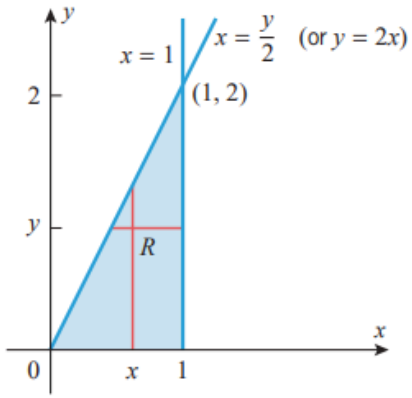
REVERSING THE ORDER OF INTEGRATION

Sometimes the evaluation of an iterated integral can be simplified by reversing the order of integration. The next example illustrates how this is done.

Example 7 Since there is no elementary antiderivative of e^{x^2}, the integral

$$\int_0^2 \int_{y/2}^1 e^{x^2}\,dx\,dy$$

cannot be evaluated by performing the x-integration first. Evaluate this integral by expressing it as an equivalent iterated integral with the order of integration reversed.



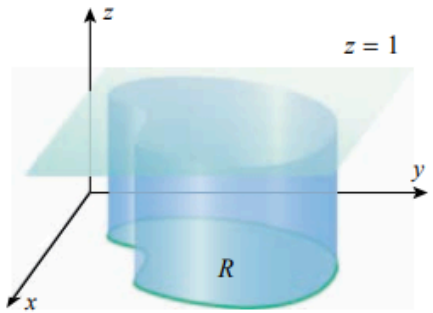
▲ Figure 14.2.12

Solution. For the inside integration, y is fixed and x varies from the line $x = y/2$ to the line $x = 1$ (Figure 14.2.12). For the outside integration, y varies from 0 to 2, so the given iterated integral is equal to a double integral over the triangular region R in Figure 14.2.12.

To reverse the order of integration, we treat R as a type I region, which enables us to write the given integral as

$$\int_0^2 \int_{y/2}^1 e^{x^2}\,dx\,dy = \iint_R e^{x^2}\,dA = \int_0^1 \int_0^{2x} e^{x^2}\,dy\,dx = \int_0^1 \left[e^{x^2} y\right]_{y=0}^{2x} dx$$

$$= \int_0^1 2xe^{x^2}\,dx = e^{x^2}\Big]_0^1 = e - 1 \quad \blacktriangleleft$$



Cylinder with base R and height 1

▲ **Figure 14.2.13**

AREA CALCULATED AS A DOUBLE INTEGRAL

Although double integrals arose in the context of calculating volumes, they can also be used to calculate areas. To see why this is so, recall that a *right cylinder* is a solid that is generated when a plane region is translated along a line that is perpendicular to the region. In Formula (2) of Section 6.2 we stated that the volume V of a right cylinder with cross-sectional area A and height h is

$$V = A \cdot h \tag{6}$$

Now suppose that we are interested in finding the area A of a region R in the xy-plane. If we translate the region R upward 1 unit, then the resulting solid will be a right cylinder that has cross-sectional area A, base R, and the plane $z = 1$ as its top (Figure 14.2.13). Thus, it follows from (6) that

$$\iint_R 1 \, dA = (\text{area of } R) \cdot 1$$

which we can rewrite as

$$\text{area of } R = \iint_R 1 \, dA = \iint_R dA \tag{7}$$

Formula (7) can be confusing because it equates an area and a volume; the formula is intended to equate only the *numerical values* of the area and volume and not the units, which must, of course, be different.

▶ **Example 8** Use a double integral to find the area of the region R enclosed between the parabola $y = \frac{1}{2}x^2$ and the line $y = 2x$.

Solution. The region R may be treated equally well as type I (Figure 14.2.14*a*) or type II (Figure 14.2.14*b*). Treating R as type I yields

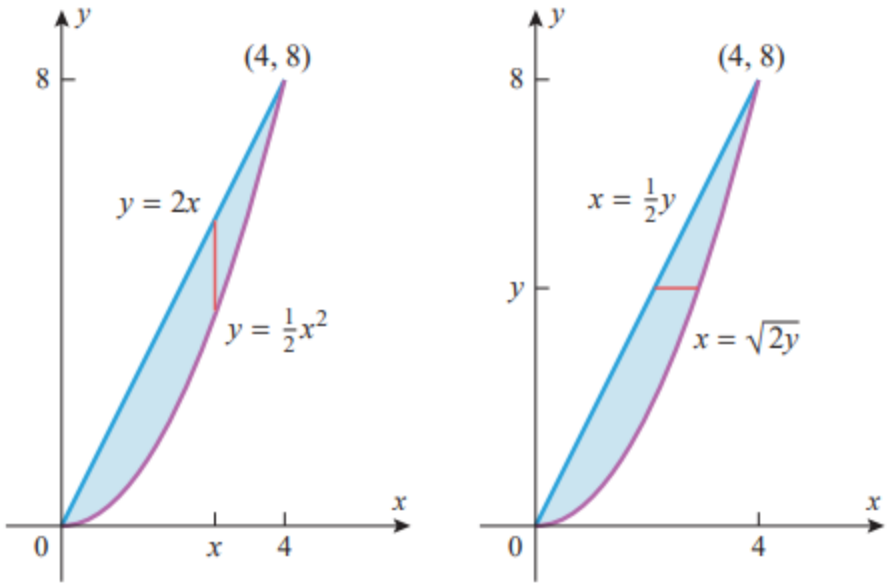
$$\text{area of } R = \iint_R dA = \int_0^4 \int_{x^2/2}^{2x} dy\, dx = \int_0^4 [y]_{y=x^2/2}^{2x}\, dx$$

$$= \int_0^4 \left(2x - \frac{1}{2}x^2\right) dx = \left[x^2 - \frac{x^3}{6}\right]_0^4 = \frac{16}{3}$$

Treating R as type II yields

$$\text{area of } R = \iint_R dA = \int_0^8 \int_{y/2}^{\sqrt{2y}} dx\, dy = \int_0^8 [x]_{x=y/2}^{\sqrt{2y}}\, dy$$

$$= \int_0^8 \left(\sqrt{2y} - \frac{1}{2}y\right) dy = \left[\frac{2\sqrt{2}}{3}y^{3/2} - \frac{y^2}{4}\right]_0^8 = \frac{16}{3}$$ ◀



▶ **Figure 14.2.14**

WATCH:

How to Set Up Double Integrals

Double Integration Example over General Regions --- two ways!

Change the order of integration to solve tricky integrals

What are double integrals? What are they for? | #SoME1 #3b1b

Use the following online solver:

Double Integral Calculator

Double Integral Calculator

Multiple (Double, Triple) Integral Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 14.2: Double Integrals over non rectangular regions: 1-26

1–8 Evaluate the iterated integral. ■

1. $\displaystyle\int_0^1 \int_{x^2}^{x} xy^2\,dy\,dx$

2. $\displaystyle\int_1^{3/2} \int_{y}^{3-y} y\,dx\,dy$

3. $\displaystyle\int_0^3 \int_0^{\sqrt{9-y^2}} y\,dx\,dy$

4. $\displaystyle\int_{1/4}^1 \int_{x^2}^{x} \sqrt{\frac{x}{y}}\,dy\,dx$

5. $\displaystyle\int_{\sqrt{\pi}}^{\sqrt{2\pi}} \int_0^{x^3} \sin\frac{y}{x}\,dy\,dx$

6. $\displaystyle\int_{-1}^1 \int_{-x^2}^{x^2} (x^2 - y)\,dy\,dx$

7. $\displaystyle\int_0^1 \int_0^x y\sqrt{x^2 - y^2}\,dy\,dx$

8. $\displaystyle\int_1^2 \int_0^{y^2} e^{x/y^2}\,dx\,dy$

FOCUS ON CONCEPTS

9. Let R be the region shown in the accompanying figure. Fill in the missing limits of integration.

(a) $\displaystyle\iint_R f(x, y)\,dA = \int_{\square}^{\square} \int_{\square}^{\square} f(x, y)\,dy\,dx$

(b) $\displaystyle\iint_R f(x, y)\,dA = \int_{\square}^{\square} \int_{\square}^{\square} f(x, y)\,dx\,dy$

11. Let R be the region shown in the accompanying figure. Fill in the missing limits of integration.

(a) $\displaystyle\iint_R f(x, y)\,dA = \int_1^2 \int_{\square}^{\square} f(x, y)\,dy\,dx + \int_2^4 \int_{\square}^{\square} f(x, y)\,dy\,dx + \int_4^5 \int_{\square}^{\square} f(x, y)\,dy\,dx$

(b) $\displaystyle\iint_R f(x, y)\,dA = \int_{\square}^{\square} \int_{\square}^{\square} f(x, y)\,dx\,dy$

12. Let R be the region shown in the accompanying figure. Fill in the missing limits of integration.

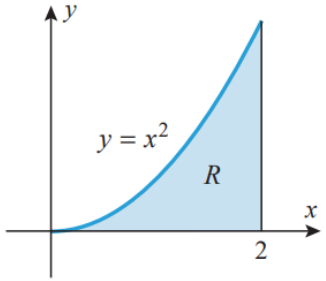
(a) $\displaystyle\iint_R f(x, y)\,dA = \int_{\square}^{\square} \int_{\square}^{\square} f(x, y)\,dy\,dx$

(b) $\displaystyle\iint_R f(x, y)\,dA = \int_{\square}^{\square} \int_{\square}^{\square} f(x, y)\,dx\,dy$

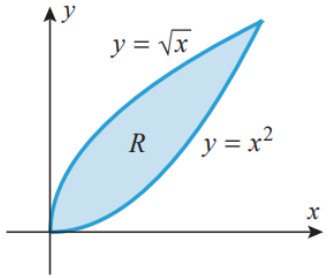
10. Let R be the region shown in the accompanying figure. Fill in the missing limits of integration.

(a) $\displaystyle\iint_R f(x, y)\,dA = \int_{\square}^{\square}\int_{\square}^{\square} f(x, y)\,dy\,dx$

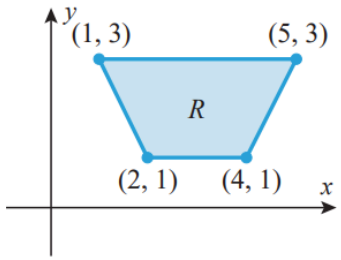
(b) $\displaystyle\iint_R f(x, y)\,dA = \int_{\square}^{\square}\int_{\square}^{\square} f(x, y)\,dx\,dy$



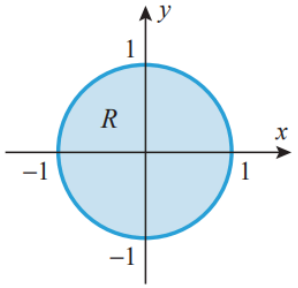
▲ **Figure Ex-9**



▲ **Figure Ex-10**



▲ **Figure Ex-11**



▲ **Figure Ex-12**

13. Evaluate $\displaystyle\iint_R xy\,dA$, where R is the region in

(a) Exercise 9 (b) Exercise 11.

14. Evaluate $\iint_R (x + y)\, dA$, where R is the region in

(a) Exercise 10 (b) Exercise 12.

15–18 Evaluate the double integral in two ways using iterated integrals: (a) viewing R as a type I region, and (b) viewing R as a type II region. ■

15. $\iint_R x^2\, dA$; R is the region bounded by $y = 16/x$, $y = x$, and $x = 8$.

16. $\iint_R xy^2\, dA$; R is the region enclosed by $y = 1$, $y = 2$, $x = 0$, and $y = x$.

17. $\iint_R (3x - 2y)\, dA$; R is the region enclosed by the circle $x^2 + y^2 = 1$.

18. $\iint_R y\,dA$; R is the region in the first quadrant enclosed between the circle $x^2 + y^2 = 25$ and the line $x + y = 5$.

19–24 Evaluate the double integral. ■

19. $\iint_R x(1 + y^2)^{-1/2}\,dA$; R is the region in the first quadrant enclosed by $y = x^2$, $y = 4$, and $x = 0$.

20. $\iint_R x \cos y\,dA$; R is the triangular region bounded by the lines $y = x$, $y = 0$, and $x = \pi$.

21. $\iint_R xy\,dA$; R is the region enclosed by $y = \sqrt{x}$, $y = 6 - x$, and $y = 0$.

22. $\iint_R x\,dA$; R is the region enclosed by $y = \sin^{-1} x$, $x = 1/\sqrt{2}$, and $y = 0$.

23. $\iint_R (x - 1)\, dA$; R is the region in the first quadrant enclosed between $y = x$ and $y = x^3$.

24. $\iint_R x^2\, dA$; R is the region in the first quadrant enclosed by $xy = 1$, $y = x$, and $y = 2x$.

25. Evaluate $\iint_R \sin(y^3)\, dA$, where R is the region bounded by $y = \sqrt{x}$, $y = 2$, and $x = 0$. [*Hint:* Choose the order of integration carefully.]

26. Evaluate $\iint_R x\, dA$, where R is the region bounded by $x = \ln y$, $x = 0$, and $y = e$.

Double Integrals in polar Coordinates

READ:

READ: HOWARD ANTON: 14.3: Double Integrals in polar Coordinates

WATCH:

READ: Polar and Cartesian Coordinates or **WATCH:** **Intro to Polar Coordinates**

Sketching Polar Curves

Double integrals and Polar integrals: Explained with 3D visualizations

Double Integration in Polar Coordinates | Example & Derivation

The Gaussian Integral // Solved Using Polar Coordinates

Use the following online solver:

Polar/Rectangular Coordinates Calculator - eMathHelp

Polar/Rectangular Equation Calculator - eMathHelp

Polar equation plotter – GeoGebra

PRACTISE: HOWARD ANTON: 14.3: Double Integrals in polar Coordinates: 1-12, 23-34

1–6 Evaluate the iterated integral. ■

1. $\int_0^{\pi/2} \int_0^{\sin\theta} r\cos\theta \, dr \, d\theta$

2. $\int_0^{\pi} \int_0^{1+\cos\theta} r \, dr \, d\theta$

3. $\int_0^{\pi/2} \int_0^{a\sin\theta} r^2 \, dr \, d\theta$

4. $\int_0^{\pi/6} \int_0^{\cos 3\theta} r \, dr \, d\theta$

5. $\int_0^{\pi} \int_0^{1-\sin\theta} r^2\cos\theta \, dr \, d\theta$

6. $\int_0^{\pi/2} \int_0^{\cos\theta} r^3 \, dr \, d\theta$

7–10 Use a double integral in polar coordinates to find the area of the region described. ■

7. The region enclosed by the cardioid $r = 1 - \cos\theta$.

8. The region enclosed by the rose $r = \sin 2\theta$.

9. The region in the first quadrant bounded by $r = 1$ and $r = \sin 2\theta$, with $\pi/4 \le \theta \le \pi/2$.

10. The region inside the circle $x^2 + y^2 = 4$ and to the right of the line $x = 1$.

FOCUS ON CONCEPTS

11–12 Let R be the region described. Sketch the region R and fill in the missing limits of integration.

$$\iint_R f(r,\theta)\,dA = \int_{\square}^{\square}\int_{\square}^{\square} f(r,\theta)r\,dr\,d\theta$$

11. The region inside the circle $r = 4\sin\theta$ and outside the circle $r = 2$.

12. The region inside the circle $r = 1$ and outside the cardioid $r = 1 + \cos\theta$.

23–26 Use polar coordinates to evaluate the double integral.

23. $\iint_R \sin(x^2 + y^2)\, dA$, where R is the region enclosed by the circle $x^2 + y^2 = 9$.

24. $\iint_R \sqrt{9 - x^2 - y^2}\, dA$, where R is the region in the first quadrant within the circle $x^2 + y^2 = 9$.

25. $\iint_R \frac{1}{1 + x^2 + y^2}\, dA$, where R is the sector in the first quadrant bounded by $y = 0$, $y = x$, and $x^2 + y^2 = 4$.

26. $\iint_R 2y\, dA$, where R is the region in the first quadrant bounded above by the circle $(x - 1)^2 + y^2 = 1$ and below by the line $y = x$.

27–34 Evaluate the iterated integral by converting to polar coordinates. ■

27. $\displaystyle\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)\, dy\, dx$

28. $\displaystyle\int_{-2}^{2} \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)}\, dx\, dy$

29. $\displaystyle\int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2 + y^2}\, dy\, dx$

30. $\displaystyle\int_0^1 \int_0^{\sqrt{1-y^2}} \cos(x^2 + y^2)\, dx\, dy$

31. $\displaystyle\int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{dy\, dx}{(1 + x^2 + y^2)^{3/2}} \quad (a > 0)$

32. $\displaystyle\int_0^1 \int_y^{\sqrt{y}} \sqrt{x^2 + y^2}\, dx\, dy$

33. $\displaystyle\int_0^{\sqrt{2}} \int_y^{\sqrt{4-y^2}} \frac{1}{\sqrt{1 + x^2 + y^2}}\, dx\, dy$

34. $\displaystyle\int_{-4}^{0} \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} 3x\, dy\, dx$

Surface Area; Parametric Surfaces

READ:

READ: HOWARD ANTON: 14.4: Surface Area; Parametric Surfaces

WATCH:

PRACTISE: HOWARD ANTON: 14.4: Surface Area; Parametric Surfaces: 1-9

1–4 Express the area of the given surface as an iterated double integral, and then find the surface area. ■

1. The portion of the cylinder $y^2 + z^2 = 9$ that is above the rectangle $R = \{(x, y) : 0 \leq x \leq 2, -3 \leq y \leq 3\}$.
2. The portion of the plane $2x + 2y + z = 8$ in the first octant.
3. The portion of the cone $z^2 = 4x^2 + 4y^2$ that is above the region in the first quadrant bounded by the line $y = x$ and the parabola $y = x^2$.
4. The portion of the surface $z = 2x + y^2$ that is above the triangular region with vertices $(0, 0)$, $(0, 1)$, and $(1, 1)$.

5–10 Express the area of the given surface as an iterated double integral in polar coordinates, and then find the surface area. ■

5. The portion of the cone $z = \sqrt{x^2 + y^2}$ that lies inside the cylinder $x^2 + y^2 = 2x$.

6. The portion of the paraboloid $z = 1 - x^2 - y^2$ that is above the xy-plane.

7. The portion of the surface $z = xy$ that is above the sector in the first quadrant bounded by the lines $y = x/\sqrt{3}$, $y = 0$, and the circle $x^2 + y^2 = 9$.

8. The portion of the paraboloid $2z = x^2 + y^2$ that is inside the cylinder $x^2 + y^2 = 8$.

9. The portion of the sphere $x^2 + y^2 + z^2 = 16$ between the planes $z = 1$ and $z = 2$.

10. The portion of the sphere $x^2 + y^2 + z^2 = 8$ that is inside the cone $z = \sqrt{x^2 + y^2}$.

- **Concept of Triple Integral, Triple integral in Cartesian, Cylindrical and Spherical Coordinate system.**

Triple Integrals

READ:

READ: HOWARD ANTON: 14.5: Triple Integrals

WATCH:

Calculus - How to find the bounds of a triple integral

Triple Integrals in Cartesian Coordinates | Volume between Surfaces

PRACTISE: HOWARD ANTON: 14.5: Triple Integrals: 1-12, 15-18

1–8 Evaluate the iterated integral. ■

1. $\int_{-1}^{1}\int_{0}^{2}\int_{0}^{1}(x^2+y^2+z^2)\,dx\,dy\,dz$

2. $\int_{1/3}^{1/2}\int_{0}^{\pi}\int_{0}^{1} zx\sin xy\,dz\,dy\,dx$

3. $\int_{0}^{2}\int_{-1}^{y^2}\int_{-1}^{z} yz\,dx\,dz\,dy$

4. $\int_{0}^{\pi/4}\int_{0}^{1}\int_{0}^{x^2} x\cos y\,dz\,dx\,dy$

5. $\int_{0}^{3}\int_{0}^{\sqrt{9-z^2}}\int_{0}^{x} xy\,dy\,dx\,dz$

6. $\int_{1}^{3}\int_{x}^{x^2}\int_{0}^{\ln z} xe^y\,dy\,dz\,dx$

7. $\int_{0}^{2}\int_{0}^{\sqrt{4-x^2}}\int_{-5+x^2+y^2}^{3-x^2-y^2} x\,dz\,dy\,dx$

8. $\int_{1}^{2}\int_{z}^{2}\int_{0}^{\sqrt{3}y}\frac{y}{x^2+y^2}\,dx\,dy\,dz$

9–12 Evaluate the triple integral. ■

9. $\iiint_G xy \sin yz \, dV$, where G is the rectangular box defined by the inequalities $0 \le x \le \pi, 0 \le y \le 1, 0 \le z \le \pi/6$.

10. $\iiint_G y \, dV$, where G is the solid enclosed by the plane $z = y$, the xy-plane, and the parabolic cylinder $y = 1 - x^2$.

11. $\iiint_G xyz \, dV$, where G is the solid in the first octant that is bounded by the parabolic cylinder $z = 2 - x^2$ and the planes $z = 0$, $y = x$, and $y = 0$.

12. $\iiint_G \cos(z/y) \, dV$, where G is the solid defined by the inequalities $\pi/6 \le y \le \pi/2$, $y \le x \le \pi/2$, $0 \le z \le xy$.

15–18 Use a triple integral to find the volume of the solid. ■

15. The solid in the first octant bounded by the coordinate planes and the plane $3x + 6y + 4z = 12$.

16. The solid bounded by the surface $z = \sqrt{y}$ and the planes $x + y = 1$, $x = 0$, and $z = 0$.

17. The solid bounded by the surface $y = x^2$ and the planes $y + z = 4$ and $z = 0$.

18. The wedge in the first octant that is cut from the solid cylinder $y^2 + z^2 \leq 1$ by the planes $y = x$ and $x = 0$.

Triple Integrals in Cylindrical and Spherical Coordinates

READ:

READ: HOWARD ANTON: 14.6: Triple Integrals in Cylindrical and Spherical Coordinates

WATCH:

Integration in Spherical Coordinates

PRACTISE: HOWARD ANTON: 14.6: Triple Integrals in Cylindrical and Spherical Coordinates: 9-12, 13-16, 17-20

9–12 Use cylindrical coordinates to find the volume of the solid.

9. The solid enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 9$.

10. The solid that is bounded above by the sphere $x^2 + y^2 + z^2 = 1$ and below by the cone $z = \sqrt{x^2 + y^2}$.

11. The solid that is inside the surface $r^2 + z^2 = 20$ but not above the surface $z = r^2$.

12. The solid enclosed between the cone $z = (hr)/a$ and the plane $z = h$.

13–16 Use spherical coordinates to find the volume of the solid.

13. The solid bounded above by the sphere $\rho = 4$ and below by the cone $\phi = \pi/3$.

14. The solid within the cone $\phi = \pi/4$ and between the spheres $\rho = 1$ and $\rho = 2$.

15. The solid enclosed by the sphere $x^2 + y^2 + z^2 = 4a^2$ and the planes $z = 0$ and $z = a$.

16. The solid within the sphere $x^2 + y^2 + z^2 = 9$, outside the cone $z = \sqrt{x^2 + y^2}$, and above the xy-plane.

17–20 Use cylindrical or spherical coordinates to evaluate the integral. ■

17. $\displaystyle\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{a^2-x^2-y^2} x^2\,dz\,dy\,dx \quad (a > 0)$

18. $\displaystyle\int_{-1}^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} e^{-(x^2+y^2+z^2)^{3/2}}\,dz\,dy\,dx$

19. $\displaystyle\int_0^2 \int_0^{\sqrt{4-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{8-x^2-y^2}} z^2\,dz\,dx\,dy$

20. $\displaystyle\int_{-3}^3 \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} \int_{-\sqrt{9-x^2-y^2}}^{\sqrt{9-x^2-y^2}} \sqrt{x^2+y^2+z^2}\,dz\,dx\,dy$

Change of Variables in Multiple Integrals, Jacobians

READ:

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Change of Variables and the Jacobian

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Jacobian Calculator - eMathHelp

PRACTISE: HOWARD ANTON: 14.7: Change of Variables in Multiple Integrals, Jacobians: 1-12, 21-24, 35-37

1–4 Find the Jacobian $\partial(x, y)/\partial(u, v)$. ■

1. $x = u + 4v,\ y = 3u - 5v$

2. $x = u + 2v^2,\ y = 2u^2 - v$

3. $x = \sin u + \cos v,\ y = -\cos u + \sin v$

4. $x = \dfrac{2u}{u^2 + v^2},\ y = -\dfrac{2v}{u^2 + v^2}$

5–8 Solve for x and y in terms of u and v, and then find the Jacobian $\partial(x, y)/\partial(u, v)$. ■

5. $u = 2x - 5y,\ v = x + 2y$

6. $u = e^x,\ v = ye^{-x}$

7. $u = x^2 - y^2,\ v = x^2 + y^2 \quad (x > 0, y > 0)$

8. $u = xy,\ v = xy^3 \quad (x > 0, y > 0)$

9–12 Find the Jacobian $\partial(x, y, z)/\partial(u, v, w)$. ■

9. $x = 3u + v,\ y = u - 2w,\ z = v + w$

10. $x = u - uv,\ y = uv - uvw,\ z = uvw$

11. $u = xy,\ v = y,\ w = x + z$

12. $u = x + y + z,\ v = x + y - z,\ w = x - y + z$

21. Use the transformation $u = x - 2y$, $v = 2x + y$ to find

$$\iint_R \frac{x - 2y}{2x + y}\, dA$$

where R is the rectangular region enclosed by the lines $x - 2y = 1$, $x - 2y = 4$, $2x + y = 1$, $2x + y = 3$.

22. Use the transformation $u = x + y$, $v = x - y$ to find

$$\iint_R (x - y)e^{x^2 - y^2}\, dA$$

over the rectangular region R enclosed by the lines $x + y = 0$, $x + y = 1$, $x - y = 1$, $x - y = 4$.

23. Use the transformation $u = \frac{1}{2}(x + y)$, $v = \frac{1}{2}(x - y)$ to find

$$\iint_R \sin \tfrac{1}{2}(x + y) \cos \tfrac{1}{2}(x - y)\, dA$$

over the triangular region R with vertices $(0, 0)$, $(2, 0)$, $(1, 1)$.

24. Use the transformation $u = y/x$, $v = xy$ to find

$$\iint_R xy^3\, dA$$

over the region R in the first quadrant enclosed by $y = x$, $y = 3x$, $xy = 1$, $xy = 4$.

35–38 Evaluate the integral by making an appropriate change of variables. ■

35. $\iint_R \frac{y-4x}{y+4x}\,dA$, where R is the region enclosed by the lines $y = 4x$, $y = 4x + 2$, $y = 2 - 4x$, $y = 5 - 4x$.

36. $\iint_R (x^2 - y^2)\,dA$, where R is the rectangular region enclosed by the lines $y = -x$, $y = 1 - x$, $y = x$, $y = x + 2$.

37. $\iint_R \frac{\sin(x-y)}{\cos(x+y)}\,dA$, where R is the triangular region enclosed by the lines $y = 0$, $y = x$, $x + y = \pi/4$.

38. $\iint_R e^{(y-x)/(y+x)}\,dA$, where R is the region in the first quadrant enclosed by the trapezoid with vertices $(0, 1)$, $(1, 0)$, $(0, 4)$, $(4, 0)$.

- **Introduction to Differential Equation, Separable Variables, Linear Equations, Integrating Factor**

Introduction to Differential Equation

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Differential Equations Solution Guide

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- **The Key Definitions of Differential Equations: ODE, order, solution, initial condition, IVP**
- **The Geometric Meaning of Differential Equations // Slope Fields, Integral Curves & Isoclines**

Separable Variables

READ:
Separation of Variables

See more: Homogeneous Differential Equations (Homogeneous Functions)

READ: DENNIS G. ZILL: 2.2: Separable Variables

DEFINITION 2.2.1 Separable Equation

A first-order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y)$$

is said to be **separable** or to have **separable variables.**

For example, the equations

$$\frac{dy}{dx} = y^2xe^{3x+4y} \quad \text{and} \quad \frac{dy}{dx} = y + \sin x$$

are separable and nonseparable, respectively. In the first equation we can factor $f(x, y) = y^2xe^{3x+4y}$ as

$$f(x, y) = y^2xe^{3x+4y} = \overset{\overset{g(x)}{\downarrow}}{(xe^{3x})}\overset{\overset{h(y)}{\downarrow}}{(y^2e^{4y})},$$

but in the second equation there is no way of expressing $y + \sin x$ as a product of a function of x times a function of y.

Observe that by dividing by the function $h(y)$, we can write a separable equation $dy/dx = g(x)h(y)$ as

$$p(y)\frac{dy}{dx} = g(x), \tag{2}$$

where, for convenience, we have denoted $1/h(y)$ by $p(y)$. From this last form we can see immediately that (2) reduces to (1) when $h(y) = 1$.

Now if $y = \phi(x)$ represents a solution of (2), we must have $p(\phi(x))\phi'(x) = g(x)$, and therefore

$$\int p(\phi(x))\phi'(x)\,dx = \int g(x)\,dx. \tag{3}$$

But $dy = \phi'(x)\,dx$, and so (3) is the same as

$$\int p(y)\,dy = \int g(x)\,dx \qquad \text{or} \qquad H(y) = G(x) + c, \tag{4}$$

where $H(y)$ and $G(x)$ are antiderivatives of $p(y) = 1/h(y)$ and $g(x)$, respectively.

METHOD OF SOLUTION Equation (4) indicates the procedure for solving separable equations. A one-parameter family of solutions, usually given implicitly, is obtained by integrating both sides of $p(y)\,dy = g(x)\,dx$.

NOTE There is no need to use two constants in the integration of a separable equation, because if we write $H(y) + c_1 = G(x) + c_2$, then the difference $c_2 - c_1$ can be replaced by a single constant c, as in (4). In many instances throughout the chapters that follow, we will relabel constants in a manner convenient to a given equation. For example, multiples of constants or combinations of constants can sometimes be replaced by a single constant.

EXAMPLE 1 Solving a Separable DE

Solve $(1 + x)\,dy - y\,dx = 0$.

SOLUTION Dividing by $(1 + x)y$, we can write $dy/y = dx/(1 + x)$, from which it follows that

$$\int \frac{dy}{y} = \int \frac{dx}{1+x}$$

$$\ln|y| = \ln|1 + x| + c_1$$

$$y = e^{\ln|1+x|+c_1} = e^{\ln|1+x|} \cdot e^{c_1} \quad \leftarrow \text{laws of exponents}$$

$$= |1 + x|\,e^{c_1}$$

$$\leftarrow \begin{cases} |1 + x| = 1 + x, & x \geq -1 \\ |1 + x| = -(1 + x), & x < -1 \end{cases}$$

$$= \pm e^{c_1}(1 + x).$$

Relabeling $\pm e^{c_1}$ as c then gives $y = c(1 + x)$.

ALTERNATIVE SOLUTION Because each integral results in a logarithm, a judicious choice for the constant of integration is $\ln|c|$ rather than c. Rewriting the second line of the solution as $\ln|y| = \ln|1 + x| + \ln|c|$ enables us to combine the terms on the right-hand side by the properties of logarithms. From $\ln|y| = \ln|c(1 + x)|$ we immediately get $y = c(1 + x)$. Even if the indefinite integrals are not *all* logarithms, it may still be advantageous to use $\ln|c|$. However, no firm rule can be given. ■

EXAMPLE 2 Solution Curve

Solve the initial-value problem $\frac{dy}{dx} = -\frac{x}{y}, \quad y(4) = -3.$

SOLUTION Rewriting the equation as $y\,dy = -x\,dx$, we get

$$\int y\,dy = -\int x\,dx \qquad \text{and} \qquad \frac{y^2}{2} = -\frac{x^2}{2} + c_1.$$

We can write the result of the integration as $x^2 + y^2 = c^2$ by replacing the constant $2c_1$ by c^2. This solution of the differential equation represents a family of concentric circles centered at the origin.

Now when $x = 4$, $y = -3$, so $16 + 9 = 25 = c^2$. Thus the initial-value problem determines the circle $x^2 + y^2 = 25$ with radius 5. Because of its simplicity we can solve this implicit solution for an explicit solution that satisfies the initial condition.

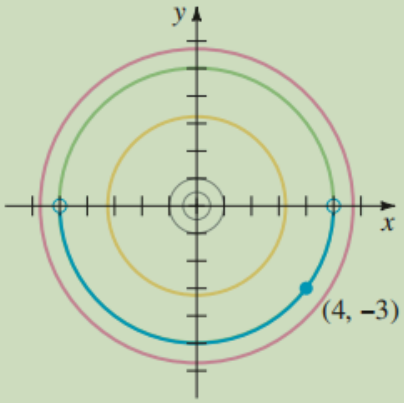

FIGURE 2.2.1 Solution curve for the IVP in Example 2

We saw this solution as $y = \phi_2(x)$ or $y = -\sqrt{25 - x^2}$, $-5 < x < 5$ in Example 3 of Section 1.1. A solution curve is the graph of a differentiable function. In this case the solution curve is the lower semicircle, shown in dark blue in Figure 2.2.1 containing the point $(4, -3)$. ■

LOSING A SOLUTION Some care should be exercised in separating variables, since the variable divisors could be zero at a point. Specifically, if r is a zero of the function $h(y)$, then substituting $y = r$ into $dy/dx = g(x)h(y)$ makes both sides zero; in other words, $y = r$ is a constant solution of the differential equation. But after variables are separated, the left-hand side of $\dfrac{dy}{h(y)} = g(x)\,dx$ is undefined at r. As a consequence, $y = r$ might not show up in the family of solutions that are obtained after integration and simplification. Recall that such a solution is called a singular solution.

EXAMPLE 3 Losing a Solution

Solve $\frac{dy}{dx} = y^2 - 4$.

SOLUTION We put the equation in the form

$$\frac{dy}{y^2 - 4} = dx \qquad \text{or} \qquad \left[\frac{\frac{1}{4}}{y - 2} - \frac{\frac{1}{4}}{y + 2}\right] dy = dx. \tag{5}$$

The second equation in (5) is the result of using partial fractions on the left-hand side of the first equation. Integrating and using the laws of logarithms gives

$$\frac{1}{4}\ln|y-2| - \frac{1}{4}\ln|y+2| = x + c_1$$

$$\text{or} \qquad \ln\left|\frac{y-2}{y+2}\right| = 4x + c_2 \qquad \text{or} \qquad \frac{y-2}{y+2} = \pm e^{4x+c_2}.$$

Here we have replaced $4c_1$ by c_2. Finally, after replacing $\pm e^{c_2}$ by c and solving the last equation for y, we get the one-parameter family of solutions

$$y = 2\frac{1 + ce^{4x}}{1 - ce^{4x}}. \tag{6}$$

Now if we factor the right-hand side of the differential equation as $dy/dx = (y-2)(y+2)$, we know from the discussion of critical points in Section 2.1 that $y = 2$ and $y = -2$ are two constant (equilibrium) solutions. The solution $y = 2$ is a member of the family of solutions defined by (6) corresponding to the value $c = 0$. However, $y = -2$ is a singular solution; it cannot be obtained from (6) for any choice of the parameter c. This latter solution was lost early on in the solution process. Inspection of (5) clearly indicates that we must preclude $y = \pm 2$ in these steps. ■

EXAMPLE 4 An Initial-Value Problem

Solve $(e^{2y} - y)\cos x \dfrac{dy}{dx} = e^y \sin 2x, \quad y(0) = 0.$

SOLUTION Dividing the equation by $e^y \cos x$ gives

$$\frac{e^{2y} - y}{e^y}\,dy = \frac{\sin 2x}{\cos x}\,dx.$$

Before integrating, we use termwise division on the left-hand side and the trigonometric identity $\sin 2x = 2 \sin x \cos x$ on the right-hand side. Then

integration by parts →
$$\int (e^y - ye^{-y})\,dy = 2\int \sin x\,dx$$

yields
$$e^y + ye^{-y} + e^{-y} = -2\cos x + c. \tag{7}$$

The initial condition $y = 0$ when $x = 0$ implies $c = 4$. Thus a solution of the initial-value problem is

$$e^y + ye^{-y} + e^{-y} = 4 - 2\cos x. \tag{8}$$ ■

WATCH:

Animated Short Story - Differential Equations

CalcGREEN 2 : Ch. 3.1 : Integrating the Linear ODE
CalcGREEN 2 : Ch. 3.2 : Affine ODEs and Integration
CalcGREEN 2 : Ch. 3.3 : Affine ODEs in Practice

Separation of Variables // Differential Equations

Newton's Law of Cooling // Separable ODE Example
The Big Theorem of Differential Equations: Existence & Uniqueness

Use the following online solver:
Calculator Ordinary Differential Equations (ODE) and Systems of ODEs
Solve differential equations online
Differential Equation Calculator - eMathHelp
Function Calculator - eMathHelp

PRACTISE: DENNIS G. ZILL: 2.2: Separable Variables: 1-14, 19-26

1. $\frac{dy}{dx} = \sin 5x$

2. $\frac{dy}{dx} = (x + 1)^2$

3. $dx + e^{3x}dy = 0$

4. $dy - (y - 1)^2 dx = 0$

5. $x\frac{dy}{dx} = 4y$

6. $\frac{dy}{dx} + 2xy^2 = 0$

7. $\frac{dy}{dx} = e^{3x+2y}$

8. $e^x y\frac{dy}{dx} = e^{-y} + e^{-2x-y}$

9. $y \ln x \frac{dx}{dy} = \left(\frac{y + 1}{x}\right)^2$

10. $\frac{dy}{dx} = \left(\frac{2y + 3}{4x + 5}\right)^2$

11. $\csc y\, dx + \sec^2 x\, dy = 0$

12. $\sin 3x\, dx + 2y \cos^3 3x\, dy = 0$

13. $(e^y + 1)^2 e^{-y}\, dx + (e^x + 1)^3 e^{-x}\, dy = 0$

14. $x(1 + y^2)^{1/2}\, dx = y(1 + x^2)^{1/2}\, dy$

19. $\dfrac{dy}{dx} = \dfrac{xy + 3x - y - 3}{xy - 2x + 4y - 8}$ **20.** $\dfrac{dy}{dx} = \dfrac{xy + 2y - x - 2}{xy - 3y + x - 3}$

21. $\dfrac{dy}{dx} = x\sqrt{1 - y^2}$ **22.** $(e^x + e^{-x})\dfrac{dy}{dx} = y^2$

In Problems 23–28 find an explicit solution of the given initial-value problem.

23. $\dfrac{dx}{dt} = 4(x^2 + 1), \quad x(\pi/4) = 1$

24. $\dfrac{dy}{dx} = \dfrac{y^2 - 1}{x^2 - 1}, \quad y(2) = 2$

25. $x^2\dfrac{dy}{dx} = y - xy, \quad y(-1) = -1$

26. $\dfrac{dy}{dt} + 2y = 1, \quad y(0) = \frac{5}{2}$

Linear Equations

READ:

First Order Linear Differential Equations

See more: The Bernoulli Differential Equation

READ: DENNIS G. ZILL: 2.3: Linear Equations

DEFINITION 2.3.1 Linear Equation

A first-order differential equation of the form

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \qquad (1)$$

is said to be a **linear equation** in the dependent variable y.

When $g(x) = 0$, the linear equation (1) is said to be **homogeneous;** otherwise, it is **nonhomogeneous.**

STANDARD FORM By dividing both sides of (1) by the lead coefficient $a_1(x)$, we obtain a more useful form, the **standard form,** of a linear equation:

$$\frac{dy}{dx} + P(x)y = f(x). \tag{2}$$

We seek a solution of (2) on an interval I for which both coefficient functions P and f are continuous.

In the discussion that follows we illustrate a property and a procedure and end up with a formula representing the form that every solution of (2) must have. But more than the formula, the property and the procedure are important, because these two concepts carry over to linear equations of higher order.

THE PROPERTY The differential equation (2) has the property that its solution is the **sum** of the two solutions: $y = y_c + y_p$, where y_c is a solution of the associated homogeneous equation

$$\frac{dy}{dx} + P(x)y = 0 \tag{3}$$

and y_p is a particular solution of the nonhomogeneous equation (2). To see this, observe that

$$\frac{d}{dx}[y_c + y_p] + P(x)[y_c + y_p] = \underbrace{\left[\frac{dy_c}{dx} + P(x)y_c\right]}_{0} + \underbrace{\left[\frac{dy_p}{dx} + P(x)y_p\right]}_{f(x)} = f(x).$$

Now the homogeneous equation (3) is also separable. This fact enables us to find y_c by writing (3) as

$$\frac{dy}{y} + P(x)\,dx = 0$$

and integrating. Solving for y gives $y_c = ce^{-\int P(x)dx}$. For convenience let us write $y_c = cy_1(x)$, where $y_1 = e^{-\int P(x)dx}$. The fact that $dy_1/dx + P(x)y_1 = 0$ will be used next to determine y_p.

THE PROCEDURE We can now find a particular solution of equation (2) by a procedure known as **variation of parameters.** The basic idea here is to find a function u so that $y_p = u(x)y_1(x) = u(x)e^{-\int P(x)\,dx}$ is a solution of (2). In other words, our assumption for y_p is the same as $y_c = cy_1(x)$ except that c is replaced by the "variable parameter" u. Substituting $y_p = uy_1$ into (2) gives

Product Rule

$$u\frac{dy_1}{dx} + y_1\frac{du}{dx} + P(x)uy_1 = f(x) \qquad \text{or} \qquad u\left[\frac{dy_1}{dx} + P(x)y_1\right] + y_1\frac{du}{dx} = f(x)$$

zero

so

$$y_1\frac{du}{dx} = f(x).$$

Separating variables and integrating then gives

$$du = \frac{f(x)}{y_1(x)}\,dx \qquad \text{and} \qquad u = \int \frac{f(x)}{y_1(x)}\,dx.$$

Since $y_1(x) = e^{-\int P(x)dx}$, we see that $1/y_1(x) = e^{\int P(x)dx}$. Therefore

$$y_p = uy_1 = \left(\int \frac{f(x)}{y_1(x)}\,dx\right)e^{-\int P(x)dx} = e^{-\int P(x)dx}\int e^{\int P(x)dx} f(x)\,dx,$$

and

$$y = \underbrace{ce^{-\int P(x)dx}}_{y_c} + \underbrace{e^{-\int P(x)dx}\int e^{\int P(x)dx} f(x)\,dx}_{y_p}. \tag{4}$$

Hence if (2) has a solution, it must be of form (4). Conversely, it is a straightforward exercise in differentiation to verify that (4) constitutes a one-parameter family of solutions of equation (2).

You should not memorize the formula given in (4). However, you should remember the special term

$$e^{\int P(x)dx} \tag{5}$$

because it is used in an equivalent but easier way of solving (2). If equation (4) is multiplied by (5),

$$e^{\int P(x)dx}y = c + \int e^{\int P(x)dx}f(x)\,dx, \tag{6}$$

and then (6) is differentiated,

$$\frac{d}{dx}\left[e^{\int P(x)dx}y\right] = e^{\int P(x)dx}f(x), \tag{7}$$

we get

$$e^{\int P(x)dx}\frac{dy}{dx} + P(x)e^{\int P(x)dx}y = e^{\int P(x)dx}f(x). \tag{8}$$

Dividing the last result by $e^{\int P(x)dx}$ gives (2).

METHOD OF SOLUTION The recommended method of solving (2) actually consists of (6)–(8) worked in reverse order. In other words, if (2) is multiplied by (5), we get (8). The left-hand side of (8) is recognized as the derivative of the product of $e^{\int P(x)dx}$ and y. This gets us to (7). We then integrate both sides of (7) to get the solution (6). Because we can solve (2) by integration after multiplication by $e^{\int P(x)dx}$, we call this function an **integrating factor** for the differential equation. For convenience we summarize these results. We again emphasize that you should not memorize formula (4) but work through the following procedure each time.

SOLVING A LINEAR FIRST-ORDER EQUATION

(*i*) Put a linear equation of form (1) into the standard form (2).

(*ii*) From the standard form identify $P(x)$ and then find the integrating factor $e^{\int P(x)dx}$.

(*iii*) Multiply the standard form of the equation by the integrating factor. The left-hand side of the resulting equation is automatically the derivative of the integrating factor and y:

$$\frac{d}{dx}\left[e^{\int P(x)dx}y\right] = e^{\int P(x)dx}f(x).$$

(*iv*) Integrate both sides of this last equation.

EXAMPLE 1 Solving a Homogeneous Linear DE

Solve $\dfrac{dy}{dx} - 3y = 0$.

SOLUTION This linear equation can be solved by separation of variables. Alternatively, since the equation is already in the standard form (2), we see that $P(x) = -3$, and so the integrating factor is $e^{\int(-3)dx} = e^{-3x}$. We multiply the equation by this factor and recognize that

$$e^{-3x}\frac{dy}{dx} - 3e^{-3x}y = 0 \qquad \text{is the same as} \qquad \frac{d}{dx}[e^{-3x}y] = 0.$$

Integrating both sides of the last equation gives $e^{-3x}y = c$. Solving for y gives us the explicit solution $y = ce^{3x}$, $-\infty < x < \infty$. ■

EXAMPLE 2 Solving a Nonhomogeneous Linear DE

Solve $\dfrac{dy}{dx} - 3y = 6.$

SOLUTION The associated homogeneous equation for this DE was solved in Example 1. Again the equation is already in the standard form (2), and the integrating factor is still $e^{\int(-3)dx} = e^{-3x}$. This time multiplying the given equation by this factor gives

$$e^{-3x}\frac{dy}{dx} - 3e^{-3x}y = 6e^{-3x}, \qquad \text{which is the same as} \qquad \frac{d}{dx}[e^{-3x}y] = 6e^{-3x}.$$

Integrating both sides of the last equation gives $e^{-3x}y = -2e^{-3x} + c$ or $y = -2 + ce^{3x}$, $-\infty < x < \infty$. ■

GENERAL SOLUTION Suppose again that the functions P and f in (2) are continuous on a common interval I. In the steps leading to (4) we showed that *if* (2) has a solution on I, then it must be of the form given in (4). Conversely, it is a straightforward exercise in differentiation to verify that any function of the form given in (4) is a solution of the differential equation (2) on I. In other words, (4) is a one-parameter family of solutions of equation (2) and *every solution of* (2) *defined on* I *is a member of this family.* Therefore we call (4) the **general solution** of the differential equation on the interval I. (See the *Remarks* at the end of Section 1.1.) Now by writing (2) in the normal form $y' = F(x, y)$, we can identify $F(x, y) = -P(x)y + f(x)$ and $\partial F/\partial y = -P(x)$. From the continuity of P and f on the interval I we see that F and $\partial F/\partial y$ are also continuous on I. With Theorem 1.2.1 as our justification, we conclude that there exists one and only one solution of the initial-value problem

$$\frac{dy}{dx} + P(x)y = f(x), \quad y(x_0) = y_0 \tag{9}$$

defined on *some* interval I_0 containing x_0. But when x_0 is in I, finding a solution of (9) is just a matter of finding an appropriate value of c in (4)—that is, to each x_0 in I there corresponds a distinct c. In other words, the interval I_0 of existence and uniqueness in Theorem 1.2.1 for the initial-value problem (9) is the entire interval I.

EXAMPLE 3 General Solution

Solve $x\dfrac{dy}{dx} - 4y = x^6e^x$.

SOLUTION Dividing by x, we get the standard form

$$\frac{dy}{dx} - \frac{4}{x}y = x^5e^x. \tag{10}$$

From this form we identify $P(x) = -4/x$ and $f(x) = x^5e^x$ and further observe that P and f are continuous on $(0, \infty)$. Hence the integrating factor is

$$e^{-4\int dx/x} = e^{-4\ln x} = e^{\ln x^{-4}} = x^{-4}.$$

we can use $\ln x$ instead of $\ln |x|$ since $x > 0$

Here we have used the basic identity $b^{\log_b N} = N, N > 0$. Now we multiply (10) by x^{-4} and rewrite

$$x^{-4}\frac{dy}{dx} - 4x^{-5}y = xe^x \qquad \text{as} \qquad \frac{d}{dx}[x^{-4}y] = xe^x.$$

It follows from integration by parts that the general solution defined on the interval $(0, \infty)$ is $x^{-4}y = xe^x - e^x + c$ or $y = x^5e^x - x^4e^x + cx^4$. ■

Except in the case in which the lead coefficient is 1, the recasting of equation (1) into the standard form (2) requires division by $a_1(x)$. Values of x for which $a_1(x) = 0$ are called **singular points** of the equation. Singular points are potentially troublesome. Specifically, in (2), if $P(x)$ (formed by dividing $a_0(x)$ by $a_1(x)$) is discontinuous at a point, the discontinuity may carry over to solutions of the differential equation.

EXAMPLE 4 General Solution

Find the general solution of $(x^2 - 9)\dfrac{dy}{dx} + xy = 0.$

SOLUTION We write the differential equation in standard form

$$\frac{dy}{dx} + \frac{x}{x^2 - 9}y = 0 \tag{11}$$

and identify $P(x) = x/(x^2 - 9)$. Although P is continuous on $(-\infty, -3)$, $(-3, 3)$, and $(3, \infty)$, we shall solve the equation on the first and third intervals. On these intervals the integrating factor is

$$e^{\int x\,dx/(x^2-9)} = e^{\frac{1}{2}\int 2x\,dx/(x^2-9)} = e^{\frac{1}{2}\ln|x^2-9|} = \sqrt{x^2 - 9}.$$

After multiplying the standard form (11) by this factor, we get

$$\frac{d}{dx}\left[\sqrt{x^2 - 9}\,y\right] = 0.$$

Integrating both sides of the last equation gives $\sqrt{x^2 - 9}\, y = c$. Thus for either $x > 3$ or $x < -3$ the general solution of the equation is $y = \dfrac{c}{\sqrt{x^2 - 9}}$. ■

Notice in Example 4 that $x = 3$ and $x = -3$ are singular points of the equation and that every function in the general solution $y = c/\sqrt{x^2 - 9}$ is discontinuous at these points. On the other hand, $x = 0$ is a singular point of the differential equation in Example 3, but the general solution $y = x^5e^x - x^4e^x + cx^4$ is noteworthy in that every function in this one-parameter family is continuous at $x = 0$ and is defined on the interval $(-\infty, \infty)$ and not just on $(0, \infty)$, as stated in the solution. However, the family $y = x^5e^x - x^4e^x + cx^4$ defined on $(-\infty, \infty)$ cannot be considered the general solution of the DE, since the singular point $x = 0$ still causes a problem. See Problem 39 in Exercises 2.3.

EXAMPLE 5 An Initial-Value Problem

Solve $\dfrac{dy}{dx} + y = x, \quad y(0) = 4.$

SOLUTION The equation is in standard form, and $P(x) = 1$ and $f(x) = x$ are continuous on $(-\infty, \infty)$. The integrating factor is $e^{\int dx} = e^x$, so integrating

$$\frac{d}{dx}[e^x y] = xe^x$$

gives $e^x y = xe^x - e^x + c$. Solving this last equation for y yields the general solution $y = x - 1 + ce^{-x}$. But from the initial condition we know that $y = 4$ when $x = 0$. Substituting these values into the general solution implies that $c = 5$. Hence the solution of the problem is

$$y = x - 1 + 5e^{-x}, \quad -\infty < x < \infty. \tag{12}$$ ■

WATCH:

- **CalcGREEN 2 : Ch. 1.5 : A Linear Differential Equation**
- **CalcGREEN 2 : Ch. 2.1 : Using the Linear ODE**

- **Linear Differential Equations & the Method of Integrating Factors**
- **The Method of Integrating Factors for Linear 1st Order ODEs **full example****
- **The Bernoulli Equation // Substitutions in Differential Equations**

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PRACTISE: DENNIS G. ZILL: 2.3: Linear Equations: 3-17,19,24,25,26,29,30

In Problems 1–24 find the general solution of the given differential equation. Give the largest interval I over which the general solution is defined. Determine whether there are any transient terms in the general solution.

1. $\frac{dy}{dx} = 5y$

2. $\frac{dy}{dx} + 2y = 0$

3. $\frac{dy}{dx} + y = e^{3x}$

4. $3\frac{dy}{dx} + 12y = 4$

5. $y' + 3x^2y = x^2$

6. $y' + 2xy = x^3$

7. $x^2y' + xy = 1$

8. $y' = 2y + x^2 + 5$

9. $x\frac{dy}{dx} - y = x^2 \sin x$

10. $x\frac{dy}{dx} + 2y = 3$

11. $x\frac{dy}{dx} + 4y = x^3 - x$

12. $(1 + x)\frac{dy}{dx} - xy = x + x^2$

13. $x^2y' + x(x + 2)y = e^x$

14. $xy' + (1 + x)y = e^{-x} \sin 2x$

15. $y\,dx - 4(x + y^6)\,dy = 0$

16. $y\,dx = (ye^y - 2x)\,dy$

17. $\cos x \dfrac{dy}{dx} + (\sin x)y = 1$

19. $(x + 1)\dfrac{dy}{dx} + (x + 2)y = 2xe^{-x}$

24. $(x^2 - 1)\dfrac{dy}{dx} + 2y = (x + 1)^2$

In Problems 25–30 solve the given initial-value problem. Give the largest interval I over which the solution is defined.

25. $xy' + y = e^x, \quad y(1) = 2$

26. $y\dfrac{dx}{dy} - x = 2y^2, \quad y(1) = 5$

29. $(x + 1)\dfrac{dy}{dx} + y = \ln x, \quad y(1) = 10$

30. $y' + (\tan x)y = \cos^2 x, \quad y(0) = -1$

- **Exact Equation, A non-Exact DE made Exact.**

Exact Equations

READ:
Exact Equations and Integrating Factors

READ: DENNIS G. ZILL: 2.4: Exact Equations

2.4 EXACT EQUATIONS

REVIEW MATERIAL

- Multivariate calculus
- Partial differentiation and partial integration
- Differential of a function of two variables

INTRODUCTION Although the simple first-order equation

$$y\,dx + x\,dy = 0$$

is separable, we can solve the equation in an alternative manner by recognizing that the expression on the left-hand side of the equality is the differential of the function $f(x, y) = xy$; that is,

$$d(xy) = y\,dx + x\,dy.$$

In this section we examine first-order equations in differential form $M(x, y)\,dx + N(x, y)\,dy = 0$. By applying a simple test to M and N, we can determine whether $M(x, y)\,dx + N(x, y)\,dy$ is a differential of a function $f(x, y)$. If the answer is yes, we can construct f by partial integration.

DIFFERENTIAL OF A FUNCTION OF TWO VARIABLES If $z = f(x, y)$ is a function of two variables with continuous first partial derivatives in a region R of the xy-plane, then its differential is

$$dz = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy. \tag{1}$$

In the special case when $f(x, y) = c$, where c is a constant, then (1) implies

$$\frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy = 0. \tag{2}$$

In other words, given a one-parameter family of functions $f(x, y) = c$, we can generate a first-order differential equation by computing the differential of both sides of the equality. For example, if $x^2 - 5xy + y^3 = c$, then (2) gives the first-order DE

$$(2x - 5y)\,dx + (-5x + 3y^2)\,dy = 0. \tag{3}$$

A DEFINITION Of course, not every first-order DE written in differential form $M(x, y)\,dx + N(x, y)\,dy = 0$ corresponds to a differential of $f(x, y) = c$. So for our purposes it is more important to turn the foregoing example around; namely, if we are given a first-order DE such as (3), is there some way we can recognize that the differential expression $(2x - 5y)\,dx + (-5x + 3y^2)\,dy$ is the differential $d(x^2 - 5xy + y^3)$? If there is, then an implicit solution of (3) is $x^2 - 5xy + y^3 = c$. We answer this question after the next definition.

DEFINITION 2.4.1 Exact Equation

A differential expression $M(x, y)\,dx + N(x, y)\,dy$ is an **exact differential** in a region R of the xy-plane if it corresponds to the differential of some function $f(x, y)$ defined in R. A first-order differential equation of the form

$$M(x, y)\,dx + N(x, y)\,dy = 0$$

is said to be an **exact equation** if the expression on the left-hand side is an exact differential.

For example, $x^2y^3\,dx + x^3y^2\,dy = 0$ is an exact equation, because its left-hand side is an exact differential:

$$d\left(\tfrac{1}{3}x^3y^3\right) = x^2y^3\,dx + x^3y^2\,dy.$$

Notice that if we make the identifications $M(x, y) = x^2y^3$ and $N(x, y) = x^3y^2$, then $\partial M/\partial y = 3x^2y^2 = \partial N/\partial x$. Theorem 2.4.1, given next, shows that the equality of the partial derivatives $\partial M/\partial y$ and $\partial N/\partial x$ is no coincidence.

THEOREM 2.4.1 Criterion for an Exact Differential

Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first partial derivatives in a rectangular region R defined by $a < x < b$, $c < y < d$. Then a necessary and sufficient condition that $M(x, y)\,dx + N(x, y)\,dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \tag{4}$$

PROOF OF THE NECESSITY For simplicity let us assume that $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives for all (x, y). Now if the expression $M(x, y)\,dx + N(x, y)\,dy$ is exact, there exists some function f such that for all x in R,

$$M(x, y)\,dx + N(x, y)\,dy = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy.$$

Therefore
$$M(x, y) = \frac{\partial f}{\partial x}, \qquad N(x, y) = \frac{\partial f}{\partial y},$$

and
$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial N}{\partial x}.$$

The equality of the mixed partials is a consequence of the continuity of the first partial derivatives of $M(x, y)$ and $N(x, y)$. ■

The sufficiency part of Theorem 2.4.1 consists of showing that there exists a function f for which $\partial f/\partial x = M(x, y)$ and $\partial f/\partial y = N(x, y)$ whenever (4) holds. The construction of the function f actually reflects a basic procedure for solving exact equations.

METHOD OF SOLUTION Given an equation in the differential form $M(x, y)\,dx + N(x, y)\,dy = 0$, determine whether the equality in (4) holds. If it does, then there exists a function f for which

$$\frac{\partial f}{\partial x} = M(x, y).$$

We can find f by integrating $M(x, y)$ with respect to x while holding y constant:

$$f(x, y) = \int M(x, y)\,dx + g(y), \tag{5}$$

where the arbitrary function $g(y)$ is the "constant" of integration. Now differentiate (5) with respect to y and assume that $\partial f/\partial y = N(x, y)$:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\int M(x, y)\,dx + g'(y) = N(x, y).$$

This gives
$$g'(y) = N(x, y) - \frac{\partial}{\partial y}\int M(x, y)\, dx. \tag{6}$$

Finally, integrate (6) with respect to y and substitute the result in (5). The implicit solution of the equation is $f(x, y) = c$.

Some observations are in order. First, it is important to realize that the expression $N(x, y) - (\partial/\partial y)\int M(x, y)\, dx$ in (6) is independent of x, because

$$\frac{\partial}{\partial x}\left[N(x, y) - \frac{\partial}{\partial y}\int M(x, y)\, dx\right] = \frac{\partial N}{\partial x} - \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}\int M(x, y)\, dx\right) = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0.$$

Second, we could just as well start the foregoing procedure with the assumption that $\partial f/\partial y = N(x, y)$. After integrating N with respect to y and then differentiating that result, we would find the analogues of (5) and (6) to be, respectively,

$$f(x, y) = \int N(x, y)\, dy + h(x) \qquad \text{and} \qquad h'(x) = M(x, y) - \frac{\partial}{\partial x}\int N(x, y)\, dy.$$

In either case *none of these formulas should be memorized.*

EXAMPLE 1 Solving an Exact DE

Solve $2xy\,dx + (x^2 - 1)\,dy = 0$.

SOLUTION With $M(x, y) = 2xy$ and $N(x, y) = x^2 - 1$ we have

$$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x}.$$

Thus the equation is exact, and so by Theorem 2.4.1 there exists a function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = 2xy \qquad \text{and} \qquad \frac{\partial f}{\partial y} = x^2 - 1.$$

From the first of these equations we obtain, after integrating,

$$f(x, y) = x^2y + g(y).$$

Taking the partial derivative of the last expression with respect to y and setting the result equal to $N(x, y)$ gives

$$\frac{\partial f}{\partial y} = x^2 + g'(y) = x^2 - 1. \quad \leftarrow N(x, y)$$

It follows that $g'(y) = -1$ and $g(y) = -y$. Hence $f(x, y) = x^2y - y$, so the solution of the equation in implicit form is $x^2y - y = c$. The explicit form of the solution is easily seen to be $y = c/(1 - x^2)$ and is defined on any interval not containing either $x = 1$ or $x = -1$. ■

NOTE The solution of the DE in Example 1 is *not* $f(x, y) = x^2y - y$. Rather, it is $f(x, y) = c$; if a constant is used in the integration of $g'(y)$, we can then write the solution as $f(x, y) = 0$. Note, too, that the equation could be solved by separation of variables.

EXAMPLE 2 Solving an Exact DE

Solve $(e^{2y} - y\cos xy)\,dx + (2xe^{2y} - x\cos xy + 2y)\,dy = 0$.

SOLUTION The equation is exact because

$$\frac{\partial M}{\partial y} = 2e^{2y} + xy\sin xy - \cos xy = \frac{\partial N}{\partial x}.$$

Hence a function $f(x, y)$ exists for which

$$M(x, y) = \frac{\partial f}{\partial x} \quad \text{and} \quad N(x, y) = \frac{\partial f}{\partial y}.$$

Now for variety we shall start with the assumption that $\partial f/\partial y = N(x, y)$; that is,

$$\frac{\partial f}{\partial y} = 2xe^{2y} - x\cos xy + 2y$$

$$f(x, y) = 2x\int e^{2y}\,dy - x\int \cos xy\,dy + 2\int y\,dy.$$

Remember, the reason x can come out in front of the symbol $\int$ is that in the integration with respect to y, x is treated as an ordinary constant. It follows that

$$f(x, y) = xe^{2y} - \sin xy + y^2 + h(x)$$

$$\frac{\partial f}{\partial x} = e^{2y} - y\cos xy + h'(x) = e^{2y} - y\cos xy, \quad \leftarrow M(x, y)$$

and so $h'(x) = 0$ or $h(x) = c$. Hence a family of solutions is

$$xe^{2y} - \sin xy + y^2 + c = 0. \qquad \blacksquare$$

EXAMPLE 3 An Initial-Value Problem

Solve $\dfrac{dy}{dx} = \dfrac{xy^2 - \cos x \sin x}{y(1 - x^2)}, \quad y(0) = 2.$

SOLUTION By writing the differential equation in the form

$$(\cos x \sin x - xy^2)\,dx + y(1 - x^2)\,dy = 0,$$

we recognize that the equation is exact because

$$\frac{\partial M}{\partial y} = -2xy = \frac{\partial N}{\partial x}.$$

Now

$$\frac{\partial f}{\partial y} = y(1 - x^2)$$

$$f(x, y) = \frac{y^2}{2}(1 - x^2) + h(x)$$

$$\frac{\partial f}{\partial x} = -xy^2 + h'(x) = \cos x \sin x - xy^2.$$

The last equation implies that $h'(x) = \cos x \sin x$. Integrating gives

$$h(x) = -\int (\cos x)(-\sin x\, dx) = -\frac{1}{2}\cos^2 x.$$

Thus $$\frac{y^2}{2}(1 - x^2) - \frac{1}{2}\cos^2 x = c_1 \quad \text{or} \quad y^2(1 - x^2) - \cos^2 x = c, \tag{7}$$

where $2c_1$ has been replaced by c. The initial condition $y = 2$ when $x = 0$ demands that $4(1) - \cos^2(0) = c$, and so $c = 3$. An implicit solution of the problem is then $y^2(1 - x^2) - \cos^2 x = 3$.

The solution curve of the IVP is the curve drawn in dark blue in Figure 2.4.1; it is part of an interesting family of curves. The graphs of the members of the one-parameter family of solutions given in (7) can be obtained in several ways, two of which are using software to graph level curves (as discussed in Section 2.2) and using a graphing utility to carefully graph the explicit functions obtained for various values of c by solving $y^2 = (c + \cos^2 x)/(1 - x^2)$ for y. ■

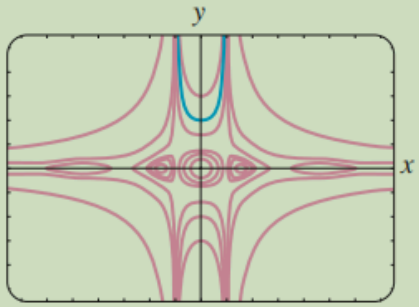


FIGURE 2.4.1 Some graphs of members of the family $y^2(1 - x^2) - \cos^2 x = c$

A non-Exact DE made Exact

INTEGRATING FACTORS Recall from Section 2.3 that the left-hand side of the linear equation $y' + P(x)y = f(x)$ can be transformed into a derivative when we multiply the equation by an integrating factor. The same basic idea sometimes works for a nonexact differential equation $M(x, y)\,dx + N(x, y)\,dy = 0$. That is, it is

sometimes possible to find an **integrating factor** $\mu(x, y)$ so that after multiplying, the left-hand side of

$$\mu(x, y)M(x, y)\,dx + \mu(x, y)N(x, y)\,dy = 0 \tag{8}$$

is an exact differential. In an attempt to find μ, we turn to the criterion (4) for exactness. Equation (8) is exact if and only if $(\mu M)_y = (\mu N)_x$, where the subscripts denote partial derivatives. By the Product Rule of differentiation the last equation is the same as $\mu M_y + \mu_y M = \mu N_x + \mu_x N$ or

$$\mu_x N - \mu_y M = (M_y - N_x)\mu. \tag{9}$$

Although M, N, M_y, and N_x are known functions of x and y, the difficulty here in determining the unknown $\mu(x, y)$ from (9) is that we must solve a partial differential equation. Since we are not prepared to do that, we make a simplifying assumption. Suppose μ is a function of one variable; for example, say that μ depends only on x. In this case, $\mu_x = d\mu/dx$ and $\mu_y = 0$, so (9) can be written as

$$\frac{d\mu}{dx} = \frac{M_y - N_x}{N}\mu. \tag{10}$$

We are still at an impasse if the quotient $(M_y - N_x)/N$ depends on both x and y. However, if after all obvious algebraic simplifications are made, the quotient $(M_y - N_x)/N$ turns out to depend solely on the variable x, then (10) is a first-order ordinary differential equation. We can finally determine μ because (10) is *separable* as well as *linear*. It follows from either Section 2.2 or Section 2.3 that $\mu(x) = e^{\int((M_y - N_x)/N)\,dx}$. In like manner, it follows from (9) that if μ depends only on the variable y, then

$$\frac{d\mu}{dy} = \frac{N_x - M_y}{M}\,\mu. \tag{11}$$

In this case, if $(N_x - M_y)/M$ is a function of y only, then we can solve (11) for μ.

We summarize the results for the differential equation

$$M(x, y)\,dx + N(x, y)\,dy = 0. \tag{12}$$

- If $(M_y - N_x)/N$ is a function of x alone, then an integrating factor for (12) is

$$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}. \tag{13}$$

- If $(N_x - M_y)/M$ is a function of y alone, then an integrating factor for (12) is

$$\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}. \tag{14}$$

EXAMPLE 4 A Nonexact DE Made Exact

The nonlinear first-order differential equation

$$xy\,dx + (2x^2 + 3y^2 - 20)\,dy = 0$$

is not exact. With the identifications $M = xy, N = 2x^2 + 3y^2 - 20$, we find the partial derivatives $M_y = x$ and $N_x = 4x$. The first quotient from (13) gets us nowhere, since

$$\frac{M_y - N_x}{N} = \frac{x - 4x}{2x^2 + 3y^2 - 20} = \frac{-3x}{2x^2 + 3y^2 - 20}$$

depends on x and y. However, (14) yields a quotient that depends only on y:

$$\frac{N_x - M_y}{M} = \frac{4x - x}{xy} = \frac{3x}{xy} = \frac{3}{y}.$$

The integrating factor is then $e^{\int 3dy/y} = e^{3\ln y} = e^{\ln y^3} = y^3$. After we multiply the given DE by $\mu(y) = y^3$, the resulting equation is

$$xy^4\,dx + (2x^2y^3 + 3y^5 - 20y^3)\,dy = 0.$$

You should verify that the last equation is now exact as well as show, using the method of this section, that a family of solutions is $\frac{1}{2}x^2y^4 + \frac{1}{2}y^6 - 5y^4 = c$. ■

REMARKS

(*i*) When testing an equation for exactness, make sure it is of the precise form $M(x, y)\,dx + N(x, y)\,dy = 0$. Sometimes a differential equation is written $G(x, y)\,dx = H(x, y)\,dy$. In this case, first rewrite it as $G(x, y)\,dx - H(x, y)\,dy = 0$ and then identify $M(x, y) = G(x, y)$ and $N(x, y) = -H(x, y)$ before using (4).

(*ii*) In some texts on differential equations the study of exact equations precedes that of linear DEs. Then the method for finding integrating factors just discussed can be used to derive an integrating factor for $y' + P(x)y = f(x)$. By rewriting the last equation in the differential form $(P(x)y - f(x))\,dx + dy = 0$, we see that

$$\frac{M_y - N_x}{N} = P(x).$$

From (13) we arrive at the already familiar integrating factor $e^{\int P(x)\,dx}$, used in Section 2.3.

WATCH:

Theory+Technique with example:

- **Exact differential equation (introduction & example)**
- **EXACT Equations (how to check for exactness!) 3 examples**
- **HOW TO CREATE YOUR OWN EXACT EQUATIONS?**

Solved Problems:

- **Exact Differential Equation, (2.4#11)**
- **Exact Differential Equation, 2.4#13**
- **Exact Differential Equation, 2.4#19**
- **Exact Differential Equation, 2.4#25**
- **Exact Differential Equation, 2.4#27a**
- **Exact Differential Equation, 2.4#28a**

Theory+Technique with example:

- **Almost Exact Differential equation & special integrating factor (introduction & example)**
- **Special integrating factor for almost exact equation, mu(y)**

Solved Problems:

- **Find a special integrating factor mu(y) for an almost-exact differential equation**
- **Almost-exact differential equation with integrating factor**
- **special integrating factor in y, sect2.5#7**
- **special integrating factor of the form x^n*y^m, sect2.5#13**
- **special integrating factor of the form x^n*y^m, sect2.5#14**

Use the following online solver:

Exact DE Solver – GeoGebra

PRACTISE: DENNIS G. ZILL: 2.4: Exact Equations: 1-20, 21-26

In Problems 1–20 determine whether the given differential equation is exact. If it is exact, solve it.

1. $(2x - 1)\,dx + (3y + 7)\,dy = 0$

2. $(2x + y)\,dx - (x + 6y)\,dy = 0$

3. $(5x + 4y)\,dx + (4x - 8y^3)\,dy = 0$

4. $(\sin y - y \sin x)\,dx + (\cos x + x \cos y - y)\,dy = 0$

5. $(2xy^2 - 3)\,dx + (2x^2y + 4)\,dy = 0$

6. $\left(2y - \frac{1}{x} + \cos 3x\right)\frac{dy}{dx} + \frac{y}{x^2} - 4x^3 + 3y \sin 3x = 0$

7. $(x^2 - y^2)\,dx + (x^2 - 2xy)\,dy = 0$

8. $\left(1 + \ln x + \frac{y}{x}\right)dx = (1 - \ln x)\,dy$

9. $(x - y^3 + y^2 \sin x)\,dx = (3xy^2 + 2y \cos x)\,dy$

10. $(x^3 + y^3)\,dx + 3xy^2\,dy = 0$

11. $(y \ln y - e^{-xy})\,dx + \left(\frac{1}{y} + x \ln y\right)dy = 0$

12. $(3x^2y + e^y)\,dx + (x^3 + xe^y - 2y)\,dy = 0$

13. $x\dfrac{dy}{dx} = 2xe^x - y + 6x^2$

14. $\left(1 - \dfrac{3}{y} + x\right)\dfrac{dy}{dx} + y = \dfrac{3}{x} - 1$

15. $\left(x^2y^3 - \dfrac{1}{1 + 9x^2}\right)\dfrac{dx}{dy} + x^3y^2 = 0$

16. $(5y - 2x)y' - 2y = 0$

17. $(\tan x - \sin x \sin y)\,dx + \cos x \cos y\,dy = 0$

18. $(2y \sin x \cos x - y + 2y^2e^{xy^2})\,dx$
$= (x - \sin^2 x - 4xye^{xy^2})\,dy$

19. $(4t^3y - 15t^2 - y)\,dt + (t^4 + 3y^2 - t)\,dy = 0$

20. $\left(\dfrac{1}{t} + \dfrac{1}{t^2} - \dfrac{y}{t^2 + y^2}\right)dt + \left(ye^y + \dfrac{t}{t^2 + y^2}\right)dy = 0$

In Problems 21–26 solve the given initial-value problem.

21. $(x + y)^2\,dx + (2xy + x^2 - 1)\,dy = 0, \quad y(1) = 1$

22. $(e^x + y)\,dx + (2 + x + ye^y)\,dy = 0, \quad y(0) = 1$

23. $(4y + 2t - 5)\,dt + (6y + 4t - 1)\,dy = 0, \quad y(-1) = 2$

24. $\left(\dfrac{3y^2 - t^2}{y^5}\right)\dfrac{dy}{dt} + \dfrac{t}{2y^4} = 0, \quad y(1) = 1$

25. $(y^2 \cos x - 3x^2y - 2x)\,dx$
$+ (2y \sin x - x^3 + \ln y)\,dy = 0, \quad y(0) = e$

26. $\left(\dfrac{1}{1 + y^2} + \cos x - 2xy\right)\dfrac{dy}{dx} = y(y + \sin x), \quad y(0) = 1$

- **Higher order DE, homogeneous and inhomogeneous cases.**

Homogeneous linear equations with constant coefficients

READ:

Second Order Differential Equations

READ: DENNIS G. ZILL: 4.3: Homogeneous linear equations with constant coefficients

4.3 HOMOGENEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

REVIEW MATERIAL

- Review Problem 27 in Exercises 1.1 and Theorem 4.1.5
- Review the algebra of solving polynomial equations (see the *Student Resource and Solutions Manual*)

INTRODUCTION As a means of motivating the discussion in this section, let us return to first-order differential equations—more specifically, to *homogeneous* linear equations $ay' + by = 0$, where the coefficients $a \neq 0$ and b are constants. This type of equation can be solved either by separation of variables or with the aid of an integrating factor, but there is another solution method, one that uses only algebra. Before illustrating this alternative method, we make one observation: Solving $ay' + by = 0$ for y' yields $y' = ky$, where k is a constant. This observation reveals the nature of the unknown solution y; the only nontrivial elementary function whose derivative is a constant multiple of itself is an exponential function e^{mx}. Now the new solution method: If we substitute $y = e^{mx}$ and $y' = me^{mx}$ into $ay' + by = 0$, we get

$$ame^{mx} + be^{mx} = 0 \quad \text{or} \quad e^{mx}(am + b) = 0.$$

Since e^{mx} is never zero for real values of x, the last equation is satisfied only when m is a solution or root of the first-degree polynomial equation $am + b = 0$. For this single value of m, $y = e^{mx}$ is a solution of the DE. To illustrate, consider the constant-coefficient equation $2y' + 5y = 0$. It is not necessary to go through the differentiation and substitution of $y = e^{mx}$ into the DE; we merely have to form the equation $2m + 5 = 0$ and solve it for m. From $m = -\frac{5}{2}$ we conclude that $y = e^{-5x/2}$ is a solution of $2y' + 5y = 0$, and its general solution on the interval $(-\infty, \infty)$ is $y = c_1 e^{-5x/2}$.

In this section we will see that the foregoing procedure can produce exponential solutions for homogeneous linear higher-order DEs,

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0, \tag{1}$$

where the coefficients a_i, $i = 0, 1, \ldots, n$ are real constants and $a_n \neq 0$.

AUXILIARY EQUATION We begin by considering the special case of the second-order equation

$$ay'' + by' + cy = 0, \tag{2}$$

where a, b, and c are constants. If we try to find a solution of the form $y = e^{mx}$, then after substitution of $y' = me^{mx}$ and $y'' = m^2e^{mx}$, equation (2) becomes

$$am^2e^{mx} + bme^{mx} + ce^{mx} = 0 \qquad \text{or} \qquad e^{mx}(am^2 + bm + c) = 0.$$

As in the introduction we argue that because $e^{mx} \neq 0$ for all x, it is apparent that the only way $y = e^{mx}$ can satisfy the differential equation (2) is when m is chosen as a root of the quadratic equation

$$am^2 + bm + c = 0. \tag{3}$$

This last equation is called the **auxiliary equation** of the differential equation (2). Since the two roots of (3) are $m_1 = (-b + \sqrt{b^2 - 4ac})/2a$ and $m_2 = (-b - \sqrt{b^2 - 4ac})/2a$, there will be three forms of the general solution of (2) corresponding to the three cases:

- m_1 and m_2 real and distinct ($b^2 - 4ac > 0$),
- m_1 and m_2 real and equal ($b^2 - 4ac = 0$), and
- m_1 and m_2 conjugate complex numbers ($b^2 - 4ac < 0$).

We discuss each of these cases in turn.

CASE I: DISTINCT REAL ROOTS Under the assumption that the auxiliary equation (3) has two unequal real roots m_1 and m_2, we find two solutions, $y_1 = e^{m_1 x}$ and $y_2 = e^{m_2 x}$. We see that these functions are linearly independent on $(-\infty, \infty)$ and hence form a fundamental set. It follows that the general solution of (2) on this interval is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}. \tag{4}$$

CASE II: REPEATED REAL ROOTS When $m_1 = m_2$, we necessarily obtain only one exponential solution, $y_1 = e^{m_1 x}$. From the quadratic formula we find that $m_1 = -b/2a$ since the only way to have $m_1 = m_2$ is to have $b^2 - 4ac = 0$. It follows from (5) in Section 4.2 that a second solution of the equation is

$$y_2 = e^{m_1 x} \int \frac{e^{2m_1 x}}{e^{2m_1 x}}\, dx = e^{m_1 x} \int dx = x e^{m_1 x}. \tag{5}$$

In (5) we have used the fact that $-b/a = 2m_1$. The general solution is then

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}. \tag{6}$$

CASE III: CONJUGATE COMPLEX ROOTS If m_1 and m_2 are complex, then we can write $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Formally, there is no difference between this case and Case I, and hence

$$y = C_1 e^{(\alpha+i\beta)x} + C_2 e^{(\alpha-i\beta)x}.$$

However, in practice we prefer to work with real functions instead of complex exponentials. To this end we use **Euler's formula:**

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

where θ is any real number.* It follows from this formula that

$$e^{i\beta x} = \cos\beta x + i\sin\beta x \qquad \text{and} \qquad e^{-i\beta x} = \cos\beta x - i\sin\beta x, \tag{7}$$

*A formal derivation of Euler's formula can be obtained from the Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ by substituting $x = i\theta$, using $i^2 = -1$, $i^3 = -i$, . . . , and then separating the series into real and imaginary parts. The plausibility thus established, we can adopt $\cos\theta + i\sin\theta$ as the *definition* of $e^{i\theta}$.

where we have used $\cos(-\beta x) = \cos \beta x$ and $\sin(-\beta x) = -\sin \beta x$. Note that by first adding and then subtracting the two equations in (7), we obtain, respectively,

$$e^{i\beta x} + e^{-i\beta x} = 2\cos \beta x \qquad \text{and} \qquad e^{i\beta x} - e^{-i\beta x} = 2i \sin \beta x.$$

Since $y = C_1 e^{(\alpha+i\beta)x} + C_2 e^{(\alpha-i\beta)x}$ is a solution of (2) for any choice of the constants C_1 and C_2, the choices $C_1 = C_2 = 1$ and $C_1 = 1$, $C_2 = -1$ give, in turn, two solutions:

$$y_1 = e^{(\alpha+i\beta)x} + e^{(\alpha-i\beta)x} \qquad \text{and} \qquad y_2 = e^{(\alpha+i\beta)x} - e^{(\alpha-i\beta)x}.$$

But
$$y_1 = e^{\alpha x}(e^{i\beta x} + e^{-i\beta x}) = 2e^{\alpha x} \cos \beta x$$

and
$$y_2 = e^{\alpha x}(e^{i\beta x} - e^{-i\beta x}) = 2ie^{\alpha x} \sin \beta x.$$

Hence from Corollary (A) of Theorem 4.1.2 the last two results show that $e^{\alpha x} \cos \beta x$ and $e^{\alpha x} \sin \beta x$ are *real* solutions of (2). Moreover, these solutions form a fundamental set on $(-\infty, \infty)$. Consequently, the general solution is

$$y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x). \tag{8}$$

EXAMPLE 1 Second-Order DEs

Solve the following differential equations.

(a) $2y'' - 5y' - 3y = 0$ **(b)** $y'' - 10y' + 25y = 0$ **(c)** $y'' + 4y' + 7y = 0$

SOLUTION We give the auxiliary equations, the roots, and the corresponding general solutions.

(a) $2m^2 - 5m - 3 = (2m + 1)(m - 3) = 0, \quad m_1 = -\frac{1}{2}, m_2 = 3$

From (4), $y = c_1e^{-x/2} + c_2e^{3x}$.

(b) $m^2 - 10m + 25 = (m - 5)^2 = 0, \quad m_1 = m_2 = 5$

From (6), $y = c_1e^{5x} + c_2xe^{5x}$.

(c) $m^2 + 4m + 7 = 0, \; m_1 = -2 + \sqrt{3}i, \quad m_2 = -2 - \sqrt{3}i$

From (8) with $\alpha = -2, \beta = \sqrt{3}$, $y = e^{-2x}\left(c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x\right)$. ■

EXAMPLE 2 An Initial-Value Problem

Solve $4y'' + 4y' + 17y = 0$, $y(0) = -1$, $y'(0) = 2$.

SOLUTION By the quadratic formula we find that the roots of the auxiliary equation $4m^2 + 4m + 17 = 0$ are $m_1 = -\frac{1}{2} + 2i$ and $m_2 = -\frac{1}{2} - 2i$. Thus from (8) we have $y = e^{-x/2}(c_1 \cos 2x + c_2 \sin 2x)$. Applying the condition $y(0) = -1$, we see from $e^0(c_1 \cos 0 + c_2 \sin 0) = -1$ that $c_1 = -1$. Differentiating $y = e^{-x/2}(-\cos 2x + c_2 \sin 2x)$ and then using $y'(0) = 2$ gives $2c_2 + \frac{1}{2} = 2$ or $c_2 = \frac{3}{4}$. Hence the solution of the IVP is $y = e^{-x/2}(-\cos 2x + \frac{3}{4} \sin 2x)$. In Figure 4.3.1 we see that the solution is oscillatory, but $y \to 0$ as $x \to \infty$ and $|y| \to \infty$ as $x \to -\infty$. ■

HIGHER-ORDER EQUATIONS In general, to solve an nth-order differential equation (1), where the a_i, $i = 0, 1, \ldots, n$ are real constants, we must solve an nth-degree polynomial equation

$$a_n m^n + a_{n-1} m^{n-1} + \cdots + a_2 m^2 + a_1 m + a_0 = 0. \tag{12}$$

If all the roots of (12) are real and distinct, then the general solution of (1) is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \cdots + c_n e^{m_n x}.$$

It is somewhat harder to summarize the analogues of Cases II and III because the roots of an auxiliary equation of degree greater than two can occur in many combinations. For example, a fifth-degree equation could have five distinct real roots, or three distinct real and two complex roots, or one real and four complex roots, or five real but equal roots, or five real roots but two of them equal, and so on. When m_1 is a root of multiplicity k of an nth-degree auxiliary equation (that is, k roots are equal to m_1), it can be shown that the linearly independent solutions are

$$e^{m_1 x}, \quad x e^{m_1 x}, \quad x^2 e^{m_1 x}, \ldots, \quad x^{k-1} e^{m_1 x}$$

and the general solution must contain the linear combination

$$c_1e^{m_1x} + c_2xe^{m_1x} + c_3x^2e^{m_1x} + \cdots + c_kx^{k-1}e^{m_1x}.$$

Finally, it should be remembered that when the coefficients are real, complex roots of an auxiliary equation always appear in conjugate pairs. Thus, for example, a cubic polynomial equation can have at most two complex roots.

EXAMPLE 3 Third-Order DE

Solve $y''' + 3y'' - 4y = 0$.

SOLUTION It should be apparent from inspection of $m^3 + 3m^2 - 4 = 0$ that one root is $m_1 = 1$, so $m - 1$ is a factor of $m^3 + 3m^2 - 4$. By division we find

$$m^3 + 3m^2 - 4 = (m - 1)(m^2 + 4m + 4) = (m - 1)(m + 2)^2,$$

so the other roots are $m_2 = m_3 = -2$. Thus the general solution of the DE is $y = c_1e^x + c_2e^{-2x} + c_3xe^{-2x}$. ■

EXAMPLE 4 Fourth-Order DE

Solve $\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0.$

SOLUTION The auxiliary equation $m^4 + 2m^2 + 1 = (m^2 + 1)^2 = 0$ has roots $m_1 = m_3 = i$ and $m_2 = m_4 = -i$. Thus from Case II the solution is

$$y = C_1e^{ix} + C_2e^{-ix} + C_3xe^{ix} + C_4xe^{-ix}.$$

By Euler's formula the grouping $C_1e^{ix} + C_2e^{-ix}$ can be rewritten as

$$c_1 \cos x + c_2 \sin x$$

after a relabeling of constants. Similarly, $x(C_3e^{ix} + C_4e^{-ix})$ can be expressed as $x(c_3 \cos x + c_4 \sin x)$. Hence the general solution is

$$y = c_1 \cos x + c_2 \sin x + c_3x \cos x + c_4x \sin x.$$ ■

WATCH:

- **The Theory of 2nd Order ODEs // Existence & Uniqueness, Superposition, & Linear Independence**
- **How to Solve Constant Coefficient Homogeneous Differential Equations**
- **Constant Coefficient ODEs: Real & Distinct vs Real & Repeated vs Complex Pair**
- **Higher Order Constant Coefficient Differential Equations: y'''+y'=0 and y''''-3y'''+3y''-y'=0**

Use the following online solver:

[Calculator Ordinary Differential Equations (ODE) and Systems of ODEs]
[Solve differential equations online]
[Differential Equation Calculator - eMathHelp]
[Function Calculator - eMathHelp]

PRACTISE: DENNIS G. ZILL: 4.3: Homogeneous linear equations with constant coefficients: 1-23, 25, 29-34, 35, 37, 39

In Problems 1–14 find the general solution of the given second-order differential equation.

1. $4y'' + y' = 0$

2. $y'' - 36y = 0$

3. $y'' - y' - 6y = 0$

4. $y'' - 3y' + 2y = 0$

5. $y'' + 8y' + 16y = 0$

6. $y'' - 10y' + 25y = 0$

7. $12y'' - 5y' - 2y = 0$

8. $y'' + 4y' - y = 0$

9. $y'' + 9y = 0$

10. $3y'' + y = 0$

11. $y'' - 4y' + 5y = 0$

12. $2y'' + 2y' + y = 0$

13. $3y'' + 2y' + y = 0$

14. $2y'' - 3y' + 4y = 0$

In Problems 15–28 find the general solution of the given higher-order differential equation.

15. $y''' - 4y'' - 5y' = 0$

16. $y''' - y = 0$

17. $y''' - 5y'' + 3y' + 9y = 0$

18. $y''' + 3y'' - 4y' - 12y = 0$

19. $\dfrac{d^3u}{dt^3} + \dfrac{d^2u}{dt^2} - 2u = 0$

20. $\dfrac{d^3x}{dt^3} - \dfrac{d^2x}{dt^2} - 4x = 0$

21. $y''' + 3y'' + 3y' + y = 0$

22. $y''' - 6y'' + 12y' - 8y = 0$

23. $y^{(4)} + y''' + y'' = 0$

25. $16\dfrac{d^4y}{dx^4} + 24\dfrac{d^2y}{dx^2} + 9y = 0$

In Problems 29–36 solve the given initial-value problem.

29. $y'' + 16y = 0, \quad y(0) = 2, y'(0) = -2$

30. $\dfrac{d^2y}{d\theta^2} + y = 0, \quad y\left(\dfrac{\pi}{3}\right) = 0, y'\left(\dfrac{\pi}{3}\right) = 2$

31. $\dfrac{d^2y}{dt^2} - 4\dfrac{dy}{dt} - 5y = 0, \quad y(1) = 0, y'(1) = 2$

32. $4y'' - 4y' - 3y = 0, \quad y(0) = 1, y'(0) = 5$

33. $y'' + y' + 2y = 0, \quad y(0) = y'(0) = 0$

34. $y'' - 2y' + y = 0, \quad y(0) = 5, y'(0) = 10$

35. $y''' + 12y'' + 36y' = 0, \quad y(0) = 0, y'(0) = 1, y''(0) = -7$

37. $y'' - 10y' + 25y = 0, \quad y(0) = 1, y(1) = 0$

39. $y'' + y = 0, \quad y'(0) = 0, y'\left(\dfrac{\pi}{2}\right) = 0$

Undetermined Coefficients-Superposition Approach

READ:
Method of Undetermined Coefficients

READ: DENNIS G. ZILL: 4.4: Undetermined Coefficients-Superposition Approach

4.4 UNDETERMINED COEFFICIENTS—SUPERPOSITION APPROACH*

REVIEW MATERIAL

- Review Theorems 4.1.6 and 4.1.7 (Section 4.1)

INTRODUCTION To solve a nonhomogeneous linear differential equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x), \tag{1}$$

we must do two things:

- find the complementary function y_c and
- find *any* particular solution y_p of the nonhomogeneous equation (1).

Then, as was discussed in Section 4.1, the general solution of (1) is $y = y_c + y_p$. The complementary function y_c is the general solution of the associated homogeneous DE of (1), that is,

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0.$$

In Section 4.3 we saw how to solve these kinds of equations when the coefficients were constants. Our goal in the present section is to develop a method for obtaining particular solutions.

METHOD OF UNDETERMINED COEFFICIENTS The first of two ways we shall consider for obtaining a particular solution y_p for a nonhomogeneous linear DE is called the **method of undetermined coefficients.** The underlying idea behind this method is a conjecture about the form of y_p, an educated guess really, that is motivated by the kinds of functions that make up the input function $g(x)$. The general method is limited to linear DEs such as (1) where

- the coefficients a_i, $i = 0, 1, \ldots, n$ are constants and
- $g(x)$ is a constant k, a polynomial function, an exponential function $e^{\alpha x}$, a sine or cosine function $\sin \beta x$ or $\cos \beta x$, or finite sums and products of these functions.

NOTE Strictly speaking, $g(x) = k$ (constant) is a polynomial function. Since a constant function is probably not the first thing that comes to mind when you think of polynomial functions, for emphasis we shall continue to use the redundancy "constant functions, polynomials, ... "

The following functions are some examples of the types of inputs $g(x)$ that are appropriate for this discussion:

$$g(x) = 10, \quad g(x) = x^2 - 5x, \qquad g(x) = 15x - 6 + 8e^{-x},$$
$$g(x) = \sin 3x - 5x\cos 2x, \qquad g(x) = xe^x \sin x + (3x^2 - 1)e^{-4x}.$$

That is, $g(x)$ is a linear combination of functions of the type

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad P(x)\, e^{\alpha x}, \quad P(x)\, e^{\alpha x} \sin \beta x, \quad \text{and} \quad P(x)\, e^{\alpha x} \cos \beta x,$$

where n is a nonnegative integer and α and β are real numbers. The method of undetermined coefficients is not applicable to equations of form (1) when

$$g(x) = \ln x, \quad g(x) = \frac{1}{x}, \quad g(x) = \tan x, \quad g(x) = \sin^{-1} x,$$

and so on. Differential equations in which the input $g(x)$ is a function of this last kind will be considered in Section 4.6.

The set of functions that consists of constants, polynomials, exponentials $e^{\alpha x}$, sines, and cosines has the remarkable property that derivatives of their sums and products are again sums and products of constants, polynomials, exponentials $e^{\alpha x}$, sines, and cosines. Because the linear combination of derivatives $a_n y_p^{(n)} + a_{n-1} y_p^{(n-1)} + \cdots + a_1 y_p' + a_0 y_p$ must be identical to $g(x)$, it seems reasonable to assume that y_p *has the same form as* $g(x)$.

The next two examples illustrate the basic method.

EXAMPLE 1 General Solution Using Undetermined Coefficients

Solve $y'' + 4y' - 2y = 2x^2 - 3x + 6.$ (2)

SOLUTION **Step 1.** We first solve the associated homogeneous equation $y'' + 4y' - 2y = 0$. From the quadratic formula we find that the roots of the auxiliary equation $m^2 + 4m - 2 = 0$ are $m_1 = -2 - \sqrt{6}$ and $m_2 = -2 + \sqrt{6}$. Hence the complementary function is

$$y_c = c_1 e^{-(2+\sqrt{6})x} + c_2 e^{(-2+\sqrt{6})x}.$$

Step 2. Now, because the function $g(x)$ is a quadratic polynomial, let us assume a particular solution that is also in the form of a quadratic polynomial:

$$y_p = Ax^2 + Bx + C.$$

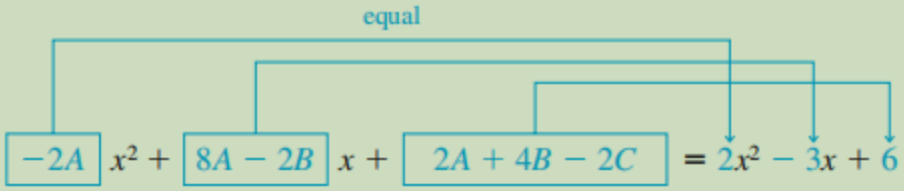
We seek to determine *specific* coefficients A, B, and C for which y_p is a solution of (2). Substituting y_p and the derivatives

$$y_p' = 2Ax + B \qquad \text{and} \qquad y_p'' = 2A$$

into the given differential equation (2), we get

$$y_p'' + 4y_p' - 2y_p = 2A + 8Ax + 4B - 2Ax^2 - 2Bx - 2C = 2x^2 - 3x + 6.$$

Because the last equation is supposed to be an identity, the coefficients of like powers of x must be equal:



That is, $\quad -2A = 2, \quad 8A - 2B = -3, \quad 2A + 4B - 2C = 6.$

Solving this system of equations leads to the values $A = -1$, $B = -\frac{5}{2}$, and $C = -9$. Thus a particular solution is

$$y_p = -x^2 - \frac{5}{2}x - 9.$$

Step 3. The general solution of the given equation is

$$y = y_c + y_p = c_1 e^{-(2+\sqrt{6})x} + c_1 e^{(-2+\sqrt{6})x} - x^2 - \frac{5}{2}x - 9. \qquad \blacksquare$$

EXAMPLE 2 Particular Solution Using Undetermined Coefficients

Find a particular solution of $y'' - y' + y = 2 \sin 3x$.

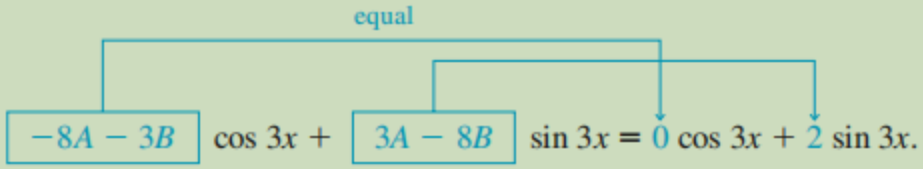
SOLUTION A natural first guess for a particular solution would be $A \sin 3x$. But because successive differentiations of $\sin 3x$ produce $\sin 3x$ *and* $\cos 3x$, we are prompted instead to assume a particular solution that includes both of these terms:

$$y_p = A \cos 3x + B \sin 3x.$$

Differentiating y_p and substituting the results into the differential equation gives, after regrouping,

$$y_p'' - y_p' + y_p = (-8A - 3B) \cos 3x + (3A - 8B) \sin 3x = 2 \sin 3x$$

or



From the resulting system of equations,

$$-8A - 3B = 0, \qquad 3A - 8B = 2,$$

we get $A = \frac{6}{73}$ and $B = -\frac{16}{73}$. A particular solution of the equation is

$$y_p = \frac{6}{73} \cos 3x - \frac{16}{73} \sin 3x.$$

■

As we mentioned, the form that we assume for the particular solution y_p is an educated guess; it is not a blind guess. This educated guess must take into consideration not only the types of functions that make up $g(x)$ but also, as we shall see in Example 4, the functions that make up the complementary function y_c.

EXAMPLE 3 Forming y_p by Superposition

Solve $y'' - 2y' - 3y = 4x - 5 + 6xe^{2x}$. (3)

SOLUTION **Step 1.** First, the solution of the associated homogeneous equation $y'' - 2y' - 3y = 0$ is found to be $y_c = c_1e^{-x} + c_2e^{3x}$.

Step 2. Next, the presence of $4x - 5$ in $g(x)$ suggests that the particular solution includes a linear polynomial. Furthermore, because the derivative of the product xe^{2x} produces $2xe^{2x}$ and e^{2x}, we also assume that the particular solution includes both xe^{2x} and e^{2x}. In other words, g is the sum of two basic kinds of functions:

$$g(x) = g_1(x) + g_2(x) = \textit{polynomial} + \textit{exponentials}.$$

Correspondingly, the superposition principle for nonhomogeneous equations (Theorem 4.1.7) suggests that we seek a particular solution

$$y_p = y_{p_1} + y_{p_2},$$

where $y_{p_1} = Ax + B$ and $y_{p_2} = Cxe^{2x} + Ee^{2x}$. Substituting

$$y_p = Ax + B + Cxe^{2x} + Ee^{2x}$$

into the given equation (3) and grouping like terms gives

$$y_p'' - 2y_p' - 3y_p = -3Ax - 2A - 3B - 3Cxe^{2x} + (2C - 3E)e^{2x} = 4x - 5 + 6xe^{2x}. \quad (4)$$

From this identity we obtain the four equations

$$-3A = 4, \qquad -2A - 3B = -5, \qquad -3C = 6, \qquad 2C - 3E = 0.$$

The last equation in this system results from the interpretation that the coefficient of e^{2x} in the right member of (4) is zero. Solving, we find $A = -\frac{4}{3}$, $B = \frac{23}{9}$, $C = -2$, and $E = -\frac{4}{3}$. Consequently,

$$y_p = -\frac{4}{3}x + \frac{23}{9} - 2xe^{2x} - \frac{4}{3}e^{2x}.$$

Step 3. The general solution of the equation is

$$y = c_1e^{-x} + c_2e^{3x} - \frac{4}{3}x + \frac{23}{9} - \left(2x + \frac{4}{3}\right)e^{2x}. \qquad \blacksquare$$

In light of the superposition principle (Theorem 4.1.7) we can also approach Example 3 from the viewpoint of solving two simpler problems. You should verify that substituting

$$y_{p_1} = Ax + B \qquad \text{into} \qquad y'' - 2y' - 3y = 4x - 5$$

and

$$y_{p_2} = Cxe^{2x} + Ee^{2x} \qquad \text{into} \qquad y'' - 2y' - 3y = 6xe^{2x}$$

yields, in turn, $y_{p_1} = -\frac{4}{3}x + \frac{23}{9}$ and $y_{p_2} = -\left(2x + \frac{4}{3}\right)e^{2x}$. A particular solution of (3) is then $y_p = y_{p_1} + y_{p_2}$.

The next example illustrates that sometimes the "obvious" assumption for the form of y_p is not a correct assumption.

EXAMPLE 4 A Glitch in the Method

Find a particular solution of $y'' - 5y' + 4y = 8e^x$.

SOLUTION Differentiation of e^x produces no new functions. Therefore proceeding as we did in the earlier examples, we can reasonably assume a particular solution of the form $y_p = Ae^x$. But substitution of this expression into the differential equation yields the contradictory statement $0 = 8e^x$, so we have clearly made the wrong guess for y_p.

The difficulty here is apparent on examining the complementary function $y_c = c_1e^x + c_2e^{4x}$. Observe that our assumption Ae^x is already present in y_c. This means that e^x is a solution of the associated homogeneous differential equation, and a constant multiple Ae^x when substituted into the differential equation necessarily produces zero.

What then should be the form of y_p? Inspired by Case II of Section 4.3, let's see whether we can find a particular solution of the form

$$y_p = Axe^x.$$

Substituting $y_p' = Axe^x + Ae^x$ and $y_p'' = Axe^x + 2Ae^x$ into the differential equation and simplifying gives

$$y_p'' - 5y_p' + 4y_p = -3Ae^x = 8e^x.$$

From the last equality we see that the value of A is now determined as $A = -\frac{8}{3}$. Therefore a particular solution of the given equation is $y_p = -\frac{8}{3}xe^x$. ■

The difference in the procedures used in Examples 1–3 and in Example 4 suggests that we consider two cases. The first case reflects the situation in Examples 1–3.

CASE I No function in the assumed particular solution is a solution of the associated homogeneous differential equation.

In Table 4.1 we illustrate some specific examples of $g(x)$ in (1) along with the corresponding form of the particular solution. We are, of course, taking for granted that no function in the assumed particular solution y_p is duplicated by a function in the complementary function y_c.

TABLE 4.1 Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A\cos 4x + B\sin 4x$
6. $\cos 4x$	$A\cos 4x + B\sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x}\sin 4x$	$Ae^{3x}\cos 4x + Be^{3x}\sin 4x$
11. $5x^2\sin 4x$	$(Ax^2 + Bx + C)\cos 4x + (Ex^2 + Fx + G)\sin 4x$
12. $xe^{3x}\cos 4x$	$(Ax + B)e^{3x}\cos 4x + (Cx + E)e^{3x}\sin 4x$

EXAMPLE 5 Forms of Particular Solutions—Case I

Determine the form of a particular solution of

(a) $y'' - 8y' + 25y = 5x^3e^{-x} - 7e^{-x}$ **(b)** $y'' + 4y = x\cos x$

SOLUTION **(a)** We can write $g(x) = (5x^3 - 7)e^{-x}$. Using entry 9 in Table 4.1 as a model, we assume a particular solution of the form

$$y_p = (Ax^3 + Bx^2 + Cx + E)e^{-x}.$$

Note that there is no duplication between the terms in y_p and the terms in the complementary function $y_c = e^{4x}(c_1 \cos 3x + c_2 \sin 3x)$.

(b) The function $g(x) = x \cos x$ is similar to entry 11 in Table 4.1 except, of course, that we use a linear rather than a quadratic polynomial and $\cos x$ and $\sin x$ instead of $\cos 4x$ and $\sin 4x$ in the form of y_p:

$$y_p = (Ax + B)\cos x + (Cx + E)\sin x.$$

Again observe that there is no duplication of terms between y_p and $y_c = c_1 \cos 2x + c_2 \sin 2x$. ■

If $g(x)$ consists of a sum of, say, m terms of the kind listed in the table, then (as in Example 3) the assumption for a particular solution y_p consists of the sum of the trial forms $y_{p_1}, y_{p_2}, \ldots, y_{p_m}$ corresponding to these terms:

$$y_p = y_{p_1} + y_{p_2} + \cdots + y_{p_m}.$$

The foregoing sentence can be put another way.

Form Rule for Case I *The form of y_p is a linear combination of all linearly independent functions that are generated by repeated differentiations of $g(x)$.*

EXAMPLE 6 Forming y_p by Superposition—Case I

Determine the form of a particular solution of

$$y'' - 9y' + 14y = 3x^2 - 5\sin 2x + 7xe^{6x}.$$

SOLUTION

Corresponding to $3x^2$ we assume $\quad y_{p_1} = Ax^2 + Bx + C.$

Corresponding to $-5\sin 2x$ we assume $\quad y_{p_2} = E\cos 2x + F\sin 2x.$

Corresponding to $7xe^{6x}$ we assume $\quad y_{p_3} = (Gx + H)e^{6x}.$

The assumption for the particular solution is then

$$y_p = y_{p_1} + y_{p_2} + y_{p_3} = Ax^2 + Bx + C + E\cos 2x + F\sin 2x + (Gx + H)e^{6x}.$$

No term in this assumption duplicates a term in $y_c = c_1e^{2x} + c_2e^{7x}$. ■

CASE II A function in the assumed particular solution is also a solution of the associated homogeneous differential equation.

The next example is similar to Example 4.

EXAMPLE 7 Particular Solution—Case II

Find a particular solution of $y'' - 2y' + y = e^x$.

SOLUTION The complementary function is $y_c = c_1e^x + c_2xe^x$. As in Example 4, the assumption $y_p = Ae^x$ will fail, since it is apparent from y_c that e^x is a solution of the associated homogeneous equation $y'' - 2y' + y = 0$. Moreover, we will not be able to find a particular solution of the form $y_p = Axe^x$, since the term xe^x is also duplicated in y_c. We next try

$$y_p = Ax^2e^x.$$

Substituting into the given differential equation yields $2Ae^x = e^x$, so $A = \frac{1}{2}$. Thus a particular solution is $y_p = \frac{1}{2}x^2e^x$. ■

Suppose again that $g(x)$ consists of m terms of the kind given in Table 4.1, and suppose further that the usual assumption for a particular solution is

$$y_p = y_{p_1} + y_{p_2} + \cdots + y_{p_m},$$

where the y_{p_i}, $i = 1, 2, \ldots, m$ are the trial particular solution forms corresponding to these terms. Under the circumstances described in Case II, we can make up the following general rule.

Multiplication Rule for Case II *If any y_{p_i} contains terms that duplicate terms in y_c, then that y_{p_i} must be multiplied by x^n, where n is the smallest positive integer that eliminates that duplication.*

EXAMPLE 8 An Initial-Value Problem

Solve $y'' + y = 4x + 10 \sin x$, $y(\pi) = 0$, $y'(\pi) = 2$.

SOLUTION The solution of the associated homogeneous equation $y'' + y = 0$ is $y_c = c_1 \cos x + c_2 \sin x$. Because $g(x) = 4x + 10 \sin x$ is the sum of a linear polynomial and a sine function, our normal assumption for y_p, from entries 2 and 5 of Table 4.1, would be the sum of $y_{p_1} = Ax + B$ and $y_{p_2} = C \cos x + E \sin x$:

$$y_p = Ax + B + C \cos x + E \sin x. \tag{5}$$

But there is an obvious duplication of the terms $\cos x$ and $\sin x$ in this assumed form and two terms in the complementary function. This duplication can be eliminated by simply multiplying y_{p_2} by x. Instead of (5) we now use

$$y_p = Ax + B + Cx \cos x + Ex \sin x. \tag{6}$$

Differentiating this expression and substituting the results into the differential equation gives

$$y_p'' + y_p = Ax + B - 2C\sin x + 2E\cos x = 4x + 10\sin x,$$

and so $A = 4$, $B = 0$, $-2C = 10$, and $2E = 0$. The solutions of the system are immediate: $A = 4$, $B = 0$, $C = -5$, and $E = 0$. Therefore from (6) we obtain $y_p = 4x - 5x\cos x$. The general solution of the given equation is

$$y = y_c + y_p = c_1\cos x + c_2\sin x + 4x - 5x\cos x.$$

We now apply the prescribed initial conditions to the general solution of the equation. First, $y(\pi) = c_1\cos\pi + c_2\sin\pi + 4\pi - 5\pi\cos\pi = 0$ yields $c_1 = 9\pi$, since $\cos\pi = -1$ and $\sin\pi = 0$. Next, from the derivative

$$y' = -9\pi\sin x + c_2\cos x + 4 + 5x\sin x - 5\cos x$$

and $$y'(\pi) = -9\pi\sin\pi + c_2\cos\pi + 4 + 5\pi\sin\pi - 5\cos\pi = 2$$

we find $c_2 = 7$. The solution of the initial-value is then

$$y = 9\pi\cos x + 7\sin x + 4x - 5x\cos x.$$ ■

EXAMPLE 9 Using the Multiplication Rule

Solve $y'' - 6y' + 9y = 6x^2 + 2 - 12e^{3x}$.

SOLUTION The complementary function is $y_c = c_1e^{3x} + c_2xe^{3x}$. And so, based on entries 3 and 7 of Table 4.1, the usual assumption for a particular solution would be

$$y_p = \underbrace{Ax^2 + Bx + C}_{y_{p_1}} + \underbrace{Ee^{3x}}_{y_{p_2}}.$$

Inspection of these functions shows that the one term in y_{p_2} is duplicated in y_c. If we multiply y_{p_2} by x, we note that the term xe^{3x} is still part of y_c. But multiplying y_{p_2} by x^2 eliminates all duplications. Thus the operative form of a particular solution is

$$y_p = Ax^2 + Bx + C + Ex^2e^{3x}.$$

Differentiating this last form, substituting into the differential equation, and collecting like terms gives

$$y_p'' - 6y_p' + 9y_p = 9Ax^2 + (-12A + 9B)x + 2A - 6B + 9C + 2Ee^{3x} = 6x^2 + 2 - 12e^{3x}.$$

It follows from this identity that $A = \frac{2}{3}$, $B = \frac{8}{9}$, $C = \frac{2}{3}$, and $E = -6$. Hence the general solution $y = y_c + y_p$ is $y = c_1e^{3x} + c_2xe^{3x} + \frac{2}{3}x^2 + \frac{8}{9}x + \frac{2}{3} - 6x^2e^{3x}$. ■

WATCH:

Undetermined Coefficients: Solving non-homogeneous ODEs

Use the following online solver:

Calculator Ordinary Differential Equations (ODE) and Systems of ODEs
Solve differential equations online
Differential Equation Calculator - eMathHelp
Function Calculator - eMathHelp

PRACTISE: DENNIS G. ZILL: 4.4: Undetermined Coefficients-Superposition Approach:
1-15, 17, 27-31

In Problems 1–26 solve the given differential equation by undetermined coefficients.

1. $y'' + 3y' + 2y = 6$

2. $4y'' + 9y = 15$

3. $y'' - 10y' + 25y = 30x + 3$

4. $y'' + y' - 6y = 2x$

5. $\frac{1}{4}y'' + y' + y = x^2 - 2x$

6. $y'' - 8y' + 20y = 100x^2 - 26xe^x$

7. $y'' + 3y = -48x^2e^{3x}$

8. $4y'' - 4y' - 3y = \cos 2x$

9. $y'' - y' = -3$

10. $y'' + 2y' = 2x + 5 - e^{-2x}$

11. $y'' - y' + \frac{1}{4}y = 3 + e^{x/2}$

12. $y'' - 16y = 2e^{4x}$

13. $y'' + 4y = 3 \sin 2x$

14. $y'' - 4y = (x^2 - 3) \sin 2x$

15. $y'' + y = 2x \sin x$

In Problems 27–36 solve the given initial-value problem.

27. $y'' + 4y = -2, \quad y\left(\frac{\pi}{8}\right) = \frac{1}{2}, y'\left(\frac{\pi}{8}\right) = 2$

28. $2y'' + 3y' - 2y = 14x^2 - 4x - 11, \quad y(0) = 0, y'(0) = 0$

29. $5y'' + y' = -6x, \quad y(0) = 0, y'(0) = -10$

30. $y'' + 4y' + 4y = (3 + x)e^{-2x}, \quad y(0) = 2, y'(0) = 5$

31. $y'' + 4y' + 5y = 35e^{-4x}, \quad y(0) = -3, y'(0) = 1$

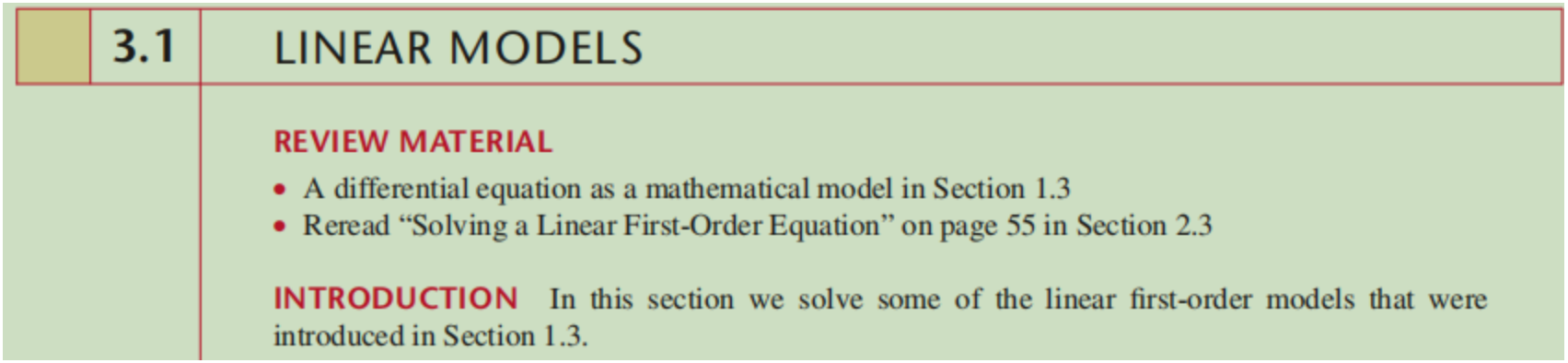
17. $y'' - 2y' + 5y = e^x \cos 2x$

- **Modeling with 1st and 2nd order DE**

Linear Models

READ:
Differential Equations - Introduction
Differential Equations Solution Guide
Solution of First Order Linear Differential Equations

READ:



3.1 LINEAR MODELS

REVIEW MATERIAL

- A differential equation as a mathematical model in Section 1.3
- Reread "Solving a Linear First-Order Equation" on page 55 in Section 2.3

INTRODUCTION In this section we solve some of the linear first-order models that were introduced in Section 1.3.

GROWTH AND DECAY The initial-value problem

$$\frac{dx}{dt} = kx, \quad x(t_0) = x_0, \tag{1}$$

where k is a constant of proportionality, serves as a model for diverse phenomena involving either growth or decay. We saw in Section 1.3 that in biological applications the rate of growth of certain populations (bacteria, small animals) over short periods of time is proportional to the population present at time t. Knowing the population at some arbitrary initial time t_0, we can then use the solution of (1) to predict the population in the future—that is, at times $t > t_0$. The constant of proportionality k in (1) can be determined from the solution of the initial-value problem, using a subsequent measurement of x at a time $t_1 > t_0$. In physics and chemistry (1) is seen in the form of a *first-order reaction*—that is, a reaction whose rate, or velocity, dx/dt is directly proportional to the amount x of a substance that is unconverted or remaining at time t. The decomposition, or decay, of U-238 (uranium) by radioactivity into Th-234 (thorium) is a first-order reaction.

EXAMPLE 1 Bacterial Growth

A culture initially has P_0 number of bacteria. At $t = 1$ h the number of bacteria is measured to be $\frac{3}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t, determine the time necessary for the number of bacteria to triple.

SOLUTION We first solve the differential equation in (1), with the symbol x replaced by P. With $t_0 = 0$ the initial condition is $P(0) = P_0$. We then use the empirical observation that $P(1) = \frac{3}{2}P_0$ to determine the constant of proportionality k.

Notice that the differential equation $dP/dt = kP$ is both separable and linear. When it is put in the standard form of a linear first-order DE,

$$\frac{dP}{dt} - kP = 0,$$

we can see by inspection that the integrating factor is e^{-kt}. Multiplying both sides of the equation by this term and integrating gives, in turn,

$$\frac{d}{dt}[e^{-kt}P] = 0 \qquad \text{and} \qquad e^{-kt}P = c.$$

Therefore $P(t) = ce^{kt}$. At $t = 0$ it follows that $P_0 = ce^0 = c$, so $P(t) = P_0e^{kt}$. At $t = 1$ we have $\frac{3}{2}P_0 = P_0e^k$ or $e^k = \frac{3}{2}$. From the last equation we get $k = \ln\frac{3}{2} = 0.4055$, so $P(t) = P_0e^{0.4055t}$. To find the time at which the number of bacteria has tripled, we solve $3P_0 = P_0e^{0.4055t}$ for t. It follows that $0.4055t = \ln 3$, or

$$t = \frac{\ln 3}{0.4055} \approx 2.71 \text{ h}.$$

See Figure 3.1.1. ■

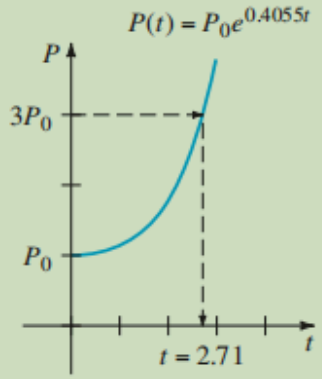


FIGURE 3.1.1 Time in which population triples

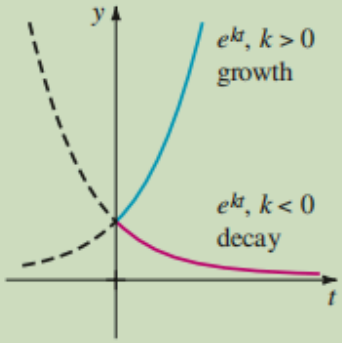


FIGURE 3.1.2 Growth ($k > 0$) and decay ($k < 0$)

Notice in Example 1 that the actual number P_0 of bacteria present at time $t = 0$ played no part in determining the time required for the number in the culture to triple. The time necessary for an initial population of, say, 100 or 1,000,000 bacteria to triple is still approximately 2.71 hours.

As shown in Figure 3.1.2, the exponential function e^{kt} increases as t increases for $k > 0$ and decreases as t increases for $k < 0$. Thus problems describing growth (whether of populations, bacteria, or even capital) are characterized by a positive value of k, whereas problems involving decay (as in radioactive disintegration) yield a negative k value. Accordingly, we say that k is either a **growth constant** ($k > 0$) or a **decay constant** ($k < 0$).

HALF-LIFE In physics the **half-life** is a measure of the stability of a radioactive substance. The half-life is simply the time it takes for one-half of the atoms in an initial amount A_0 to disintegrate, or transmute, into the atoms of another element. The longer the half-life of a substance, the more stable it is. For example, the half-life of highly radioactive radium, Ra-226, is about 1700 years. In 1700 years one-half of a given quantity of Ra-226 is transmuted into radon, Rn-222. The most commonly occurring uranium isotope, U-238, has a half-life of approximately 4,500,000,000 years. In about 4.5 billion years, one-half of a quantity of U-238 is transmuted into lead, Pb-206.

EXAMPLE 2 Half-Life of Plutonium

A breeder reactor converts relatively stable uranium 238 into the isotope plutonium 239. After 15 years it is determined that 0.043% of the initial amount A_0 of plutonium has disintegrated. Find the half-life of this isotope if the rate of disintegration is proportional to the amount remaining.

SOLUTION Let $A(t)$ denote the amount of plutonium remaining at time t. As in Example 1 the solution of the initial-value problem

$$\frac{dA}{dt} = kA, \quad A(0) = A_0$$

is $A(t) = A_0e^{kt}$. If 0.043% of the atoms of A_0 have disintegrated, then 99.957% of the substance remains. To find the decay constant k, we use $0.99957A_0 = A(15)$—that is, $0.99957A_0 = A_0e^{15k}$. Solving for k then gives $k = \frac{1}{15}\ln 0.99957 = -0.00002867$. Hence $A(t) = A_0e^{-0.00002867t}$. Now the half-life is the corresponding value of time at which $A(t) = \frac{1}{2}A_0$. Solving for t gives $\frac{1}{2}A_0 = A_0e^{-0.00002867t}$, or $\frac{1}{2} = e^{-0.00002867t}$. The last equation yields

$$t = \frac{\ln 2}{0.00002867} \approx 24{,}180 \text{ yr}. \quad \blacksquare$$

CARBON DATING About 1950 the chemist Willard Libby devised a method of using radioactive carbon as a means of determining the approximate ages of fossils. The theory of **carbon dating** is based on the fact that the isotope carbon 14 is produced in the atmosphere by the action of cosmic radiation on nitrogen. The ratio of the amount of C-14 to ordinary carbon in the atmosphere appears to be a constant, and as a consequence the proportionate amount of the isotope present in all living organisms is the same as that in the atmosphere. When an organism dies, the absorption of C-14, by either breathing or eating, ceases. Thus by comparing the proportionate amount of C-14 present, say, in a fossil with the constant ratio found in the atmosphere, it is possible to obtain a reasonable estimation of the fossil's age. The method is based on the knowledge that the half-life of radioactive C-14 is approximately 5600 years. For his work Libby won the Nobel Prize for chemistry in

1960. Libby's method has been used to date wooden furniture in Egyptian tombs, the woven flax wrappings of the Dead Sea scrolls, and the cloth of the enigmatic shroud of Turin.

EXAMPLE 3 Age of a Fossil

A fossilized bone is found to contain one-thousandth of the C-14 level found in living matter. Estimate the age of the fossil.

SOLUTION The starting point is again $A(t) = A_0e^{kt}$. To determine the value of the decay constant k, we use the fact that $\frac{1}{2}A_0 = A(5600)$ or $\frac{1}{2}A_0 = A_0e^{5600k}$. From $5600k = \ln\frac{1}{2} = -\ln 2$ we then get $k = -(\ln 2)/5600 = -0.00012378$. Therefore $A(t) = A_0e^{-0.00012378t}$. With $A(t) = \frac{1}{1000}A_0$ we have $\frac{1}{1000}A_0 = A_0e^{-0.00012378t}$, so $-0.00012378t = \ln\frac{1}{1000} = -\ln 1000$. Thus the age of the fossil is about

$$t = \frac{\ln 1000}{0.00012378} \approx 55{,}800 \text{ yr.}$$

■

The age found in Example 3 is really at the border of accuracy for this method. The usual carbon-14 technique is limited to about 9 half-lives of the isotope, or about 50,000 years. One reason for this limitation is that the chemical analysis needed to obtain an accurate measurement of the remaining C-14 becomes somewhat formidable around the point of $\frac{1}{1000}A_0$. Also, this analysis demands the destruction of a rather large sample of the specimen. If this measurement is accomplished indirectly, based on the actual radioactivity of the specimen, then it is very difficult to distinguish between the radiation from the fossil and the normal background radiation.* But recently, the use of a particle accelerator has enabled scientists to separate C-14 from stable C-12 directly. When the precise value of the ratio of C-14 to C-12 is computed, the accuracy of this method can be extended to 70,000–100,000 years. Other isotopic techniques such as using potassium 40 and argon 40 can give ages of several million years.† Nonisotopic methods based on the use of amino acids are also sometimes possible.

*The number of disintegrations per minute per gram of carbon is recorded by using a Geiger counter. The lower level of detectability is about 0.1 disintegrations per minute per gram.

†Potassium-argon dating is used in dating terrestrial materials such as minerals, rocks, and lava and extraterrestrial materials such as meteorites and lunar rocks. The age of a fossil can be estimated by determining the age of the rock stratum in which it was found.

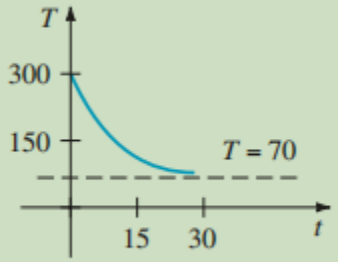
NEWTON'S LAW OF COOLING/WARMING In equation (3) of Section 1.3 we saw that the mathematical formulation of Newton's empirical law of cooling/warming of an object is given by the linear first-order differential equation

$$\frac{dT}{dt} = k(T - T_m), \tag{2}$$

where k is a constant of proportionality, $T(t)$ is the temperature of the object for $t > 0$, and T_m is the ambient temperature—that is, the temperature of the medium around the object. In Example 4 we assume that T_m is constant.

EXAMPLE 4 Cooling of a Cake

When a cake is removed from an oven, its temperature is measured at 300° F. Three minutes later its temperature is 200° F. How long will it take for the cake to cool off to a room temperature of 70° F?



(a)

SOLUTION In (2) we make the identification $T_m = 70$. We must then solve the initial-value problem

$$\frac{dT}{dt} = k(T - 70), \quad T(0) = 300 \tag{3}$$

and determine the value of k so that $T(3) = 200$.

Equation (3) is both linear and separable. If we separate variables,

(a)

$T(t)$	t (min)
75°	20.1
74°	21.3
73°	22.8
72°	24.9
71°	28.6
70.5°	32.3

(b)

FIGURE 3.1.3 Temperature of cooling cake approaches room temperature

$$\frac{dT}{T-70} = k\,dt,$$

yields $\ln|T-70| = kt + c_1$, and so $T = 70 + c_2e^{kt}$. When $t = 0$, $T = 300$, so $300 = 70 + c_2$ gives $c_2 = 230$; therefore $T = 70 + 230e^{kt}$. Finally, the measurement $T(3) = 200$ leads to $e^{3k} = \frac{13}{23}$, or $k = \frac{1}{3}\ln\frac{13}{23} = -0.19018$. Thus

$$T(t) = 70 + 230e^{-0.19018t}. \tag{4}$$

We note that (4) furnishes no finite solution to $T(t) = 70$, since $\lim_{t\to\infty} T(t) = 70$. Yet we intuitively expect the cake to reach room temperature after a reasonably long period of time. How long is "long"? Of course, we should not be disturbed by the fact that the model (3) does not quite live up to our physical intuition. Parts (a) and (b) of Figure 3.1.3 clearly show that the cake will be approximately at room temperature in about one-half hour. ■

The ambient temperature in (2) need not be a constant but could be a function $T_m(t)$ of time t. See Problem 18 in Exercises 3.1.

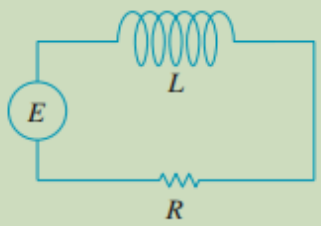


FIGURE 3.1.5 *LR* series circuit

SERIES CIRCUITS For a series circuit containing only a resistor and an inductor, Kirchhoff's second law states that the sum of the voltage drop across the inductor ($L(di/dt)$) and the voltage drop across the resistor (iR) is the same as the impressed voltage ($E(t)$) on the circuit. See Figure 3.1.5.

Thus we obtain the linear differential equation for the current $i(t)$,

$$L\frac{di}{dt} + Ri = E(t), \tag{7}$$

where L and R are constants known as the inductance and the resistance, respectively. The current $i(t)$ is also called the **response** of the system.

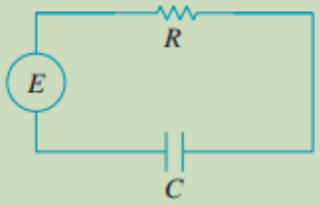


FIGURE 3.1.6 *RC* series circuit

The voltage drop across a capacitor with capacitance C is given by $q(t)/C$, where q is the charge on the capacitor. Hence, for the series circuit shown in Figure 3.1.6, Kirchhoff's second law gives

$$Ri + \frac{1}{C}q = E(t). \tag{8}$$

But current i and charge q are related by $i = dq/dt$, so (8) becomes the linear differential equation

$$R\frac{dq}{dt} + \frac{1}{C}q = E(t). \tag{9}$$

EXAMPLE 6 Series Circuit

A 12-volt battery is connected to a series circuit in which the inductance is $\frac{1}{2}$ henry and the resistance is 10 ohms. Determine the current i if the initial current is zero.

SOLUTION From (7) we see that we must solve

$$\frac{1}{2}\frac{di}{dt} + 10i = 12,$$

subject to $i(0) = 0$. First, we multiply the differential equation by 2 and read off the integrating factor e^{20t}. We then obtain

$$\frac{d}{dt}[e^{20t}i] = 24e^{20t}.$$

Integrating each side of the last equation and solving for i gives $i(t) = \frac{6}{5} + ce^{-20t}$. Now $i(0) = 0$ implies that $0 = \frac{6}{5} + c$ or $c = -\frac{6}{5}$. Therefore the response is $i(t) = \frac{6}{5} - \frac{6}{5}e^{-20t}$. ■

From (4) of Section 2.3 we can write a general solution of (7):

$$i(t) = \frac{e^{-(R/L)t}}{L} \int e^{(R/L)t} E(t)\, dt + ce^{-(R/L)t}. \tag{10}$$

In particular, when $E(t) = E_0$ is a constant, (10) becomes

$$i(t) = \frac{E_0}{R} + ce^{-(R/L)t}. \tag{11}$$

Note that as $t \to \infty$, the second term in equation (11) approaches zero. Such a term is usually called a **transient term;** any remaining terms are called the **steady-state** part of the solution. In this case E_0/R is also called the **steady-state current;** for large values of time it appears that the current in the circuit is simply governed by Ohm's law ($E = iR$).

WATCH:

- **CalcGREEN 2 : Ch. 3.3 : Affine ODEs in Practice**
- **Newton's Law of Cooling // Separable ODE Example**
- **The Logistic Growth Differential Equation**
- **CalcGREEN 2 : Ch. 4.4 : So Many ODEs**

PRACTISE: DENNIS G. ZILL: 3.1: Linear Models: Example 1-4, 7(supposed to be 6) and exercise 1-7, 13-17, 29

Growth and Decay

1. The population of a community is known to increase at a rate proportional to the number of people present at time t. If an initial population P_0 has doubled in 5 years, how long will it take to triple? To quadruple?

2. Suppose it is known that the population of the community in Problem 1 is 10,000 after 3 years. What was the initial population P_0? What will be the population in 10 years? How fast is the population growing at $t = 10$?

3. The population of a town grows at a rate proportional to the population present at time t. The initial population of 500 increases by 15% in 10 years. What will be the population in 30 years? How fast is the population growing at $t = 30$?

4. The population of bacteria in a culture grows at a rate proportional to the number of bacteria present at time t. After 3 hours it is observed that 400 bacteria are present. After 10 hours 2000 bacteria are present. What was the initial number of bacteria?

5. The radioactive isotope of lead, Pb-209, decays at a rate proportional to the amount present at time t and has a half-life of 3.3 hours. If 1 gram of this isotope is present initially, how long will it take for 90% of the lead to decay?

6. Initially 100 milligrams of a radioactive substance was present. After 6 hours the mass had decreased by 3%. If the rate of decay is proportional to the amount of the substance present at time t, find the amount remaining after 24 hours.

7. Determine the half-life of the radioactive substance described in Problem 6.

Newton's Law of Cooling/Warming

13. A thermometer is removed from a room where the temperature is 70° F and is taken outside, where the air temperature is 10° F. After one-half minute the thermometer reads 50° F. What is the reading of the thermometer at $t = 1$ min? How long will it take for the thermometer to reach 15° F?

14. A thermometer is taken from an inside room to the outside, where the air temperature is 5° F. After 1 minute the thermometer reads 55° F, and after 5 minutes it reads 30° F. What is the initial temperature of the inside room?

15. A small metal bar, whose initial temperature was 20° C, is dropped into a large container of boiling water. How long will it take the bar to reach 90° C if it is known that its temperature increases 2° in 1 second? How long will it take the bar to reach 98° C?

16. Two large containers A and B of the same size are filled with different fluids. The fluids in containers A and B are maintained at 0° C and 100° C, respectively. A small metal bar, whose initial temperature is 100° C, is lowered into container A. After 1 minute the temperature of the bar is 90° C. After 2 minutes the bar is removed and instantly transferred to the other container. After 1 minute in container B the temperature of the bar rises 10°. How long, measured from the start of the entire process, will it take the bar to reach 99.9° C?

17. A thermometer reading 70° F is placed in an oven preheated to a constant temperature. Through a glass window in the oven door, an observer records that the thermometer reads 110° F after $\frac{1}{2}$ minute and 145° F after 1 minute. How hot is the oven?

Series Circuits

29. A 30-volt electromotive force is applied to an LR series circuit in which the inductance is 0.1 henry and the resistance is 50 ohms. Find the current $i(t)$ if $i(0) = 0$. Determine the current as $t \to \infty$.